

# CHAPTER SEVEN

## *partial differential equations*

### 7-1 INTRODUCTION

In many physical problems the fundamental object under investigation is a vector or scalar field and the desired end is to be able to predict the behavior of this field throughout all space and for all time given initially only the local behavior of the field.

One method of specifying this local behavior is to give a relation which must hold at each point of space and time and which interrelates, at any given point, the rates of change of the field in different directions.

Such a relation is known as a "partial differential equation," and the problem of predicting the behavior of the field throughout all space and for all time amounts to integrating this partial differential equation.

In this chapter we shall not treat the general theory of partial differential equations at all. Rather, we shall confine our attention to certain partial differential equations of interest in physics and to the methods used to integrate them.

### 7-2 THE ROLE OF THE LAPLACIAN

The Laplacian operator  $\nabla^2$  appears in a partial differential equation whenever the difference between the local value of a function at a point and the average value of the function in a neighborhood of the point governs the further space-time evolution of the function.

To see why this is true let us consider a cube with sides of length  $a$  centered at the origin of coordinates  $x = 0$ ,  $y = 0$ ,  $z = 0$ . A scalar field defined throughout space may take on different values throughout the interior of the cube, and in particular the value  $\phi_0 = \phi(0,0,0)$  is taken on at the center of the cube. The average value  $\bar{\phi}$  of  $\phi$  in the interior of the cube is given by the mean-value theorem:

$$\bar{\phi} = \frac{1}{a^3} \int_{-a/2}^{a/2} \int_{-a/2}^{a/2} \int_{-a/2}^{a/2} \phi(x,y,z) dx dy dz \quad (7-1)$$

If the function  $\phi(x,y,z)$  is now expanded in a Taylor series about

$x = 0, y = 0, z = 0$ , then

$$\begin{aligned}\phi(x, y, z) = & \phi_0 + \left(\frac{\partial\phi}{\partial x}\right)_0 x + \left(\frac{\partial\phi}{\partial y}\right)_0 y + \left(\frac{\partial\phi}{\partial z}\right)_0 z \\ & + \frac{1}{2} \left[ \left(\frac{\partial^2\phi}{\partial x^2}\right)_0 x^2 + \left(\frac{\partial^2\phi}{\partial y^2}\right)_0 y^2 + \left(\frac{\partial^2\phi}{\partial z^2}\right)_0 z^2 \right] \\ & + \left(\frac{\partial^2\phi}{\partial x \partial y}\right)_0 xy + \left(\frac{\partial^2\phi}{\partial y \partial z}\right)_0 yz + \left(\frac{\partial^2\phi}{\partial z \partial x}\right)_0 zx + \cdots \quad (7-2)\end{aligned}$$

Inserting Eq. (7-2) into Eq. (7-1) and carrying out the integrations from  $-a/2$  to  $a/2$ , we find that the odd functions cancel out while from the squared terms we obtain

$$\frac{1}{a^3} \int_{-a/2}^{+a/2} \int_{-a/2}^{a/2} \int_{-a/2}^{a/2} \frac{1}{2} \left(\frac{\partial^2\phi}{\partial x^2}\right)_0 x^2 dx dy dz = \frac{a^2}{24} \left(\frac{\partial^2\phi}{\partial x^2}\right)_0 \quad (7-3)$$

etc. Hence,

$$\bar{\phi} = \phi_0 + \frac{a^2}{24} \left( \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} \right)_0 + \text{higher-order terms} \quad (7-4)$$

For an infinitesimal cube we have to have a good approximation,

$$\bar{\phi} - \phi_0 = \frac{a^2}{24} (\nabla^2\phi)_0 \quad (7-5)$$

and therefore the quantity  $\nabla^2\phi$  is a measure of the difference between the value of the scalar  $\phi$  at a point and the average value of  $\phi$  in an infinitesimal neighborhood of the point.

### 7-3 LAPLACE'S EQUATION

Whenever a scalar field  $\phi(x, y, z)$  has the property that its value at a point equals its average value throughout a neighborhood of the point, then according to Eq. (7-5), that field must satisfy Laplace's equation:

$$\nabla^2\phi = 0 \quad (7-6)$$

The scalar potentials associated with gravitational, electrostatic, and magnetostatic fields are examples of functions which must satisfy Laplace's equation.

### 7-4 POISSON'S EQUATION

According to Gauss' law, the total electric charge contained within a volume  $V$  enclosed by a surface  $S(V)$  is given by the flux of electric displacement through the surface  $S(V)$ , or, in other words,

$$\int_{S(V)} \mathbf{D} \cdot d\mathbf{S} = q \quad (7-7)$$

In this case the flux of  $\mathbf{D}$  must have the dimensions of electric charge. Using the three-dimensional divergence theorem, we can also write

$$\int_V \nabla \cdot \mathbf{D} dV = \int_{S(V)} \mathbf{D} \cdot d\mathbf{S} = q \quad (7-8)$$

where the divergence of  $\mathbf{D}$  represents the flux per unit volume in  $V$ . Thus  $\nabla \cdot \mathbf{D}$  has the dimensions of charge per unit volume and therefore represents the charge density in  $V$ . This is expressed by writing

$$\int_V \nabla \cdot \mathbf{D} dV = \int_V \rho dV = q \quad (7-9)$$

which, if it is to hold for all arbitrary volumes contained within  $V$ , implies that

$$\nabla \cdot \mathbf{D} = \rho \quad (7-10)$$

throughout  $V$ . If the electric displacement field is now expressed as the gradient of a scalar function  $\phi$  then

$$\mathbf{D} = -\epsilon_0 \nabla \phi \quad (7-11)$$

and consequently the scalar function  $\phi$  must satisfy Poisson's equation,

$$\nabla^2\phi = -\frac{\rho}{\epsilon_0} \quad (7-12)$$

throughout  $V$ . For example, the presence of a net positive electric charge smeared out through the interior of volume  $V$  forces the value of the potential at any point in  $V$  to be greater than its average value over a small neighborhood of the point. At any point at which the charge density vanishes, Poisson's equation reduces to Laplace's equation.

### 7-5 THE DIFFUSION EQUATION

The scalar fields which satisfy Laplace's or Poisson's equation are generally, though not always, time-independent. However, there are a number of physical processes in which the value of a scalar function  $\phi$  at any point may be different from its average value in the neighborhood at some time, but not for all values of time. In stable situations this difference causes a tendency toward equalization with time. In other words, for stable behavior the local value of  $\phi$  will approach the average value of  $\phi$  at a rate proportional to the difference between the average and local values. In quantitative terms, this relationship is expressed by means of the *diffusion equation*,

$$\frac{\partial\phi}{\partial t} = \alpha \nabla^2\phi \quad (7-13)$$

where the parameter  $\alpha$  is known as the *diffusivity*. Thus the larger the diffusivity, the more rapidly the difference between the average and local values of  $\phi$  is equalized.

The conduction of heat through a solid is a classical example of such a tendency toward equalization. Since heat flows from higher to lower temperatures in a solid, the heat current density due to a temperature field  $T$  is given by

$$\mathbf{q} = -k \nabla T \quad (7-14)$$

where  $k$  is called the "thermal conductivity" of the solid and is measured in calories per centimeter per second per degree centigrade. If  $H$  represents the heat per unit volume in the solid, then thermodynamics tells us that

$$dH = \rho c dT \quad (7-15)$$

where  $\rho$  is the density in grams per cubic centimeter, and  $c$  is the specific heat of the solid in calories per gram per degree centigrade. This equation means that a change in temperature  $dT$  is produced by the addition of an amount of heat  $dH/\rho c$  to the solid; thus,

$$\frac{dH}{dt} = \rho c \frac{dT}{dt} \quad (7-16)$$

If the solid remains fixed and rigid, then  $\mathbf{v} = 0$ , and Eq. (7-16) reduces to

$$\frac{\partial H}{\partial t} = \rho c \frac{\partial T}{\partial t} \quad (7-17)$$

The total decrease in heat in the region  $V$  per unit time is given by

$$- \int_V \frac{\partial H}{\partial t} dV \quad (7-18)$$

and if heat is to be conserved, then

$$- \int_V \frac{\partial H}{\partial t} dV = \int_{S(V)} d\mathbf{S} \cdot \mathbf{q} \quad (7-19)$$

which states that the rate of decrease of heat in region  $V$  is equal to the flux of heat current emerging from the boundary of  $V$ . Inserting Eqs. (7-14) and (7-17) into (7-19) and applying the three-dimensional divergence theorem yields

$$\begin{aligned} - \int_V \rho c \frac{\partial T}{\partial t} dV &= - \int_{S(V)} d\mathbf{S} \cdot (k \nabla T) \\ &= - \int_V \nabla \cdot (k \nabla T) dV \end{aligned} \quad (7-20)$$

or

$$\int_V \left[ \rho c \frac{\partial T}{\partial t} - \nabla \cdot (k \nabla T) \right] dV = 0 \quad (7-21)$$

and if Eq. (7-21) is to hold for all volumes contained in  $V$ , then at each point in  $V$ ,

$$\rho c \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) \quad (7-22)$$

which is the equation governing the temperature distribution in the solid. When the thermal conductivity  $k$  is a constant, Eq. (7-22) reduces to

$$\nabla^2 T = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad (7-23)$$

in which the diffusivity  $\alpha$  is given by

$$\alpha = \frac{k}{\rho c} \quad (7-24)$$

Thus we see that, as expected, the greater the thermal conductivity of a solid, the more rapidly temperature differences are smoothed out.

## 7-6 THE WAVE EQUATION

In an elastic medium which is at rest, each point may be thought of as occupying a position of stable equilibrium. Then wave motion takes place when the particles comprising the medium undergo the following kind of collective interaction. Suppose a given particle is disturbed from its position of stable equilibrium. As a result of elastic coupling, its nearest neighbors will also be displaced from their equilibrium positions. They in turn will displace their nearest neighbors, and so on, causing the original disturbance to spread or propagate through the medium. The remarkable feature of this phenomenon is that despite the essentially unlimited distance over which it can propagate energy, no particle of the medium undergoes more than a small deviation from its position of stable equilibrium!

For example, consider a stretched elastic membrane such as a drumhead. If a point on the drumhead, say  $x = 0, y = 0$ , is given a displacement  $\phi_0 = \phi(0, 0, t)$  above or below its equilibrium plane, then because of elastic coupling, the points near  $x = 0, y = 0$  will also be displaced from their equilibrium positions, though not by the same amount  $\phi_0$ . Let the average displacement in an infinitesimal neighborhood of  $x = 0, y = 0$  be called  $\bar{\phi}$ . Then  $\bar{\phi}$  and  $\phi_0$  are unequal, and the membrane acts in such a way as to reduce the difference between  $\bar{\phi}$  and  $\phi_0$ . As a result, the mass at  $x = 0, y = 0$  will be accelerated back toward the plane in which  $\phi_0 = \bar{\phi}$ . This acceleration continues, decreasing to zero as the difference between  $\phi_0$  and  $\bar{\phi}$  vanishes. As the mass at  $x = 0, y = 0$  passes through the plane in which  $\bar{\phi}$  and  $\phi_0$  are equal, its acceleration vanishes. However, its velocity is now a maximum, and therefore inertia will cause it to overshoot this plane. Consequently,  $\phi_0$  will deviate from  $\bar{\phi}$  once again. Apparently  $\phi_0$  will oscillate about the value  $\bar{\phi}$  with some characteristic frequency  $\omega$ . We can write for the acceleration of the mass at  $x = 0, y = 0$ ,

$$\frac{1}{\omega^2} \frac{\partial^2 \phi_0}{\partial t^2} + \phi_0 = \bar{\phi} \quad (7-25)$$

and using Eq. (7-5), this becomes

$$\frac{\partial^2 \phi_0}{\partial t^2} = \omega^2 (\bar{\phi} - \phi_0) = \frac{\omega^2 a^2}{24} (\nabla^2 \phi)_0 \quad (7-26)$$

Since  $\omega^2 a^2 / 24 = c^2$  has the dimensions of a velocity squared, we shorten the expression for the acceleration of the piece of membrane

at  $x = 0, y = 0$  to

$$\left(\frac{\partial^2 \phi}{\partial t^2}\right)_0 = c^2(\nabla^2 \phi)_0 \quad (7-27)$$

However, the point  $x = 0, y = 0$  is arbitrary, and therefore the acceleration of the mass at any point  $(x, y)$  must satisfy the *wave equation*

$$\frac{\partial^2 \phi}{\partial t^2} = c^2 \nabla^2 \phi \quad (7-28)$$

A more precise analysis of the vibrations of a membrane shows that

$$c = \sqrt{\frac{T}{\rho}} \quad (7-29)$$

where  $T$  is the tension or force per unit length in the membrane, and  $\rho$  is the surface density in grams per square centimeter. We see that, as expected, the greater the tension in the membrane, the more rapidly its particles oscillate about their equilibrium positions.

## 7-7 A FEW GENERAL REMARKS

The four partial differential equations which we have just derived in a more or less heuristic manner are of major importance in classical theoretical physics, and for this reason we shall give a detailed discussion of how they can be solved. One point to be emphasized is that the same general procedures apply to each of the four equations.

What do we really wish to do? For Laplace's equation and Poisson's equation we want to construct time-independent solutions in two and three space dimensions. For the time-dependent wave and diffusion equations we seek solutions which depend on one time coordinate and one, two, or three space coordinates. Furthermore, we want to construct not just any solution but rather a solution which satisfies certain specified conditions. For example, in the vibrating drumhead problem we may wish to find a function which not only satisfies the two-dimensional wave equation but also represents, together with its time derivative, the initial displacement and velocity of each point in the drumhead at  $t = 0$ . Even this may not be enough. If the drumhead is finite and rigidly clamped along the edge of the drum, then we may also demand that our displacement function, in addition to satisfying the wave equation and initial conditions, also reduce to zero on the edge of the drum for all time. A problem of this type is called a "mixed initial- and boundary-value problem" for the two-dimensional wave equation. Similarly, if the ends of a metal rod of finite length are maintained at two different temperatures, and the initial temperature distribution in the rod is specified at  $t = 0$ , then we obtain a mixed initial- and boundary-value problem for the one-dimensional diffusion equation.

Finally, the problem of the electric potential due to a point charge external to a grounded conducting spherical shell leads to a boundary-value problem for the three-dimensional Laplace equation. Here the potential must vanish on the boundary of the shell.

The solution of a partial differential equation subject to initial and boundary conditions is a complete problem as opposed to an incomplete problem of merely writing down arbitrary solutions of a given equation. Physical phenomena invariably lead to complete problems for the corresponding partial differential equations. The student will do well to get accustomed from the beginning to thinking in terms of solving not merely the partial differential equation but also the complete problem, together with all of its accessory conditions.

For the equations of potential theory, wave propagation, and diffusion, the process of solving a complete problem can always be broken down into two steps. The first step consists of representing the complete solution in terms of integrals involving the product of the accessory data with a Green's function. The Green's function is a particular solution of the given partial differential equation. It represents the response to a delta-function source term. The first step is generally accomplished with the aid of an appropriate Green's identity. The second step involves the actual construction of the Green's function, and it is here that the true practical difficulties are encountered.

## 7-8 SOLUTION OF POTENTIAL PROBLEMS IN TWO DIMENSIONS

As our first example, consider the problem of computing the electric potential due to a line charge of linear density  $q$  coul/m located between a pair of grounded parallel metal plates (Fig. 7-1). The line charge runs from  $z = -\infty$  to  $z = +\infty$  parallel to the  $z$  axis and passes through the  $x, y$  plane at  $x = 0, y = d$ . The lower plate is at  $y = 0$ , and the upper plate is at  $y = a$ . This example leads us to a complete problem for Poisson's equation in which the charge density due to the line charge is  $\rho(x, y) = q\delta(x)\delta(y - d)$  coul/m<sup>3</sup>, and  $\delta$  is the Dirac delta function.

We are to solve Poisson's equation,

$$\nabla^2 \phi = -\frac{\rho(x, y)}{\epsilon_0} \quad (7-30)$$

subject to the boundary conditions

$$\phi(x, 0) = 0 \quad -\infty \leq x \leq +\infty \quad (7-31)$$

$$\phi(x, a) = 0 \quad -\infty \leq x \leq +\infty \quad (7-32)$$

stating that the two parallel plates are grounded and therefore at zero potential. The problem will be solved by executing steps 1 and 2 of Sec. 7-7. Step 1 calls for us to express the solution of Eqs. (7-30) to (7-32) in terms of integrals involving the product of the accessory

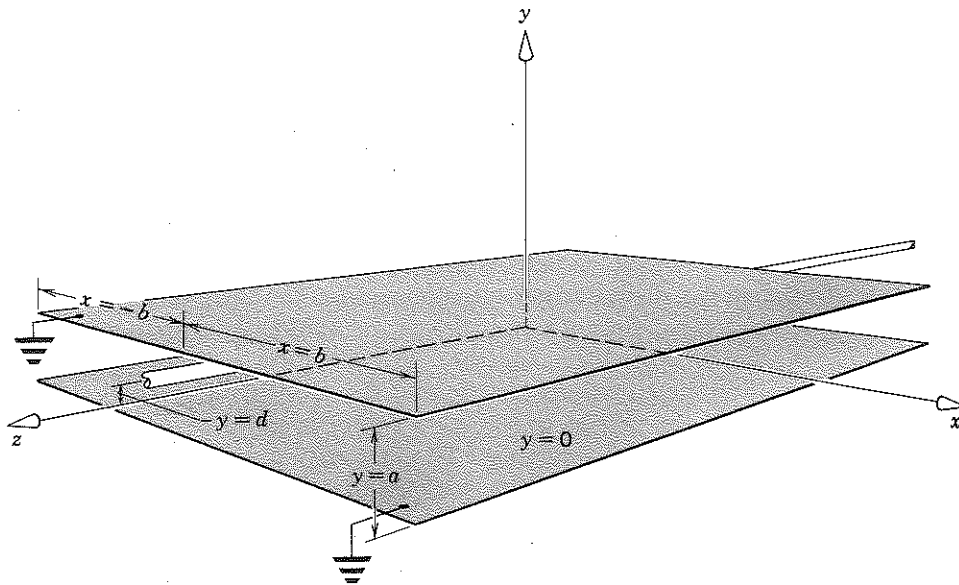


Figure 7-1 A line charge located between a pair of grounded parallel plates.

data (7-31) and (7-32) with a Green's function which will be defined below. To do this we need to develop an appropriate form of Green's identity. Since this is a two-dimensional problem, we begin with the two-dimensional divergence theorem:

$$\int_S \nabla \cdot \mathbf{A} \, dS = \int_{C(S)} d\mathbf{s} \cdot \mathbf{A} \quad (7-33)$$

Let  $\mathbf{A} = G \nabla \phi$  in Eq. (7-33), and then let  $\mathbf{A} = \phi \nabla G$  as well. Subtracting the second result from the first gives the desired form of Green's symmetric identity,

$$\int_S \{G \nabla^2 \phi - \phi \nabla^2 G\} \, dS = \int_{C(S)} \left\{ G \frac{\partial \phi}{\partial n} - \phi \frac{\partial G}{\partial n} \right\} \, ds \quad (7-34)$$

Here  $\partial \phi / \partial n$  represents the rate of change of  $\phi$  along the outward normal to  $C(S)$ , and  $ds$  represents the element of arc length on  $C(S)$ . In the problem at hand,  $S$  represents the interior of the rectangle given by

$$0 \leq y \leq a \quad (7-35)$$

$$-b \leq x \leq +b \quad (7-36)$$

and  $C(S)$  represents its perimeter. Eventually, the sides of the rectangle at  $x = \pm b$  will be allowed to recede to infinity.

The Green's function for Poisson's equation is defined as the solution of

$$\nabla^2 G = -\delta(x - x') \delta(y - y') \quad (7-37)$$

and using Eqs. (7-37), (7-34), and (7-30) together gives

$$\begin{aligned} \int_0^a \int_{-b}^b \phi(x, y) \delta(x - x') \delta(y - y') \, dx \, dy \\ = \int_0^a \int_{-b}^b G(x, x'; y, y') \frac{\rho(x, y)}{\epsilon_0} \, dx \, dy \\ + \int_{C(S)} \left\{ G \frac{\partial \phi}{\partial n} - \phi \frac{\partial G}{\partial n} \right\} \, ds \end{aligned} \quad (7-38)$$

The first term in Eq. (7-38) gives

$$\begin{aligned} \int_0^a \int_{-b}^b \phi(x, y) \delta(x - x') \delta(y - y') \, dx \, dy \\ = \phi(x', y') \int_0^a \int_{-b}^b \delta(x - x') \delta(y - y') \, dx \, dy \\ = \begin{cases} \phi(x', y') & (x', y') \text{ contained in } S \\ 0 & (x', y') \text{ outside } S \end{cases} \end{aligned} \quad (7-39)$$

Thus, assuming that  $(x', y')$  is a point contained in the rectangle  $S$ , Eq. (7-38) becomes

$$\begin{aligned} \phi(x', y') = \int_0^a \int_{-b}^b G(x, x'; y, y') \frac{\rho(x, y)}{\epsilon_0} \, dx \, dy \\ + \int_{C(S)} \left\{ G \frac{\partial \phi}{\partial n} - \phi \frac{\partial G}{\partial n} \right\} \, ds \end{aligned} \quad (7-40)$$

We must now simplify the line-integral contribution to Eq. (7-40) as much as possible, and if possible, eliminate it entirely. The boundary conditions (7-31) and (7-32) eliminate the term  $\phi \partial G / \partial n$  on the portions  $y = 0$  and  $y = a$  of  $C(S)$ . We do not have any information regarding the behavior of  $\phi(x, y)$  on  $x = \pm b$ ;  $0 \leq y \leq a$ . However, since the line charge is located at  $x = 0, y = d$ , it is reasonable to assume that, provided there are no line sources at infinity, the greater  $b$  is, the smaller the potential at  $(b, y)$  will be. Thus we assume that

$$\lim_{\substack{x \rightarrow \pm \infty \\ 0 < y < a}} \phi(x, y) = 0 \quad (7-41)$$

and that as a result, the contributions to the second line-integral term in Eq. (7-40) arising from integrating  $\phi \partial G / \partial n$  over the ends of the rectangle at  $x = \pm b$  vanishes as these ends recede to infinity. Under these circumstances Eq. (7-40) may be replaced by

$$\phi(x', y') = \int_0^a \int_{-b}^b G(x, x'; y, y') \frac{\rho(x, y)}{\epsilon_0} \, dx \, dy + \int_{C(S)} G \frac{\partial \phi}{\partial n} \, ds \quad (7-42)$$

and if we assume that

$$G(x, 0) = 0 \quad -\infty \leq x \leq +\infty \quad (7-43)$$

$$G(x, a) = 0 \quad -\infty \leq x \leq +\infty \quad (7-44)$$

$$\lim_{\substack{x \rightarrow \pm \infty \\ 0 < y < a}} G(x, y) = 0 \quad (7-45)$$

as well, then the remaining line integral also vanishes as the ends of the rectangle recede to infinity, leaving us with

$$\phi(x', y') = \int_0^a \int_{-\infty}^{+\infty} G(x, x'; y, y') \frac{\rho(x, y)}{\epsilon_0} dx dy \quad (7-46)$$

Equation (7-46) is the formal solution of problem (7-30) to (7-32), and we can simplify (7-46) even further by recalling that in our special case

$$\rho(x, y) = q \delta(x) \delta(y - d) \quad (7-47)$$

where  $0 < d < a$ . Therefore (7-46) reduces to

$$\begin{aligned} \phi(x', y') &= \frac{q}{\epsilon_0} \int_0^a \int_{-\infty}^{+\infty} G(x, x'; y, y') \delta(x) \delta(y - d) dx dy \\ &= \frac{q}{\epsilon_0} G(0, x'; d, y') \int_0^a \int_{-\infty}^{+\infty} \delta(x) \delta(y - d) dx dy \\ &= q \frac{G(0, x'; d, y')}{\epsilon_0} \end{aligned} \quad (7-48)$$

Consequently,

$$\phi(x', y') = q \frac{G(0, x'; d, y')}{\epsilon_0} \quad (7-49)$$

represents the electric potential at the point  $(x', y')$  due to a line source of charge of linear density  $q$  located at  $x = 0, y = d$ . By analogy,  $qG(x, x'; y, y')/\epsilon_0$  must represent the electric potential at  $(x', y')$  due to a line source of charge of linear density  $q$  located at  $(x, y)$ . This observation permits us to interpret (7-46). Equation (7-46) states that the potential at the point  $(x', y')$  due to an arbitrary charge density distribution  $\rho(x, y)$  may be represented as a superposition of the potentials generated by an infinity of line sources, the line source at the point  $(x, y)$  having a linear charge density  $\rho(x, y) dx dy$  coul/m. It is only because Poisson's equation is a linear partial differential equation that we may represent the potential due to  $\rho(x, y)$  as the superposition of the potentials due to appropriately placed line charges having the correct linear charge density.

To complete the solution of problem (7-30) to (7-32) we must perform step 2 of Sec. 7-7, which requires that we actually construct the Green's function  $G(x, x'; y, y')$  which satisfies

$$\nabla^2 G = -\delta(x - x') \delta(y - y') \quad (7-50)$$

$$G = 0 \text{ on } y = 0 \quad -\infty \leq x \leq +\infty \quad (7-51)$$

$$G = 0 \text{ on } y = a \quad -\infty \leq x \leq +\infty \quad (7-52)$$

$$\lim_{|x| \rightarrow \infty} G = 0 \quad 0 < y < a \quad (7-53)$$

At this point the student may wonder why we bothered with step 1 at all, since we are left with the task of solving Eqs. (7-50) to (7-53), which is the original problem. There are three reasons. First of all, if instead of a line charge, we are given distributed charge density  $\rho(x, y)$ , then step 1 is necessary to obtain Eq. (7-46). Secondly,

although in this problem the two line integrals in Eq. (7-40) vanish, there are many other problems in which they do not both vanish, and this possibility must be kept in mind. Finally, we are interested in developing a general technique of attacking boundary-value problems in potential theory, wave propagation, and diffusion, and the Green's function method is one of the most powerful and most general techniques available.

### Construction of the Green's function

The easiest and fastest way to construct the Green's function which satisfies Eqs. (7-50) to (7-53) is to employ integral transforms, particularly since these have already been introduced in Chap. 5. Another more classical but clumsier way to construct the solution is based on the method of separation of variables, and we shall illustrate both methods for purposes of comparison. The integral transform method can immediately take the boundary conditions (7-51) and (7-52) into account. Thus the vanishing of  $G$  on  $y = 0$  and  $y = a$  suggests that we take a finite sine transform of Eq. (7-50). Multiplying (7-50) through by  $\sin n\pi y/a$ , integrating twice by parts with respect to  $y$  from 0 to  $a$ , and then eliminating the integrated parts with the aid of Eqs. (7-51) and (7-52) gives an ordinary differential equation,

$$\frac{d^2 \bar{G}}{dx^2} - \left(\frac{n\pi}{a}\right)^2 \bar{G} = -\delta(x - x') \sin \frac{n\pi y'}{a} \quad (7-54)$$

for the determination of  $\bar{G}$ . Here

$$\bar{G} = \int_0^a G \sin \frac{n\pi y}{a} dy \quad (7-55)$$

is the finite sine transform of  $G$ . The ordinary differential equation (7-54) has already been encountered in both Chaps. 5 and 6, and its general solution is

$$\bar{G} = A e^{n\pi x/a} + B e^{-n\pi x/a} + \frac{e^{-n\pi x > /a} e^{n\pi x < /a}}{W\{e^{-n\pi x'/a}, e^{n\pi x'/a}; x'\}} \sin \frac{n\pi y'}{a} \quad (7-56)$$

where the reader will recall that

$$x_{>} = \text{greater of } (x, x') \quad (7-57)$$

$$x_{<} = \text{lesser of } (x, x') \quad (7-58)$$

and that  $W\{e^{-n\pi x'/a}, e^{n\pi x'/a}; x'\}$  is the Wronskian of  $e^{-n\pi x'/a}$  and  $e^{n\pi x'/a}$  which is defined by

$$W = \begin{vmatrix} e^{-n\pi x'/a} & e^{n\pi x'/a} \\ -\frac{n\pi}{a} e^{-n\pi x'/a} & \frac{n\pi}{a} e^{n\pi x'/a} \end{vmatrix} = \frac{2\pi n}{a} \quad (7-59)$$

The boundary condition (7-53) stating that  $\bar{G}$  must vanish as  $x \rightarrow \pm \infty$  means that we must set both  $A$  and  $B$  equal to zero in Eq. (7-56), leaving

$$\bar{G} = \frac{a}{2\pi n} \sin \frac{n\pi y'}{a} e^{-n\pi x_{>}/a} e^{n\pi x_{<}/a} \quad (7-60)$$

The inversion theorem for the finite sine transform,

$$G = \frac{2}{a} \sum_{n=1}^{\infty} \bar{G} \sin \frac{n\pi y}{a} \quad (7-61)$$

then gives

$$G(x, x'; y, y') = \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi y'}{a} \sin \frac{n\pi y}{a} e^{-n\pi x > / a} e^{n\pi x < / a} \quad (7-62)$$

which is the desired Green's-function solution of Eqs. (7-50) to (7-53). From Eq. (7-49),

$$\phi(x', y') = \frac{q}{\epsilon_0} G(0, x'; d, y') \quad (7-63)$$

we recover the expression for the potential at  $(x', y')$  due to the line source at  $x = 0, y = d$ . Thus, putting  $x = 0, y = d$  in Eq. (7-62), we find

$$\phi(x', y') = \frac{q}{\pi \epsilon_0} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi d}{a} e^{-n\pi x' / a} \sin \frac{n\pi y'}{a} \quad (7-64)$$

for  $x' > 0$  and

$$\phi(x', y') = \frac{q}{\pi \epsilon_0} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi d}{a} e^{n\pi x' / a} \sin \frac{n\pi y'}{a} \quad (7-65)$$

for  $x' < 0$ . Notice that the character of the solution of Eq. (7-54) has forced a division of the space between the plates into two regions according to whether  $x'$  is greater or less than the coordinate  $x = 0$  of the source.

This is not the only way in which the space between the plates may be subdivided. We may also separate the space between the plates into two regions according to whether  $y'$  is greater or less than the  $y$  coordinate of the source.

To do this we observe that the  $x$  subdivision occurs when a finite sine transform on  $y$  is used to solve Eq. (7-50). Hence we expect that a subdivision in  $y$  should occur when a transform with respect to  $x$  is used to solve (7-50). Since, in this problem, the  $x$  coordinate has the infinite range  $-\infty \leq x \leq +\infty$ , the complex Fourier transform is the most appropriate one to use.

Multiplying Eq. (7-50) through by  $e^{-ikx}$  and integrating twice by parts with respect to  $x$  from  $x = -\infty$  to  $x = +\infty$  gives us

$$\frac{d^2 \bar{G}}{dy^2} - k^2 \bar{G} = -e^{-ikx'} \delta(y - y') \quad (7-66)$$

where  $\bar{G} = \int_{-\infty}^{+\infty} G e^{-ikx} dx$  and provided that

$$\lim_{|x| \rightarrow \infty} \left( \frac{\partial G}{\partial x} + ikG \right) = 0 \quad (7-67)$$

Equation (7-67) may be reasonably expected to hold because Eq. (7-53) implies that the potential very far from any sources is zero, and therefore the electric field, which is the gradient of the potential, should also vanish at such remote places.

The general solution of Eq. (7-66) is given by

$$\bar{G} = A \sinh ky + B \cosh ky + \frac{\sinh ky > \cosh ky <}{W\{\sinh ky', \cosh ky'; y'\}} e^{-ikx'} \quad (7-68)$$

and the Wronskian of  $\sinh ky'$  and  $\cosh ky'$  is

$$W = \begin{vmatrix} \sinh ky' & \cosh ky' \\ k \cosh ky' & k \sinh ky' \end{vmatrix} = -k \quad (7-69)$$

Therefore

$$\bar{G} = A \sinh ky + B \cosh ky - (\sinh ky > \cosh ky <) \frac{e^{-ikx'}}{k} \quad (7-70)$$

We must now satisfy the boundary conditions (7-51) and (7-52). On  $y = 0, \bar{G} = 0$ , and so

$$B = \frac{\sinh ky'}{k} e^{-ikx'} \quad (7-71)$$

On  $y = a, \bar{G} = 0$  gives

$$A \sinh ka + B \cosh ka - \sinh ka \cosh ky' \frac{e^{-ikx'}}{k} = 0 \quad (7-72)$$

and combining Eqs. (7-71) and (7-72), we find

$$A = \frac{\sinh k(a - y')}{k \sinh ka} e^{-ikx'} \quad (7-73)$$

$$B = \sinh ky' \frac{e^{-ikx'}}{k} \quad (7-74)$$

Therefore  $\bar{G}$  becomes

$$\bar{G} = \frac{e^{-ikx'}}{k} \left[ \frac{\sinh k(a - y')}{\sinh ka} \sinh ky + \sinh ky' \cosh ky - \sinh ky > \cosh ky < \right] \quad (7-75)$$

Equation (7-75) provides us with a natural subdivision of the space between the two plates into two regions according to whether  $y$  is greater or less than  $y'$ .

For  $y < y'$ ,

$$\bar{G} = \frac{e^{-ikx'}}{k} \left[ \frac{\sinh k(a - y')}{\sinh ka} \sinh ky \right] \quad (7-76)$$

whereas for  $y > y'$ ,

$$\bar{G} = \frac{e^{-ikx'}}{k} \left[ \frac{\sinh k(a - y')}{\sinh ka} \sinh ky - \sinh k(y - y') \right] \quad (7-77)$$

The inversion theorem (5-320) for Fourier transforms now gives us the following expressions for the Green's function:

$$G(x, x'; y, y') = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{ik(x-x')} \frac{\sinh k(a-y')}{k \sinh ka} \sinh ky \, dk \quad (7-78)$$

for  $y < y'$ , and

$$G(x, x'; y, y') = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{ik(x-x')} \, dk \left[ \frac{\sinh k(a-y')}{k \sinh ka} \sinh ky - \frac{\sinh k(y-y')}{k} \right] \quad (7-79)$$

for  $y > y'$ . The potentials for  $y' > d$  and  $y' < d$  are obtained from Eq. (7-49), and they are given by

$$\phi(x', y') = \frac{q}{2\pi\epsilon_0} \int_{-\infty}^{+\infty} e^{-ikx'} \frac{\sinh k(a-y')}{k \sinh ka} \sinh kd \, dk \quad (7-80)$$

for  $d \leq y' \leq a$  and

$$\phi(x', y') = \frac{q}{2\pi\epsilon_0} \int_{-\infty}^{+\infty} e^{-ikx'} \, dk \left[ \frac{\sinh k(a-y')}{k \sinh ka} \sinh kd - \frac{\sinh k(d-y')}{k} \right] \quad (7-81)$$

for  $0 \leq y' \leq d$ .

The expressions (7-80) and (7-81) represent the electric potential at the point  $(x', y')$  due to a line source at  $x = 0$ ,  $y = d$ . These expressions for the potentials can be simplified even further. For example, the integrand in (7-81) can be shortened by using the identity

$$\begin{aligned} \frac{\sinh k(a-y')}{k \sinh ka} \sinh kd - \frac{\sinh k(d-y')}{k} \\ = \frac{\sinh k(a-y') \sinh kd - \sinh ka \sinh k(d-y')}{k \sinh ka} \\ = \frac{\sinh k(a-d)}{k \sinh ka} \sinh ky' \end{aligned} \quad (7-82)$$

Next we recall that  $e^{-ikx'} = \cos kx' - i \sin kx'$ , and observing that the integrands in Eqs. (7-80) and (7-81), exclusive of the factor  $e^{-ikx'}$ , are strictly even in  $k$ , we conclude that only the real parts of (7-80) and (7-81) are nonzero. This happens because the  $i \sin kx'$  contributions to (7-80) and (7-81), being odd, integrate to zero over the symmetric  $k$  interval  $[-\infty, +\infty]$ . Thus we are left with

$$\phi(x', y') = \frac{q}{2\pi\epsilon_0} \int_{-\infty}^{+\infty} \frac{\sinh kd}{k \sinh ka} \cos kx' \sinh k(a-y') \, dk \quad (7-83)$$

for  $d \leq y' \leq a$ , and

$$\phi(x', y') = \frac{q}{2\pi\epsilon_0} \int_{-\infty}^{+\infty} \frac{\sinh k(a-d)}{k \sinh ka} \cos kx' \sinh ky' \, dk \quad (7-84)$$

which holds for  $0 \leq y' \leq d$ .

**REMARKS:** The existence of two alternative ways to divide the space between the plates is not unique to this problem. In fact, most two-dimensional problems in potential theory, wave propagation, and diffusion exhibit a similar phenomenon when a Green's function must be determined.

How can we decide which of the two forms of the solution (7-64) and (7-65) or (7-83) and (7-84) is better? The solution (7-64) and (7-65) leaves us with the problem of summing a pair of infinite series, while (7-83) and (7-84) asks us to evaluate two complicated integrals.

When the coordinate  $x = x'$  of the observer differs markedly from the coordinate  $x = 0$  of the source, then the exponential factor  $e^{-\pi|x'|/a}$  occurring in Eqs. (7-64) and (7-65) is very small to begin with, and grows even smaller as  $n$  increases. Thus (7-64) and (7-65) are rapidly converging alternating series, the first few terms of which will give an excellent approximation to  $\phi(x', y')$ . On the other hand, when  $|x'|$  is large, the  $\cos kx'$  factor appearing in both (7-83) and (7-84) causes the integrands to oscillate very rapidly with changing  $k$ . As a result, the integrals will converge only very slowly to  $\phi(x', y')$ . Obviously, when the  $x$  coordinate of the observer differs markedly from the  $x$  coordinate of the source, the best solution to use is (7-64) and (7-65). If instead of the  $x$  coordinate, it is the  $y$  coordinate of the observer which differs markedly from the  $y$  coordinate of the source, then the situation is reversed. The exponential terms in (7-64) and (7-65), while still assisting convergence, are not so effective as before, and the two series will converge to  $\phi(x', y')$  much more slowly than previously. At the same time, the factor  $\cos kx'$  no longer oscillates as rapidly with changing  $k$  as it did before. Consequently, the integrals (7-83) and (7-84) will converge more rapidly to  $\phi(x', y')$  than before. Hence, when the  $y$  coordinate of the observer differs markedly from the  $y$  coordinate of the source, the best solution to use is (7-83) and (7-84). We can summarize these results in the form of a rule:

In two-dimensional problems in potential theory, wave propagation, and diffusion, there will exist two distinct representations of the Green's function. One such representation will be found by taking an integral transform with respect to the  $y$  coordinate, and the other one by taking an integral transform with respect to the  $x$  coordinate.

When the  $x$  coordinate of the observer differs markedly from the  $x$  coordinate of the source, the Green's function found by using the integral transform on  $y$  converges the most rapidly. When the  $y$  coordinate of the observer differs markedly from the  $y$  coordinate of the source, the Green's function found by using the integral transform on  $x$  converges the most rapidly.

### Reciprocity principle

The Green's function  $G(x, x'; y, y')$  represents, apart from the constant factor of  $q/\epsilon_0$ , the potential at the point  $(x', y')$  due to a line source at

$(x, y)$ . We shall see that the Green's function satisfies the *reciprocity relation*

$$G(x, x'; y, y') = G(x', x; y', y) \quad (7-85)$$

and consequently  $G(x, x'; y, y')$  must also represent the potential at  $(x, y)$  due to a line source at  $(x', y')$ .

In other words, the points  $(x, y)$  and  $(x', y')$  are completely reciprocal in the sense that the potential at the second point due to a source at the first point is the same as the potential at the first point due to the same source placed at the second point.

To see that (Eq. 7-85) is true, let us begin with the identity

$$2 \sin \frac{n\pi y'}{a} \sin \frac{n\pi y}{a} = \cos \frac{n\pi}{a} (y - y') - \cos \frac{n\pi}{a} (y + y') \quad (7-86)$$

Inserting Eq. (7-86) into (7-62) and making use of the fact that

$$e^{-n\pi x} > /a e^{n\pi x} < /a = e^{(-n\pi/a)|x-x'|} \quad (7-87)$$

we can reduce (7-62) to

$$G(x, x'; y, y') = \frac{1}{2\pi} \sum_{n=1}^{\infty} \left[ \cos \frac{n\pi}{a} (y - y') - \cos \frac{n\pi}{a} (y + y') \right] e^{(-n\pi/a)|x-x'|} \quad (7-88)$$

and since  $|x - x'| = |x' - x|$  and the cosine is an even function, we have shown that

$$G(x, x'; y, y') = G(x', x; y', y)$$

Similarly, Eqs. (7-78) and (7-79) can be reduced to the single equation,

$$G(x, x'; y, y') = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{\sinh k(a - y_{>})}{k \sinh ka} \sinh ky_{<} \cos k(x - x') dk \quad (7-89)$$

by using the identity

$$\frac{\sinh k(a - y) \sinh ky}{k \sinh ka} - \frac{\sinh k(y - y')}{k} = \frac{\sinh ky' \sinh k(a - y)}{k \sinh ka} \quad (7-90)$$

and remembering that since the integrand, apart from the factor  $e^{ik(x-x')} = \cos k(x - x') + i \sin k(x - x')$ , is even in  $k$ , only the cosine term will survive the integration from  $-\infty$  to  $+\infty$ . Finally, we observe that since

$$\sinh k(a - y_{>}) \sinh ky_{<} = \cosh k[a - (y_{>} - y_{<})] - \cosh k[a - (y_{>} + y_{<})] \quad (7-91)$$

and

$$y_{>} - y_{<} = |y - y'| \quad (7-92)$$

Eq. (7-89) becomes

$$G(x, x'; y, y') = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{\cosh k(a - |y - y'|) - \cosh k[a - (y + y')]}{k \sinh ka} \cosh k(x - x') dk \quad (7-93)$$

and once again we have demonstrated that

$$G(x, x'; y, y') = G(x', x; y', y)$$

## 7-9 SEPARATION OF VARIABLES

An alternative to using integral transforms to construct the Green's function defined by

$$\nabla^2 G = -\delta(x - x') \delta(y - y') \quad (7-94)$$

$$G = 0 \text{ on } y = 0 \quad -\infty \leq x \leq +\infty \quad (7-95)$$

$$G = 0 \text{ on } y = a \quad -\infty \leq x \leq +\infty \quad (7-96)$$

$$\lim_{|x| \rightarrow \infty} G = 0 \quad 0 < y < a \quad (7-97)$$

is to use the classical method of separation of variables.

This method begins with the observation that except at the point  $x = x'$ ,  $y = y'$  the Green's function  $G$  must satisfy Laplace's equation:

$$\nabla^2 G = 0 \quad (7-98)$$

One then seeks a trial solution of Eq. (7-98) of the form

$$G = X(x)Y(y) \quad (7-99)$$

Substituting Eq. (7-99) into Eq. (7-98) gives

$$XY'' + XY'' = 0 \quad (7-100)$$

and on dividing both sides of Eq. (7-100) by Eq. (7-99), we find

$$\frac{X''(x)}{X(x)} + \frac{Y''(y)}{Y(y)} = 0 \quad (7-101)$$

Since  $x$  and  $y$  vary independently, the only condition under which Eq. (7-101) can hold is that  $X''(x)/X(x)$  and  $Y''(y)/Y(y)$  are both constants. In particular, we can write

$$\frac{X''(x)}{X(x)} = -\frac{Y''(y)}{Y(y)} = C \quad (7-102)$$

where the parameter  $C$  is called the separation constant. This separation of variables gives rise to two ordinary differential equations,

$$X''(x) - CX(x) = 0 \quad (7-103)$$

and

$$Y''(y) + CY(y) = 0 \quad (7-104)$$

which determine  $X$  and  $Y$ .

If there are no restrictions on the separation constant  $C$ , the product of the general solutions of the ordinary differential equations forms a fairly general solution of the two-dimensional Laplace equation.

As we shall see, the boundary conditions of the problem limit both the nature of the solution and the values of the separation constant  $C$ . For our purposes, it is convenient to let

$$C = \pm k^2 \quad (7-105)$$

so that the ordinary differential equations become

$$X'' \pm k^2 X = 0 \quad (7-106)$$

and

$$Y'' \pm k^2 Y = 0 \quad (7-107)$$

When  $C = +k^2$ , the general solutions of Eqs. (7-105) and (7-106) are

$$X(x) = D(k)e^{kx} + E(k)e^{-kx} \quad (7-108)$$

$$Y(y) = A(k) \sin ky + B(k) \cos ky \quad (7-109)$$

and for  $C = -k^2$ , the corresponding general solutions may be written in the form

$$X(x) = D(k) \cos kx + E(k) \sin kx \quad (7-110)$$

$$Y(y) = A(k) \sinh ky + B(k) \cosh ky \quad (7-111)$$

Thus both

$$G = X(x)Y(y) = \{D(k)e^{kx} + E(k)e^{-kx}\} \{A(k) \sin ky + B(k) \cos ky\} \quad (7-112)$$

and

$$G = X(x)Y(y) = \{D(k) \cos kx + E(k) \sin kx\} \{A(k) \sinh ky + B(k) \cosh ky\} \quad (7-113)$$

are possible solutions of Laplace's equation  $\nabla^2 G = 0$ . The arbitrary parameters  $A$ ,  $B$ ,  $D$ , and  $E$  may depend on the separation constant  $k$ , and on  $x'$  and  $y'$ , but not on  $x$  or  $y$ .

Both solutions (7-112) and (7-113) can be tailored to satisfy the boundary conditions at  $y = 0$  and  $y = a$ . For example, consider Eq. (7-112). Let  $k = n\pi/a$  where  $n$  is an integer, and choose  $B = 0$ . Then  $G_n$  vanishes at the upper and lower plates, and (7-112) reduces to

$$G_n = X_n(x) \sin \frac{n\pi y}{a} \quad (7-114)$$

where

$$X_n(x) = A \{De^{n\pi x/a} + Ee^{-n\pi x/a}\} \quad (7-115)$$

Since Laplace's equation is linear and Eq. (7-114) is a solution of it for arbitrary integer values of  $n$ , a superposition

$$G = \sum_{n=1}^{\infty} X_n(x) \sin \frac{n\pi y}{a} \quad (7-116)$$

of the solutions (7-114) will also satisfy  $\nabla^2 G = 0$ .

The functions  $X_n(x)$  and  $Y_n(y) = \sin n\pi y/a$  are called the "eigenfunctions" of the separated equations,

$$X_n'' = \left(\frac{n\pi}{a}\right)^2 X_n \quad (7-117)$$

$$Y_n'' = -\left(\frac{n\pi}{a}\right)^2 Y_n \quad (7-118)$$

obtained by applying the method of separation of variables to Laplace's equation. The numbers  $n\pi/a$  are the corresponding eigenvalues associated with the eigenfunctions  $X_n(x)$  and  $Y_n(y)$ . Whenever a linear operator acting on a function yields a constant times the function, the constant is called an "eigenvalue," the function is called an "eigenfunction," and the resulting equation is known as an "eigenvalue equation."

In the method of separation of variables, the allowed values of the separation constant generate the eigenvalues, the separation process gives the eigenvalue equations, and their solutions provide the corresponding eigenfunctions. The allowed values of the separation constant are usually determined by the boundary conditions. Here, for example, setting  $k = n\pi/a$ , where  $n$  is an integer, allows  $G$  to vanish at the upper and lower plates.

Equation (7-116) represents a solution of Laplace's equation as an expansion in products of eigenfunctions of the separated equations. Although (7-116) vanishes at  $y = 0$  and  $y = a$ , it does not yet satisfy the boundary condition (7-97) at infinity. Furthermore, (7-116) is a solution of the homogeneous equation  $\nabla^2 G = 0$ , and what we really want is a solution of the inhomogeneous equation  $\nabla^2 G = -\delta(x - x')\delta(y - y')$ . Our problem, therefore, is to choose  $X_n(x)$  so that it satisfies three conditions. These are that it must be a solution of Eq. (7-117), except perhaps at  $x = x'$ ; it must be such that

$$G = \sum_{n=1}^{\infty} X_n(x) \sin \frac{n\pi y}{a} \quad (7-119)$$

is a solution of  $\nabla^2 G = -\delta(x - x')\delta(y - y')$ ; and finally,  $X_n(x)$  must satisfy

$$\lim_{|x| \rightarrow \infty} X_n(x) = 0 \quad (7-120)$$

To fix these  $X_n$ 's, let us treat them as essentially undetermined coefficients in the expansion (7-119). If we can expand  $-\delta(x - x')\delta(y - y')$  in terms of  $\sin n\pi y/a$ , then we can use Eq. (7-94) to determine the  $X_n$ 's. Assuming an expansion

$$-\delta(x - x')\delta(y - y') = -\sum_{n=1}^{\infty} \delta(x - x')A_n \sin \frac{n\pi y}{a} \quad (7-121)$$

we multiply Eq. (7-121) through by  $\sin m\pi y/a$  and integrate with respect to  $y$  from 0 to  $a$  using

$$\int_0^a \sin \frac{n\pi y}{a} \sin \frac{m\pi y}{a} dy = \frac{a}{2} \delta_{nm} \quad (7-122)$$

to find

$$A_n = \frac{2}{a} \sin \frac{n\pi y'}{a} \quad (7-123)$$

and as a result

$$-\delta(x - x') \delta(y - y') = \frac{-2}{a} \sum_{n=1}^{\infty} \delta(x - x') \sin \frac{n\pi y'}{a} \sin \frac{n\pi y}{a} \quad (7-124)$$

It follows from Eqs. (7-119) and (7-124) that  $\nabla^2 G = -\delta(x - x') \delta(y - y')$  yields

$$\begin{aligned} \sum_{n=1}^{\infty} \left[ \frac{d^2 X_n}{dx^2} - \left( \frac{n\pi}{a} \right)^2 X_n \right] \sin \frac{n\pi y}{a} \\ = \frac{-2}{a} \sum_{n=1}^{\infty} \delta(x - x') \sin \frac{n\pi y'}{a} \sin \frac{n\pi y}{a} \end{aligned} \quad (7-125)$$

and on equating the coefficients of  $\sin n\pi y/a$  on both sides of Eq. (7-125), we obtain

$$\frac{d^2 X_n}{dx^2} - \left( \frac{n\pi}{a} \right)^2 X_n = \frac{-2}{a} \sin \frac{n\pi y'}{a} \delta(x - x') \quad (7-126)$$

This equation is essentially Eq. (7-54), and its particular integral is

$$X_n(x) = \frac{1}{n\pi} \sin \frac{n\pi y'}{a} e^{-n\pi x > /a} e^{n\pi x < /a} \quad (7-127)$$

We see from Eq. (7-127) that except at  $x = x'$ ,  $X_n(x)$  satisfies Eq. (7-117) and has the correct behavior (7-120) at infinity. Inserting (7-127) into Eq. (7-119) gives

$$G(x, x'; y, y') = \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi y'}{a} \sin \frac{n\pi y}{a} e^{-n\pi x > /a} e^{n\pi x < /a} \quad (7-128)$$

and we have recovered Eq. (7-62).

Instead of beginning with Eq. (7-112), we could have used (7-113). However, since the hyperbolic sine and cosine are not periodic functions, we can no longer make  $G$  vanish on  $y = 0$  and  $y = a$  by choosing  $k = n\pi/a$ . Thus the separation constant  $k$  must remain a continuous parameter. Application of the principle of superposition permits us to write solutions of Laplace's equation like

$$G = \int_{-\infty}^{+\infty} \{ D(k) \cos kx + E(k) \sin kx \} \\ \{ A(k) \sinh ky + B(k) \cosh ky \} dk \quad (7-129)$$

wherein the summation over a discrete separation parameter such as  $n\pi/a$  is replaced by an integration over the continuous separation parameter  $k$ .

Our problem now is to determine the coefficients  $A$ ,  $B$ ,  $D$ , and  $E$  in such a way that Eqs. (7-94) through (7-97) are satisfied. One

clue as to how to proceed is based on the Fourier integral representation of the Dirac delta function:

$$\delta(x - x') = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \cos k(x - x') dk \quad (7-130)$$

Using Eq. (7-130), we can write  $-\delta(x - x') \delta(y - y')$  as

$$-\delta(x - x') \delta(y - y') = -\frac{1}{2\pi} \int_{-\infty}^{+\infty} \delta(y - y') \cos k(x - x') dk \quad (7-131)$$

and this is the present equivalent of (7-124). Since Eq. (7-131) depends on  $\cos k(x - x')$ , we choose  $D$  and  $E$  in Eq. (7-129) in such a way that Eq. (7-129) takes the form

$$G = \int_{-\infty}^{+\infty} Y(k, y) \cos k(x - x') dk \quad (7-132)$$

where

$$Y(k, y) = \{ A(k) \sinh ky + B(k) \cosh ky \} \quad (7-133)$$

It follows from Eqs. (7-130) and (7-132) that  $\nabla^2 G = -\delta(x - x') \delta(y - y')$  yields

$$\begin{aligned} \int_{-\infty}^{+\infty} \left( \frac{d^2 Y}{dy^2} - k^2 Y \right) \cos k(x - x') dk \\ = \frac{-1}{2\pi} \int_{-\infty}^{+\infty} \delta(y - y') \cos k(x - x') dk \end{aligned} \quad (7-134)$$

as the analogue of Eq. (7-125). Equating the coefficients of  $\cos k(x - x')$  on each side of Eq. (7-134) gives us

$$\frac{d^2 Y}{dy^2} - k^2 Y = -\frac{1}{2\pi} \delta(y - y') \quad (7-135)$$

for the determination of  $Y(k, y)$ .

The solution of Eq. (7-135) which satisfies the boundary conditions

$$Y = 0 \text{ on } y = 0 \quad (7-136)$$

and

$$Y = 0 \text{ on } y = a \quad (7-137)$$

can be written as

$$Y(k, y) = \frac{\sinh k(a - y >) \sinh ky <}{2\pi k \sinh ka} \quad (7-138)$$

and inserting this expression into Eq. (7-132) gives

$$G(x, x'; y, y') = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{\sinh k(a - y >) \sinh ky <}{k \sinh ka} \cos k(x - x') dk \quad (7-139)$$

As expected, we have recovered the result of Eq. (7-89).

## 7-10 THE SOLUTION OF LAPLACE'S EQUATION IN A HALF SPACE

Consider the problem of computing a potential function  $\phi(x, y)$  throughout a half space  $x \geq 0$ , in such a way that it reduces to a specific function, say  $f(y)$ , on  $x = 0$ . In mathematical terms, we want to find a solution of

$$\nabla^2 \phi = 0 \quad (7-140)$$

for  $x \geq 0$  that satisfies the following conditions:

$$\phi(0, y) = f(y) \text{ on } x = 0 \quad (7-141)$$

$$\lim_{x \rightarrow \infty} \phi(x, y) = 0 \quad (7-142)$$

for any real  $y$ , and

$$\lim_{|y| \rightarrow \infty} \phi(x, y) = 0 \quad (7-143)$$

for all  $x \geq 0$ .

This problem can be solved with the general Green's-function technique outlined in Sec. 7-8. The computation begins with Green's symmetric identity:

$$\int_S \{g \nabla^2 \phi - \phi \nabla^2 g\} dS = \int_{C(S)} \left\{ g \frac{\partial \phi}{\partial n} - \phi \frac{\partial g}{\partial n} \right\} ds \quad (7-144)$$

In Eq. (7-144),  $S$  represents the portion of the half plane  $x \geq 0$  enclosed by the curve  $C(S)$ . This curve is formed by the  $y$  axis and the circumference of a semicircle of very large (actually infinite) radius located in the right half plane. It is reasonable to assume that the potential vanishes everywhere along the infinite semicircle. Under these circumstances only the line integral along  $y$  contributes to Eq. (7-144), which becomes

$$\int_0^{+\infty} \int_{-\infty}^{+\infty} \{g \nabla^2 \phi - \phi \nabla^2 g\} dx dy = \int_{-\infty}^{+\infty} \left\{ g \frac{\partial \phi}{\partial n} - \phi \frac{\partial g}{\partial n} \right\} dy \quad (7-145)$$

The line integral on the right side of Eq. (7-145) is understood to be computed along the line  $x = 0$ , in other words, along the  $y$  axis. Since we have not been given any information about  $\partial \phi / \partial n$  on  $x = 0$ , the term in which it appears must be eliminated. We eliminate it by choosing  $g$  so that

$$g = 0 \text{ on } x = 0 \quad (7-146)$$

One way of satisfying Eq. (7-146) is to let  $g$  be a special "half-space" Green's function which has odd symmetry with respect to  $x$ . If  $g$  is such a function, then it must satisfy

$$g(-x, x'; y, y') = -g(x, x'; y, y') \quad (7-147)$$

and consequently  $g$  will vanish on  $x = 0$ . Suppose that

$$g(x, x'; y, y') = G(x, x'; y, y') - G(-x, x'; y, y') \quad (7-148)$$

and that  $G(x, x'; y, y')$  satisfies

$$\nabla^2 G = -\delta(x - x') \delta(y - y') \quad (7-149)$$

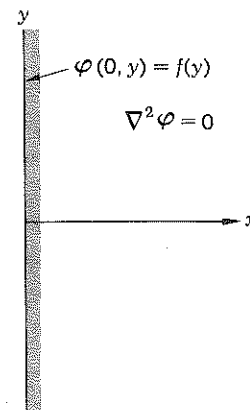


Figure 7-2 The potential problem for a half space.

Then  $G$  represents the potential at  $(x', y')$  due to a line source of unit positive charge density located at  $(x, y)$ . Obviously  $g$ , as given by Eq. (7-148), has property (7-147) and thus satisfies Eq. (7-146). Furthermore, (7-148) gives us a physical interpretation for  $g$ . Because of Eq. (7-149),  $g$  must satisfy

$$\nabla^2 g = -\delta(x - x') \delta(y - y') + \delta(x + x') \delta(y - y') \quad (7-150)$$

with

$$g = 0 \text{ on } x = 0 \quad (7-151)$$

and therefore  $g(x, x'; y, y')$  must represent the potential at the point  $(x', y')$  when there is a grounded conducting plane, halfway between a line source of unit positive charge density located at  $(x, y)$  and a line source of unit negative charge density located at  $(-x, y)$ . In electromagnetic theory, this solution is described by saying that in order to determine the potential due to a positive line charge in the presence of a grounded conducting plane, it is necessary to introduce a *fictitious image line charge* of the opposite sign. The location of the image charge is chosen so that it is the geometrical reflection in the grounded plane of the original line charge. It follows that, by symmetry, the grounded plane must be at zero potential.

We are now ready to return to Eq. (7-145). Since  $\partial g / \partial n$  represents the rate of change of  $g$  along the *outward* normal to  $C(S)$ , we must have

$$\frac{\partial g}{\partial n} = -\frac{\partial g}{\partial x} \text{ on } x = 0 \quad (7-152)$$

Taking note of (7-140), (7-141), and (7-150) to (7-152), Eq. (7-145) gives us

$$\begin{aligned} \int_0^{+\infty} \int_{-\infty}^{+\infty} \phi(x, y) [\delta(x - x') \delta(y - y') - \delta(x + x') \delta(y - y')] dx dy \\ = \int_{-\infty}^{+\infty} f(y) \frac{\partial g}{\partial x} dy \end{aligned} \quad (7-153)$$

Only the first term in Eq. (7-153) contributes to the left side because  $x$  is always positive, and as a result

$$\phi(x', y') = \int_{-\infty}^{+\infty} f(y) \frac{\partial g}{\partial x} dy \quad (7-154)$$

is the formal solution of our problem. Equation (7-154) represents  $\phi(x', y')$  in terms of the half-space Green's function. As usual, the real problem is to construct the Green's function itself.

### Construction of the half-space Green's function

Once again let us consider the problem of solving

$$\nabla^2 G = -\delta(x - x') \delta(y - y') \quad (7-155)$$

subject to the condition that very far away from the source,  $G$  and its derivatives approach zero. Multiplying Eq. (7-155) by  $e^{-iky'}$  and integrating over  $y$  from  $y = -\infty$  to  $y = +\infty$  gives

$$\frac{d^2 \bar{G}}{dx^2} - k^2 \bar{G} = -\delta(x - x') e^{-iky'} \quad (7-156)$$

where

$$\bar{G} = \int_{-\infty}^{+\infty} G e^{-iky'} dy \quad (7-157)$$

assuming that

$$\lim_{|y| \rightarrow \infty} \left( \frac{\partial G}{\partial y} + ikG \right) = 0 \quad (7-158)$$

We have already encountered both Eq. (7-156) and its general solution in Eqs. (7-66) and (7-68). However, it is also enlightening to obtain a particular integral of (7-156) by a different technique. Thus, consider

$$\bar{G} = A e^{-|k|(x-x')} \quad x > x' \quad (7-159)$$

and

$$\bar{G} = B e^{k|x-x'|} \quad x < x' \quad (7-160)$$

Both Eqs. (7-159) and (7-160) satisfy (7-156) when  $x \neq x'$ . The solution (7-159) has the property that as the source coordinate  $x \rightarrow +\infty$ , the disturbance  $\bar{G}$  at  $x'$  goes to zero as it should, because then the source and observer are infinitely removed from each other. Similarly, as  $x \rightarrow -\infty$  in Eq. (7-160), the disturbance at  $x'$  also vanishes. Each "half" of the solution gives the correct behavior at  $x'$  as the source recedes to  $\infty$  in the appropriate direction. Out of these two pieces, (7-159) and (7-160), we shall manufacture a single continuous solution of Eq. (7-156).

Continuity of Eqs. (7-159) and (7-160) at  $x = x'$  requires that  $A = B$ , and therefore

$$\bar{G} = A e^{-|k|(x-x')} \quad x > x' \quad (7-161)$$

and

$$\bar{G} = A e^{k|x-x'|} \quad x < x' \quad (7-162)$$

together, form a single continuous solution of Eq. (7-156). The object now is to choose  $A$  so that Eqs. (7-161) and (7-162) satisfy (7-156) not only for  $x > x'$  and  $x < x'$ , respectively, but for  $x \geq x'$  and  $x \leq x'$  as well. To do this, suppose that we integrate (7-156) about a small interval of length  $2\epsilon$  centered on  $x = x'$ ,

$$\int_{x=x'-\epsilon}^{x=x'+\epsilon} \frac{d^2 \bar{G}}{dx^2} dx - k^2 \int_{x=x'-\epsilon}^{x=x'+\epsilon} \bar{G} dx = -e^{-iky'} \int_{x=x'-\epsilon}^{x=x'+\epsilon} \delta(x - x') dx \quad (7-163)$$

and then take the limit of Eq. (7-163) as  $\epsilon \rightarrow 0$ . The result is

$$\lim_{\epsilon \rightarrow 0} \frac{d\bar{G}}{dx} \bigg|_{x=x'-\epsilon}^{x=x'+\epsilon} = -e^{-iky'} \quad (7-164)$$

The second term on the left side of Eq. (7-163) vanishes as  $\epsilon \rightarrow 0$  because  $\bar{G}$  is constructed to be continuous at  $x = x'$ . The result (7-164) tells us that the derivative of  $\bar{G}$  must be discontinuous at  $x = x'$  and that the *jump* or *saltus* in  $d\bar{G}/dx$  must be equal to  $-e^{-iky'}$ . It is this condition which permits us to determine  $A$  in Eqs. (7-161) and (7-162), for by differentiation we find

$$\frac{d}{dx} A e^{-|k|(x-x')} = -|k| A e^{-|k|(x-x')} \quad x > x' \quad (7-165)$$

and

$$\frac{d}{dx} A e^{k|x-x'|} = |k| A e^{k|x-x'|} \quad x < x' \quad (7-166)$$

Therefore

$$\frac{d\bar{G}}{dx} \bigg|_{x=x'-\epsilon}^{x=x'+\epsilon} = -|k| A e^{-|k|\epsilon} - |k| A e^{-|k|\epsilon} = -2|k| A e^{-|k|\epsilon} \quad (7-167)$$

gives us

$$\lim_{\epsilon \rightarrow 0} \frac{d\bar{G}}{dx} \bigg|_{x=x'-\epsilon}^{x=x'+\epsilon} = -2|k| A = -e^{-iky'} \quad (7-168)$$

so that

$$A = \frac{e^{-iky'}}{2|k|} \quad (7-169)$$

We can represent the two formulas (7-161) and (7-162) as a single equation by using  $|x - x'|$  in the exponential; then

$$\bar{G} = \frac{e^{-iky' - |k||x-x'|}}{2|k|} \quad (7-170)$$

is the desired solution of Eq. (7-156). The Fourier transform of the half-space Green's function (7-148) must satisfy

$$\bar{g}(x, x'; y'; k) = \bar{G}(x, x'; y'; k) - \bar{G}(-x, x'; y'; k) \quad (7-171)$$

where

$$\bar{g}(x, x'; y'; k) = \int_{-\infty}^{+\infty} g(x, x'; y, y') e^{-iky} dy \quad (7-172)$$

Using Eq. (7-170), we conclude that

$$\bar{g}(x, x'; y'; k) = e^{-ik y'} \left( \frac{e^{-|k||x-x'|} - e^{-|k||x+x'|}}{2|k|} \right) \quad (7-173)$$

and applying Fourier's inversion formula (5-348) to Eq. (7-173) gives

$$g(x, x'; y, y') = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{ik(y-y')} \frac{e^{-|k||x-x'|} - e^{-|k||x+x'|}}{2|k|} dk \quad (7-174)$$

The formula (7-154) for  $\phi(x', y')$ ,

$$\phi(x', y') = \int_{-\infty}^{+\infty} f(y) \frac{\partial g}{\partial x} dy \quad (7-175)$$

can now be simplified if we remember that  $\partial g / \partial x$  is to be evaluated on  $x = 0$ .

Differentiating Eq. (7-174) under the integral sign gives us

$$\frac{\partial g}{\partial x} = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-ik(y-y')} \frac{-|k|e^{-|k||x-x'|} + |k|e^{-|k||x+x'|}}{2|k|} dk \quad x > x' > 0 \quad (7-176)$$

and

$$\frac{\partial g}{\partial x} = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{ik(y-y')} \frac{|k|e^{-|k||x-x'|} + |k|e^{-|k||x+x'|}}{2|k|} dk \quad x' > x > 0 \quad (7-177)$$

We must select the appropriate expression for  $\partial g / \partial x$  as  $x \rightarrow 0$  from the two possible choices (7-176) and (7-177). Since the observer is assumed to be located in the physical half space  $x' > 0$ , the inequality  $x > x' > 0$  associated with (7-176) eliminates this expression from contention when  $x = 0$ . Therefore we are left with (7-177), which reduces to

$$\left( \frac{\partial g}{\partial x} \right)_{x=0} = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{ik(y-y')-|k||x'|} dk \quad \text{on } x = 0 \quad (7-178)$$

The integral in Eq. (7-178) can be integrated directly to give

$$\begin{aligned} \int_{-\infty}^{+\infty} e^{ik(y-y')-|k||x'|} dk &= \int_{-\infty}^0 e^{ik(y-y')-|k||x'|} dk \\ &+ \int_0^{\infty} e^{ik(y-y')-|k||x'|} dk = \int_0^{\infty} e^{-ik(y-y')-|k||x'|} dk \\ &+ \int_0^{\infty} e^{ik(y-y')-|k||x'|} dk = \frac{-1}{-i(y-y')-|x'|} \\ &- \frac{1}{i(y-y')-|x'|} = \frac{+2|x'|}{(x')^2 + (y-y')^2} \end{aligned} \quad (7-179)$$

and consequently

$$\left( \frac{\partial g}{\partial x} \right)_{x=0} = \frac{1}{\pi} \frac{|x'|}{(x')^2 + (y-y')^2} \quad (7-180)$$

Thus, the problem of computing a potential function which satisfies Laplace's equation in the half space  $x' \geq 0$  and which reduces to

$f(y')$  on  $x' = 0$  is solved by the formula

$$\phi(x', y') = \frac{x'}{\pi} \int_{-\infty}^{+\infty} \frac{f(y)}{(x')^2 + (y-y')^2} dy \quad (7-181)$$

**REMARK:** The problem of solving  $\nabla^2 \phi = 0$  for some domain in such a way that  $\phi$  reduces to a given function on the boundary of the domain is known as "Dirichlet's problem." One says that  $\phi$  obeys an inhomogeneous Dirichlet condition on the boundary. For example, Eq. (7-181) is the solution of Dirichlet's problem for the half space  $x' \geq 0$ , and  $\phi(x', y')$  satisfies the inhomogeneous Dirichlet condition  $\phi(0, y') = f(y')$  on the boundary  $x' = 0$  of the half space.

## 7-11 LAPLACE'S EQUATION IN POLAR COORDINATES

In plane polar coordinates  $(r, \theta)$ , Laplace's equation becomes

$$\nabla^2 \phi = \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} = 0 \quad (7-182)$$

The Green's-function technique of solving boundary-value problems for Eq. (7-182) applies in general. It is only in constructing the Green's function itself, that one must deal with particular coordinate systems. Thus Eq. (7-34) gives us the formal solution of (7-182),

$$\phi(r', \theta') = \int_{C(S)} \left\{ G \frac{\partial \phi}{\partial n} - \phi \frac{\partial G}{\partial n} \right\} ds \quad (7-183)$$

where now the Green's function  $G(r, r'; \theta, \theta')$  appearing in Eq. (7-183) is defined as the solution of

$$\nabla^2 G = \frac{-\delta(r-r') \delta(\theta-\theta')}{r} \quad (7-184)$$

A few comments regarding Eqs. (7-183) and (7-184) are in order. Formula (7-183) represents the potential  $\phi$  at the point  $(r', \theta')$  in terms of the boundary values of  $\phi$  and  $\partial \phi / \partial n$  on a curve  $C(S)$  enclosing  $(r', \theta')$ .

This actually overdetermines the problem, because  $\phi$  and  $\partial \phi / \partial n$  cannot be independently assigned on  $C(S)$ . To see why this is true, consider the one-dimensional analogue of Laplace's equation,

$$\frac{d^2 \phi}{dx^2} = 0 \quad \text{on } a \leq x \leq b \quad (7-185)$$

Corresponding to Eq. (7-183), we have

$$\phi(x') = \left\{ G \frac{d\phi}{dx} - \phi \frac{dG}{dx} \right\}_{x=a}^{x=b} \quad (7-186)$$

where  $G(x|x')$  is the solution of

$$\frac{d^2 G}{dx^2} = -\delta(x-x') \quad (7-187)$$

Formula (7-186) asks us to specify both  $\phi$  and  $d\phi/dx$  at  $x = a$  and at  $x = b$ . However, the solution of Eq. (7-185) must be a straight line. Having specified  $\phi$  and  $d\phi/dx$  at  $x = a$ , these quantities are completely determined at  $x = b$  and cannot be arbitrarily assigned. One remedy for this impasse is to choose  $G$  so that it vanishes at  $x = a$  and  $x = b$ . Then Eq. (7-186) involves only the values of  $\phi$  at  $a$  and  $b$ , and these can indeed be arbitrarily assigned.

The same sort of argument shows that  $\phi$  and  $\partial\phi/\partial n$  cannot both be arbitrarily specified on  $C(S)$ . This difficulty is now resolved as before. One solves Eq. (7-184) subject to a *homogeneous Dirichlet condition*,  $G = 0$  on  $C(S)$ , obtaining in the process

$$\phi(r', \theta') = - \int_{C(S)} \phi \frac{\partial G}{\partial n} ds \quad (7-188)$$

It is also possible to solve Eq. (7-184) subject to a *homogeneous Neumann condition*,  $\partial G/\partial n = 0$  on  $C(S)$ ; then

$$\phi(r', \theta') = \int_{C(S)} G \frac{\partial \phi}{\partial n} ds \quad (7-189)$$

Other possibilities exist as well, but we shall not treat these here.

Returning to (7-184), we notice that the right side contains the expression  $-\delta(\theta - \theta') \delta(r - r')/r$ . The  $1/r$  factor arises because the element of area in polar coordinates is  $dS = r dr d\theta$ , and we must have

$$\begin{aligned} \int_S \delta(\mathbf{r} - \mathbf{r}') dS &= \iint_S \delta(x - x') \delta(y - y') dx dy \\ &= \iint_S \frac{\delta(r - r')}{r} \delta(\theta - \theta') r dr d\theta \\ &= \begin{cases} 1 & (r', \theta') \text{ contained in area } S \\ 0 & (r', \theta') \text{ outside of area } S \end{cases} \end{aligned} \quad (7-190)$$

## 7-12 CONSTRUCTION OF A GREEN'S FUNCTION IN POLAR COORDINATES

The real problem in using Eq. (7-188) or (7-189) is the actual construction of an appropriate Green's function. We shall apply the method of separation of variables for this purpose. One sets

$$\phi = R(r)\Theta(\theta) \quad (7-191)$$

into

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial r^2} + \frac{1}{r} \frac{\partial \phi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} = 0 \quad (7-192)$$

to obtain

$$r^2 \frac{d^2 R/dr^2 + (1/r) dR/dr}{R} = - \frac{d^2 \Theta/d\theta^2}{\Theta} \quad (7-193)$$

Since the left side of Eq. (7-193) is a function of  $r$  alone, while the right side depends only on  $\theta$ , both sides must be equal to one and

the same constant, say  $n^2$ . Then

$$- \frac{d^2 \Theta/d\theta^2}{\Theta} = n^2 \quad (7-194)$$

$$r^2 \frac{d^2 R/dr^2 + (1/r) dR/dr}{R} = n^2 \quad (7-195)$$

give the separated equations

$$\frac{d^2 \Theta}{d\theta^2} + n^2 \Theta = 0 \quad (7-196)$$

and

$$\frac{d^2 R}{dr^2} + \frac{1}{r} \frac{dR}{dr} - \frac{n^2}{r^2} R = 0 \quad (7-197)$$

When  $n$  is an integer,  $\Theta$  is periodic in  $\theta$  with period  $2\pi$ . The solutions of Eqs. (7-196) and (7-197) always separate into two groups according to whether  $n$  is zero or not.

If  $n$  is not zero,

$$\Theta_n(\theta) = A_n \cos n\theta + B_n \sin n\theta \quad (7-198)$$

and

$$R_n(r) = C_n r^n + D_n r^{-n} \quad (7-199)$$

are the general solutions of Eqs. (7-196) and (7-197). For  $n = 0$  the solutions become

$$\Theta_0 = A_0 \theta + B_0 \quad (7-200)$$

$$R_0(r) = C_0 \log r + D_0 \quad (7-201)$$

Assuming that  $n$  is an integer, an infinite sum of these separated solutions gives the general solution of  $\nabla^2 \phi = 0$  in polar coordinates:

$$\begin{aligned} \phi(r, \theta) &= \sum_{n=0}^{\infty} R_n(r) \Theta_n(\theta) = (A_0 \theta + B_0)(C_0 \log r + D_0) \\ &\quad + \sum_{n=1}^{\infty} \{A_n \cos n\theta + B_n \sin n\theta\} \{C_n r^n + D_n r^{-n}\} \end{aligned} \quad (7-202)$$

The term  $\{A_n \cos n\theta + B_n \sin n\theta\} \{C_n r^n + D_n r^{-n}\}$  is called a "circular harmonic of degree  $n$ ," and  $\{A_0 \theta + B_0\} \{C_0 \log r + D_0\}$  is called a "circular harmonic of degree zero."

### Construction of the Green's function

The Green's function must satisfy

$$\nabla^2 G = - \frac{\delta(r - r') \delta(\theta - \theta')}{r} \quad (7-203)$$

and if we want to solve Dirichlet's problem for a circle using Eq. (7-188), then  $G$  must also satisfy the boundary condition

$$G = 0 \text{ on } r = a \quad 0 \leq \theta \leq 2\pi \quad (7-204)$$

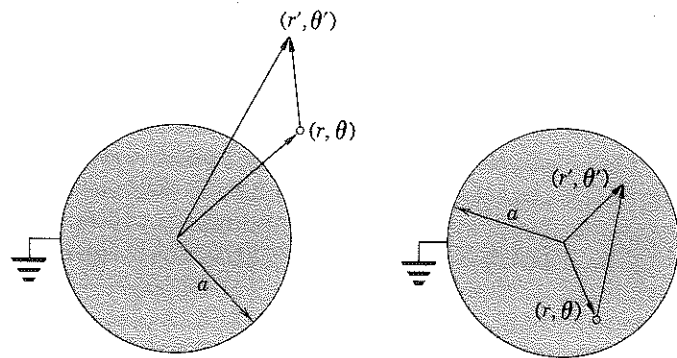


Figure 7-3 The exterior and interior Dirichlet problems for a circle.

The Green's function  $G(r, r'; \theta, \theta')$  defined by Eq. (7-203) represents the potential at  $(r', \theta')$  due to a line source of positive charge located at  $(r, \theta)$ . Equation (7-204) expresses the presence of a grounded conducting cylinder of radius  $r = a$ . If  $r < a$ ,  $r' < a$ , both source and observer are inside the circle  $r = a$ , and (7-203) and (7-204) form an interior Dirichlet problem. If  $r > a$ ,  $r' > a$ , then the reverse is true, and (7-203) and (7-204) constitute an exterior Dirichlet problem for the circle (Fig. 7-3).

Following the procedure of Eqs. (7-119) to (7-128), let us represent the solution of

$$\nabla^2 G = \frac{-\delta(r - r') \delta(\theta - \theta')}{r} \quad (7-205)$$

in terms of an expansion in eigenfunctions of the separated equation (7-196):

$$G = \sum_{n=-\infty}^{+\infty} R_n(r) e^{in\theta} \quad (7-206)$$

The right side of Eq. (7-205) must also be expanded in these eigenfunctions, and we write

$$\frac{\delta(r - r') \delta(\theta - \theta')}{r} = \sum_{n=-\infty}^{+\infty} \frac{\delta(r - r')}{r} C_n e^{in\theta} \quad (7-207)$$

To compute the  $C_n$ 's, multiply Eq. (7-207) through by  $e^{-im\theta}$ , and integrate both sides with respect to  $\theta$ , using

$$\int_{-\pi}^{\pi} e^{i(n-m)\theta} d\theta = 2\pi \delta_{nm} \quad (7-208)$$

This gives  $C_n = (1/2\pi) e^{-in\theta'}$ , and therefore

$$\frac{\delta(r - r') \delta(\theta - \theta')}{r} = \frac{1}{2\pi} \sum_{n=-\infty}^{+\infty} \frac{\delta(r - r')}{r} e^{in(\theta - \theta')} \quad (7-209)$$

Equations (7-206) and (7-209) are now substituted into (7-205) to give us

$$\sum_{n=-\infty}^{+\infty} \left( \frac{d^2 R_n}{dr^2} + \frac{1}{r} \frac{dR_n}{dr} - \frac{n^2}{r^2} R_n \right) e^{in\theta} = \frac{-1}{2\pi} \frac{\delta(r - r')}{r} \sum_{n=-\infty}^{+\infty} e^{in(\theta - \theta')} \quad (7-210)$$

Equating the coefficients of  $e^{in\theta}$  on both sides of Eq. (7-210) determines  $R_n$ . It must be a solution of

$$\frac{d^2 R_n}{dr^2} + \frac{1}{r} \frac{dR_n}{dr} - \frac{n^2}{r^2} R_n = -\frac{1}{2\pi} \frac{\delta(r - r')}{r} e^{-in\theta'} \quad (7-211)$$

We shall solve Eq. (7-211) using the technique of Eq. (7-159). Let  $r'$  be a fixed positive number, and suppose that  $n \neq 0$  and  $r \neq r'$ . Then (7-211) is satisfied by

$$R_n = A r^{|n|} \quad 0 \leq r < r' < \infty \quad (7-212)$$

and

$$R_n = B r^{-|n|} \quad r' < r < \infty \quad (7-213)$$

The absolute-value signs around  $n$  in Eqs. (7-212) and (7-213) guarantee that  $R_n$  will always remain finite for  $r = 0$  and  $r = \infty$ , regardless of whether  $n$  is a positive or a negative integer.

Continuity of  $R_n$  at  $r = r'$  requires that

$$A(r')^{|n|} = B(r')^{-|n|} \quad (7-214)$$

The determination of  $A$  and  $B$  is completed by integrating Eq. (7-211) with respect to  $r$  from  $r = r' - \epsilon$  to  $r = r' + \epsilon$ , where  $\epsilon$  is some small positive number. We have

$$\frac{1}{r} \frac{d}{dr} \left( r \frac{dR_n}{dr} \right) - \frac{n^2}{r^2} R_n = -\frac{1}{2\pi} \frac{\delta(r - r')}{r} e^{-in\theta'} \quad (7-215)$$

and therefore

$$\int_{r'-\epsilon}^{r'+\epsilon} \frac{d}{dr} \left( r \frac{dR_n}{dr} \right) dr - n^2 \int_{r'-\epsilon}^{r'+\epsilon} \frac{R_n}{r} dr = -\frac{1}{2\pi} e^{-in\theta'} \quad (7-216)$$

Suppose we take the limit in Eq. (7-216) as  $\epsilon \rightarrow 0$ . Since  $R_n$  has been made continuous at  $r = r'$ , the second term in (7-216) vanishes, and we are left with

$$\lim_{\epsilon \rightarrow 0} r \frac{dR_n}{dr} \Big|_{r=r'-\epsilon}^{r=r'+\epsilon} = -\frac{1}{2\pi} e^{-in\theta'} \quad (7-217)$$

Combined with Eqs. (7-212) and (7-213), Eq. (7-217) gives

$$-|n| B(r')^{-|n|} - |n| A(r')^{|n|} = -\frac{1}{2\pi} e^{-in\theta'} \quad (7-218)$$

Solving Eqs. (7-218) and (7-214) simultaneously determines  $A$  and  $B$ :

$$A = \frac{e^{-in\theta'}}{4\pi |n| (r')^{|n|}} \quad (7-219)$$

$$B = \frac{(r')^{|n|} e^{-in\theta'}}{4\pi |n|} \quad (7-220)$$

For  $n \neq 0$ ,

$$R_n = \frac{1}{4\pi|n|} \left(\frac{r}{r'}\right)^{|n|} e^{-in\theta'} \quad 0 \leq r \leq r' \quad (7-221)$$

and

$$R_n = \frac{1}{4\pi|n|} \left(\frac{r'}{r}\right)^{|n|} e^{-in\theta'} \quad r' \leq r < \infty \quad (7-222)$$

These two formulas can be written as usual as a single equation:

$$R_n = \frac{1}{4\pi|n|} \left(\frac{r_{<}}{r_{>}}\right)^{|n|} e^{-in\theta'} \quad n \neq 0 \quad (7-223)$$

For the case where  $n = 0$ , Eq. (7-215) reduces to

$$\frac{1}{r} \frac{d}{dr} \left( r \frac{dR_0}{dr} \right) = -\frac{1}{2\pi} \frac{\delta(r-r')}{r} e^{-in\theta} \quad (7-224)$$

In order to construct the desired particular integral of Eq. (7-224) we need two linearly independent solutions of the homogeneous version of (7-224). These solutions are  $C_0 \log r$  and a constant,  $D_0$ . If  $G$  is to remain finite at the origin  $r = 0$ , then one must choose

$$R_0(r) = D_0 \quad 0 \leq r < r' \quad (7-225)$$

$$R_0(r) = C_0 \log r \quad r' < r < \infty \quad (7-226)$$

Continuity of  $R_0(r)$  at  $r = r'$  requires

$$C_0 \log r' = D_0 \quad (7-227)$$

and for  $n = 0$ , Eq. (7-217) becomes

$$\lim_{\epsilon \rightarrow 0} r \frac{dR_0}{dr} \Big|_{r=r'+\epsilon} - \lim_{\epsilon \rightarrow 0} r \frac{dR_0}{dr} \Big|_{r=r'-\epsilon} = -\frac{1}{2\pi} \quad (7-228)$$

which gives  $C_0 = -1/2\pi$ . Thus

$$R_0(r) = -\frac{1}{2\pi} \log r' \quad 0 \leq r \leq r' \quad (7-229)$$

$$R_0(r) = -\frac{1}{2\pi} \log r \quad r' \leq r < \infty \quad (7-230)$$

is the solution of Eq. (7-215) for  $n = 0$ . As before, we can express Eqs. (7-229) and (7-230) as a single equation:

$$R_0(r) = -\frac{1}{2\pi} \log r_{>} \quad (7-231)$$

From Eq. (7-206), using (7-223) and (7-231), we obtain the eigenfunction expansion of the infinite-medium Green's function:

$$G_{\infty}(r, r'; \theta, \theta') = -\frac{1}{2\pi} \log r_{>} + \frac{1}{4\pi} \sum_{n=-\infty}^{+\infty} \left(\frac{r_{<}}{r_{>}}\right)^{|n|} \frac{e^{in(\theta-\theta')}}{|n|} \quad (7-232)$$

The prime on the summation sign in Eq. (7-232) means that the value  $n = 0$  is excluded. Using Euler's identity, (7-232) reduces to

$$G_{\infty}(r, r'; \theta, \theta') = -\frac{1}{2\pi} \log r_{>} + \frac{1}{2\pi} \sum_{n=1}^{\infty} \left(\frac{r_{<}}{r_{>}}\right)^n \frac{\cos n(\theta - \theta')}{n} \quad (7-233)$$

where as usual

$$r_{>} = \text{greater of } (r, r') \quad (7-234)$$

$$r_{<} = \text{lesser of } (r, r') \quad (7-235)$$

The infinite-medium Green's function (7-233) is a particular integral of Eq. (7-205). It represents the potential due to a line source of unit positive charge density, provided no boundaries are present.

### Green's function for the interior Dirichlet problem

The Green's function for the interior Dirichlet problem is the solution of Eq. (7-205) corresponding to a line source of positive charge contained inside the grounded conducting cylinder  $r = a$ . We write it as

$$G = G_i + G_{\infty} \quad (7-236)$$

where  $G_{\infty}$  represents the potential in the absence of the cylinder, and the induced potential  $G_i$  is a solution of

$$\nabla^2 G_i = 0 \quad (7-237)$$

which remains finite inside the cylinder. The condition that the cylinder be grounded requires that

$$G = G_i + G_{\infty} = 0 \text{ on } r = a \quad (7-238)$$

From our previous analysis it follows that

$$G_i = \sum_{n=-\infty}^{+\infty} C_n r^{|n|} e^{in\theta} + \frac{1}{2\pi} \log a \quad (7-239)$$

where the constants  $C_n$  are to be determined from Eq. (7-238). For the interior Dirichlet problem, when  $r = a$ , we must have  $r' < r = a$ . Using Eqs. (7-232) and (7-239) in (7-238) gives

$$C_n a^{|n|} + \frac{1}{4\pi} \left(\frac{r'}{a}\right)^{|n|} \frac{e^{-in\theta'}}{|n|} = 0 \quad (7-240)$$

or

$$C_n = -\frac{1}{4\pi} \left(\frac{r'}{a^2}\right)^{|n|} \frac{e^{-in\theta'}}{|n|} \quad (7-241)$$

The additive constant  $(1/2\pi) \log a$  in Eq. (7-239) is so chosen that when it is combined with  $-(1/2\pi) \log r_{>}$  in  $G_{\infty}$ , it reduces to  $(1/2\pi) \log a/r_{>}$  which vanishes on  $r = a$ .

Our solution for the interior Green's function is

$$G(r, r'; \theta, \theta') = \frac{1}{2\pi} \log \frac{a}{r_{>}} + \frac{1}{4\pi} \sum_{n=-\infty}^{+\infty} \left[ \left(\frac{r_{<}}{r_{>}}\right)^{|n|} - \left(\frac{r_{>} r_{<}}{a^2}\right)^{|n|} \right] \frac{e^{in(\theta-\theta')}}{|n|} \quad (7-242)$$

or

$$G(r, r'; \theta, \theta') = \frac{1}{2\pi} \log \frac{a}{r} + \frac{1}{2\pi} \sum_{n=1}^{\infty} \left[ \left( \frac{r}{a} \right)^n - \left( \frac{a}{r} \right)^n \right] \frac{\cos n(\theta - \theta')}{n} \quad (7-243)$$

One can readily verify that  $G$  vanishes on  $r = a$  when  $r' < a$ .

#### Interpretation of $G(r, r'; \theta, \theta')$

Suppose that  $r' < r$ ; then from Eq. (7-233),

$$G_{\infty}(r, r'; \theta, \theta') = -\frac{1}{2\pi} \log r + \frac{1}{2\pi} \sum_{n=1}^{\infty} \left( \frac{r'}{r} \right)^n \frac{\cos n(\theta - \theta')}{n} \quad (7-244)$$

and replacing  $r$  by  $a^2/r$  in Eq. (7-244), we find

$$G_{\infty}\left(\frac{a^2}{r}, r'; \theta, \theta'\right) = -\frac{1}{2\pi} \log \frac{a^2}{r} + \frac{1}{2\pi} \sum_{n=1}^{\infty} \left( \frac{rr'}{a^2} \right)^n \frac{\cos n(\theta - \theta')}{n} \quad (7-245)$$

Subtracting Eq. (7-245) from (7-244) gives

$$G_{\infty}(r, r'; \theta, \theta') - G_{\infty}\left(\frac{a^2}{r}, r'; \theta, \theta'\right) = \frac{1}{2\pi} \log \left( \frac{a}{r} \right)^2 + \frac{1}{2\pi} \sum_{n=1}^{\infty} \left[ \left( \frac{r'}{r} \right)^n - \left( \frac{rr'}{a^2} \right)^n \right] \frac{\cos n(\theta - \theta')}{n} \quad (7-246)$$

and from Eq. (7-243) we find

$$G(r, r'; \theta, \theta') = \frac{1}{2\pi} \log \frac{a}{r} + \frac{1}{2\pi} \sum_{n=1}^{\infty} \left[ \left( \frac{r'}{r} \right)^n - \left( \frac{rr'}{a^2} \right)^n \right] \frac{\cos n(\theta - \theta')}{n} \quad (7-247)$$

Now comparing Eqs. (7-246) and (7-247), we see that

$$G(r, r'; \theta, \theta') = G_{\infty}(r, r'; \theta, \theta') - G_{\infty}\left(\frac{a^2}{r}, r'; \theta, \theta'\right) - \frac{1}{2\pi} \log \frac{a}{r} \quad (7-248)$$

for  $r' < r \leq a$ . The first two terms in Eq. (7-248) represent the potentials at  $(r', \theta')$  due to a positive line source at  $(r, \theta)$  and a negative line source at  $(a^2/r, \theta)$ . The points  $(r, \theta)$  and  $(a^2/r, \theta)$  are images of each other in the circle  $r = a$ . When the original positive line charge is inside the circle  $r = a$ , then its negative image charge lies outside the circle, on the same diameter, and at a distance  $a^2/r$  from the center ( $r = 0$ ) of the circle.

If we set  $r' = 0$  in Eq. (7-244), then

$$G_{\infty} = -\frac{1}{2\pi} \log r \quad (7-249)$$

represents the potential at a distance  $r$  units away from a unit posi-

tive line charge. Therefore, we may also write

$$G_{\infty} = -\frac{1}{2\pi} \log R \quad (7-250)$$

in which

$$R = |\mathbf{r}' - \mathbf{r}| = [(r')^2 + (r)^2 - 2rr' \cos(\theta - \theta')]^{1/2} \quad (7-251)$$

measures the distance between the line source and the observer. One immediate consequence of Eqs. (7-250) and (7-233) is the expansion

$$G_{\infty}(r, r'; \theta, \theta') = \frac{1}{2\pi} \log \frac{1}{R} = -\frac{1}{2\pi} \log r + \frac{1}{2\pi} \sum_{n=1}^{\infty} \left( \frac{r'}{r} \right)^n \frac{\cos n(\theta - \theta')}{n} \quad (7-252)$$

With the aid of (7-252), Eq. (7-248) reduces to

$$G(r, r'; \theta, \theta') = \frac{1}{2\pi} \log \frac{rR'}{aR} \quad (7-253)$$

where

$$R' = \left| \mathbf{r}' - \frac{a^2}{r^2} \mathbf{r} \right| = \left[ \frac{a^4}{r^2} + (r')^2 - 2a^2 \frac{r'}{r} \cos(\theta - \theta') \right]^{1/2} \quad (7-254)$$

determines the distance between the point of observation  $\mathbf{r}'$  and the image charge at  $(a^2/r^2)\mathbf{r}$ . Combining Eqs. (7-251), (7-253), and (7-254) we obtain

$$G(r, r'; \theta, \theta') = \frac{1}{4\pi} \log \frac{a^2 + (rr'/a)^2 - 2rr' \cos(\theta - \theta')}{(r')^2 + (r)^2 - 2rr' \cos(\theta - \theta')} \quad (7-255)$$

for the closed-form expression of the interior Green's function.

The potential  $\phi$  at  $(r', \theta')$  inside the circle is given in terms of its boundary values on the circle  $r = a$  by Eq. (7-188):

$$\phi(r', \theta') = - \int_{r=a} \phi(a, \theta) \frac{\partial G}{\partial n} ds \quad (7-256)$$

Since the outward normal to the circle  $r = a$  is along the radius  $r$ , we have

$$\left( \frac{\partial G}{\partial n} \right)_{r=a} = \left( \frac{\partial G}{\partial r} \right)_{r=a} = \frac{1}{2\pi} \frac{(r')^2 - a^2}{a[a^2 - 2ar' \cos(\theta - \theta') + (r')^2]} \quad (7-257)$$

On the circle  $r = a$ , the element of arc length is  $ds = a d\theta$ ; therefore Eqs. (7-256) and (7-257) yield

$$\phi(r', \theta') = \frac{a^2 - (r')^2}{2\pi} \int_0^{2\pi} \frac{\phi(a, \theta) d\theta}{a^2 - 2ar' \cos(\theta - \theta') + (r')^2} \quad (7-258)$$

which solves the problem. Formula (7-258) is known as Poisson's integral solution of the interior Dirichlet problem for a circle.

For the interior Dirichlet problem, we constructed an infinite-medium Green's function  $G_\infty$  with the property that if  $r \rightarrow 0$  and  $r' \neq 0$ , the  $G_\infty$  remains finite. This is physically reasonable. It says that the potential  $r'$  units away from a line source is finite. For the exterior Dirichlet problem, both source and observer are located outside the circle  $r = a$ . If one allows the source to recede to infinity, i.e.,  $r \rightarrow \infty$ , while keeping the point of observation fixed at a distance  $r'$  from the origin of coordinates, then the potential at  $r'$  must also remain finite. This condition requires an alteration of Eq. (7-233) which otherwise diverges logarithmically as  $r \rightarrow \infty$ . Instead of (7-233), one has

$$G_\infty(r, r'; \theta, \theta') = -\frac{1}{2\pi} \log r < + \frac{1}{2\pi} \sum_{n=1}^{\infty} \left(\frac{r <}{r >}\right)^n \frac{\cos n(\theta - \theta')}{n} \quad (7-259)$$

This form of the infinite-medium Green's function satisfies Eq. (7-205) and remains finite as  $r \rightarrow \infty$ .

The Green's function for the exterior Dirichlet problem is the solution of Eq. (7-205) corresponding to a line source of positive charge contained outside the grounded conducting cylinder  $r = a$ . The exterior Green's function is written as

$$G = G_i + G_\infty \quad (7-260)$$

where  $G_\infty$  is given by Eq. (7-259) and represents the potential in the absence of the cylinder. The induced potential  $G_i$  is a solution

$$\nabla^2 G_i = 0 \quad (7-261)$$

which remains finite as  $r \rightarrow \infty$  and which satisfies the boundary condition

$$G = G_i + G_\infty = 0 \text{ on } r = a \quad (7-262)$$

The function  $G_i$  must be of the form

$$G_i = \sum_{n=1}^{\infty} C_n r^{-n} \cos n(\theta - \theta') + \frac{1}{2\pi} \log a \quad (7-263)$$

since Eq. (7-263) satisfies Eq. (7-261) and behaves correctly as  $r \rightarrow \infty$ . The constant term in (7-263) is chosen to cancel the term  $-(1/2\pi) \log r <$  in  $G_\infty$  when  $r = a$ . It follows, using Eqs. (7-259), (7-263), and (7-262) to compute the  $C_n$ 's, that the exterior Green's function for a circle is

$$G(r, r'; \theta, \theta') = \frac{1}{2\pi} \log \left(\frac{r <}{a}\right) + \frac{1}{2\pi} \sum_{n=1}^{\infty} \left[ \left(\frac{r <}{r >}\right)^n - \left(\frac{a^2}{r > r <}\right)^n \right] \frac{\cos n(\theta - \theta')}{n} \quad (7-264)$$

Suppose that  $a \leq r < r'$ ; then Eq. (7-264) gives us

$$G(r, r'; \theta, \theta') = \frac{1}{2\pi} \log \frac{r}{a} + \frac{1}{2\pi} \sum_{n=1}^{\infty} \left[ \left(\frac{r}{r'}\right)^n - \left(\frac{a^2}{rr'}\right)^n \right] \frac{\cos n(\theta - \theta')}{n} \quad (7-265)$$

Recalling Eq. (7-252), we have

$$\frac{1}{2\pi} \log R = -\frac{1}{2\pi} \log r' = \frac{1}{2\pi} \sum_{n=1}^{\infty} \left(\frac{r}{r'}\right)^n \frac{\cos n(\theta - \theta')}{n} \quad (7-266)$$

and

$$\frac{1}{2\pi} \log R' = -\frac{1}{2\pi} \log r' + \frac{1}{2\pi} \sum_{n=1}^{\infty} \left(\frac{a^2}{rr'}\right)^n \frac{\cos n(\theta - \theta')}{n} \quad (7-267)$$

where

$$R = |\mathbf{r}' - \mathbf{r}| \quad (7-268)$$

The distance  $R'$  is formed from  $R$  by replacing  $r$  with  $a^2/r$ :

$$R' = \left| \mathbf{r}' - \frac{a^2}{r^2} \mathbf{r} \right| \quad (7-269)$$

Comparing Eqs. (7-265) through (7-267) shows that

$$G(r, r'; \theta, \theta') = \frac{1}{2\pi} \log \frac{1}{R} - \frac{1}{2\pi} \log \frac{1}{R'} + \frac{1}{2\pi} \log \frac{r}{a} \quad (7-270)$$

Hence

$$G(r, r'; \theta, \theta') = \frac{1}{2\pi} \log \frac{rR'}{aR} \quad (7-271)$$

and this result coincides with Eq. (7-253). The potential  $\phi$  at a point  $(r', \theta')$  outside of the circle  $r = a$  is given in terms of its boundary values on the circle by Eq. (7-188):

$$\phi(r', \theta') = - \int_{r=a} \phi(a, \theta) \frac{\partial G}{\partial n} ds \quad (7-272)$$

One must now observe that since we are calculating the potential in the exterior region  $r' > a$ , the outward normal on the boundary of the circle  $r = a$  points toward the center of the circle. With this remark, Eq. (7-272) gives us

$$\phi(r', \theta') = \frac{(r')^2 - a^2}{2\pi} \int_0^{2\pi} \frac{\phi(a, \theta) d\theta}{(r')^2 - 2ar' \cos(\theta - \theta') + a^2} \quad (7-273)$$

for  $a < r'$ . The integrand in Eq. (7-273) can be expanded in a power series in  $a/r'$  when  $a < r'$ . The result is

$$\begin{aligned} \frac{(r')^2 - a^2}{(r')^2 - 2ar' \cos(\theta - \theta') + a^2} &= 1 + 2 \sum_{n=1}^{\infty} \left(\frac{a}{r'}\right)^n \cos n(\theta - \theta') \\ &= \sum_{n=1}^{\infty} \epsilon_n \left(\frac{a}{r'}\right)^n \cos n(\theta - \theta') \end{aligned} \quad (7-274)$$

where

$$\epsilon_n = \begin{cases} 2 & n = 1, 2, \dots \\ 1 & n = 0 \end{cases} \quad (7-275)$$

With the aid of Eq. (7-274) the potential  $\phi(r', \theta')$  can be expressed as

$$\phi(r', \theta') = \frac{1}{2\pi} \sum_{n=0}^{\infty} \left(\frac{a}{r'}\right)^n \epsilon_n \int_0^{2\pi} \phi(a, \theta) \cos n(\theta - \theta') d\theta \quad (7-276)$$

and this completes our solution of the problem.

## 7-14 LAPLACE'S EQUATION IN CYLINDRICAL COORDINATES

In cylindrical coordinates  $(r, \theta, z)$ , Laplace's equation takes the form

$$\nabla^2 \phi = \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} + \frac{\partial^2 \phi}{\partial z^2} = 0 \quad (7-277)$$

The Green's function corresponding to Eq. (7-277) is the solution of

$$\nabla^2 G = \frac{-\delta(r - r') \delta(\theta - \theta') \delta(z - z')}{r} \quad (7-278)$$

and  $G(r, r'; \theta, \theta'; z, z')$  represents the potential at  $(r', \theta', z')$  due to a unit positive point charge located at  $(r, \theta, z)$ .

As usual, we begin by applying the method of separation of variables to (7-277). Let  $\phi = R(r)\Theta(\theta)Z(z)$  in Eq. (7-277). Then divide the result by  $R\Theta Z/r^2$  to obtain

$$\frac{r}{R} \frac{d}{dr} \left( r \frac{dR}{dr} \right) + \frac{r^2}{Z} \frac{d^2 Z}{dz^2} = -\frac{1}{\Theta} \frac{d^2 \Theta}{d\theta^2} \quad (7-279)$$

The right side of Eq. (7-279) depends only on  $\theta$  while the left side depends on  $r$  and  $z$ . Consequently, the right side of (7-279) must be constant, and one writes

$$-\frac{1}{\Theta} \frac{d^2 \Theta}{d\theta^2} = n^2 \quad (7-280)$$

We are left with

$$\frac{1}{rR} \frac{d}{dr} \left( r \frac{dR}{dr} \right) - \frac{n^2}{r^2} = \frac{-1}{Z} \frac{d^2 Z}{dz^2} \quad (7-281)$$

where again the left side of Eq. (7-281) depends only on  $r$  while the right side depends only on  $z$ . It follows that both sides of (7-281) must be equal to one and the same constant. This we write as  $k^2$ , and therefore

$$\frac{1}{Z} \frac{d^2 Z}{dz^2} = k^2 \quad (7-282)$$

$$\frac{1}{r} \frac{d}{dr} \left( r \frac{dR}{dr} \right) + \left( k^2 - \frac{n^2}{r^2} \right) R = 0 \quad (7-283)$$

By separating variables, we have reduced Eq. (7-277) to three ordinary differential equations,

$$\frac{d^2 Z}{dz^2} - k^2 Z = 0 \quad (7-284)$$

$$\frac{d^2 \Theta}{d\theta^2} + n^2 \Theta = 0 \quad (7-285)$$

$$\frac{d^2 R}{dr^2} + \frac{1}{r} \frac{dR}{dr} + \left( k^2 - \frac{n^2}{r^2} \right) R = 0 \quad (7-286)$$

containing two separation parameters  $n$  and  $k$ . The solution of Eq. (7-285) for the  $\Theta$  factor is either  $\sin n\theta$  or  $\cos n\theta$  or some linear combination of these such as  $e^{in\theta}$ . Ultimately, the allowed values of the separation parameters  $n$  and  $k$  will be determined by the nature of the boundary conditions which  $\phi$  must satisfy. For example, if there are no plane boundaries for  $\theta = \text{constant}$  but  $\theta$  is allowed to go from 0 to  $2\pi$ , then  $n$  must be zero or an integer; otherwise  $\phi$  will not have  $2\pi$  as a period. Equation (7-284) has as solutions either trigonometric or hyperbolic functions, depending on whether  $k$  is real or imaginary.

If one wishes  $\phi$  or  $\partial\phi/\partial z$  to vanish on the flat ends of a cylinder while taking on a specified  $z$  dependence along the curved portion, then  $\phi$  should be periodic in  $z$ , and  $k$  must be imaginary. On the other hand, if one wishes to specify the dependence of  $\phi$  on  $r$  at one or both flat ends of the cylinder while  $\phi$  remains constant on the curved part, then  $k$  should be real.

The equation for  $R(r)$  is a Bessel equation with solutions  $J_n(kr)$  and  $N_n(kr)$  when  $k$  is real. The Bessel function  $J_n(kr)$  is finite at the origin; so it must be used if the cylindrical axis is inside the boundary surface, where the solution is to be computed. The Neumann function  $N_n(kr)$  is the second solution. It is infinite at the origin  $r = 0$ . When  $k$  is imaginary, the solutions of Eq. (7-286) are the hyperbolic Bessel functions  $I_n(kr)$  and  $K_n(kr)$ . Of these,  $I_n(kr)$  is finite at the origin and infinite at infinity, while  $K_n(kr)$  is infinite at the origin and vanishes at infinity.

Complete solutions of Eqs. (7-284) through (7-286) for real nonzero  $k$  are

$$Z(kz) = Ce^{kz} + De^{-kz} = C' \cosh kz + D' \sinh kz \quad (7-287)$$

$$\Theta(n\theta) = Ae^{in\theta} + Be^{-in\theta} = A' \cos n\theta + B' \sin n\theta \quad (7-288)$$

$$R_n(kr) = EJ_n(kr) + FN_n(kr) = E'H_n^{(1)}(kr) + F'H_n^{(2)}(kr) \quad (7-289)$$

while for imaginary nonzero  $k$ ,

$$Z(kz) = C \cos kz + D \sin kz = C'e^{ikz} + D'e^{-ikz} \quad (7-290)$$

$$\Theta(n\theta) = Ae^{in\theta} + Be^{-in\theta} = A' \cos n\theta + B' \sin n\theta \quad (7-291)$$

$$R_n(kr) = EK_n(kr) + FI_n(kr) \quad (7-292)$$

If  $k = 0$ , and  $n$  is real and nonzero,

$$Z(z) = Cz + D \quad (7-293)$$

$$\Theta(n\theta) = A \cos n\theta + B \sin n\theta \quad (7-294)$$

$$R_n(r) = Er^n + Fr^{-n} \quad (7-295)$$

whereas if  $k = 0$ , and  $n$  is imaginary and nonzero,

$$Z(z) = Cz + D \quad (7-296)$$

$$\Theta(n\theta) = A \cosh n\theta + B \sinh n\theta \quad (7-297)$$

$$R(r) = E \cos(n \log r) + F \sin(n \log r) \quad (7-298)$$

Finally, if both  $k$  and  $n$  are zero,

$$Z(z) = Cz + D \quad (7-299)$$

$$\Theta(\theta) = A\theta + B \quad (7-300)$$

$$R(r) = E \log r + F \quad (7-301)$$

Each expression of the form

$$\phi = R_n(kr)\Theta(n\theta)Z(kz) \quad (7-302)$$

is a solution of Laplace's equation in cylindrical coordinates. For this reason, expressions such as Eq. (7-302) are called "cylindrical harmonics." By summing or integrating over the separation parameters  $k$  and  $n$ , one can form more general cylindrical harmonics which satisfy Laplace's equation.

## 7-15 CONSTRUCTION OF THE GREEN'S FUNCTION

The Green's function must satisfy

$$\nabla^2 G = \frac{-\delta(r-r')\delta(\theta-\theta')\delta(z-z')}{r} \quad (7-303)$$

and we shall construct  $G$  by expanding it in terms of the separated solutions of Eq. (7-277); thus let

$$G = \sum_{n=-\infty}^{\infty} \int_{-\infty}^{+\infty} R_n(kr) e^{i(n\theta+kz)} dk \quad (7-304)$$

The right side of Eq. (7-303) must also be expanded in terms of the same separated solutions, and therefore we write

$$\begin{aligned} \frac{\delta(r-r')\delta(\theta-\theta')\delta(z-z')}{r} \\ = \sum_{n=-\infty}^{\infty} \int_{-\infty}^{+\infty} \frac{\delta(r-r')}{r} C_n(k) e^{i(n\theta+kz)} dk \end{aligned} \quad (7-305)$$

To calculate the coefficients  $C_n(k)$ , multiply both sides of Eq. (7-305) by  $e^{-i(m\theta+kz)}$ , and then integrate with respect to  $\theta$  and  $z$ , using

$$\int_{-\pi}^{\pi} e^{i(n-m)\theta} d\theta = 2\pi\delta_{nm} \quad (7-306)$$

and

$$\int_{-\infty}^{+\infty} e^{i(k-h)z} dz = 2\pi\delta(k-h) \quad (7-307)$$

This gives

$$C_n(k) = \frac{1}{4\pi^2} e^{-i(n\theta'+kz')} \quad (7-308)$$

and consequently

$$\begin{aligned} \frac{\delta(r-r')\delta(\theta-\theta')\delta(z-z')}{r} \\ = \frac{1}{4\pi^2} \sum_{n=-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{\delta(r-r')}{r} e^{i[n(\theta-\theta')+k(z-z')]} dk \end{aligned} \quad (7-309)$$

Now we substitute Eqs. (7-304) and (7-309) into (7-303) to obtain

$$\begin{aligned} \sum_{n=-\infty}^{+\infty} \int_{-\infty}^{+\infty} \left[ \frac{d^2 R_n}{dr^2} + \frac{1}{r} \frac{dR_n}{dr} - \left( k^2 + \frac{n^2}{r^2} \right) R_n \right] e^{i(n\theta+kz)} dk \\ = - \frac{1}{4\pi^2} \sum_{n=-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{\delta(r-r')}{r} e^{i[n(\theta-\theta')+k(z-z')]} dk \end{aligned} \quad (7-310)$$

and on equating the coefficients of  $e^{i(n\theta+kz)}$  on both sides of Eq. (7-310), we obtain the equation for  $R_n(kr)$ :

$$\frac{d^2 R_n}{dr^2} + \frac{1}{r} \frac{dR_n}{dr} - \left( k^2 + \frac{n^2}{r^2} \right) R_n = \frac{-\delta(r-r')}{4\pi^2 r^2} e^{-i(n\theta'+kz')} \quad (7-311)$$

To solve Eq. (7-311) for  $k \neq 0$ , we notice that when  $r \neq r'$ ,

$$R_n(kr) = \begin{cases} AI_n(kr) \\ BK_n(kr) \end{cases} \quad (7-312)$$

are linearly independent solutions of Eq. (7-311). Since the source is presumably neither at the origin  $r = 0$  nor at infinity, the Green's function should remain finite at  $r = 0$  and  $r = \infty$ . This forces us to assign the two solutions (7-312) so that

$$R_n = AI_n(kr) \quad 0 \leq r < r' \quad (7-313)$$

$$R_n = BK_n(kr) \quad r' < r < \infty \quad (7-314)$$

Continuity of Eqs. (7-313) and (7-314) at  $r = r'$  requires that

$$AI_n(kr') = BK_n(kr') \quad (7-315)$$

which yields one equation for  $A$  and  $B$ . A second equation is obtained by integrating Eq. (7-311) with respect to  $r$  from  $r = r' - \epsilon$  to  $r = r' + \epsilon$  where  $\epsilon$  is a small positive number. We have

$$\frac{1}{r} \frac{d}{dr} \left( r \frac{dR_n}{dr} \right) - \left( k^2 + \frac{n^2}{r^2} \right) R_n = - \frac{\delta(r-r')}{4\pi^2 r} e^{-i(n\theta'+kz')} \quad (7-316)$$

and therefore

$$\begin{aligned} \int_{r'-\epsilon}^{r'+\epsilon} \frac{d}{dr} \left( r \frac{dR_n}{dr} \right) dr - \int_{r'-\epsilon}^{r'+\epsilon} \left( k^2 + \frac{n^2}{r^2} \right) R_n dr \\ = - \frac{1}{4\pi^2} e^{-i(n\theta'+kz')} \end{aligned} \quad (7-317)$$

Taking the limit of Eq. (7-317) as  $\epsilon \rightarrow 0$  and remembering that the second term in (7-317) vanishes as  $\epsilon \rightarrow 0$  because of continuity, it follows that

$$\lim_{\epsilon \rightarrow 0} r \frac{dR_n}{dr} \Big|_{r=r'+\epsilon}^{r=r'-\epsilon} = -\frac{1}{4\pi^2} e^{-i(n\theta'+kz')} \quad (7-318)$$

Equations (7-318), (7-313), and (7-314) give us our second equation for  $A$  and  $B$ ,

$$\lim_{\epsilon \rightarrow 0} (kr) \frac{dR_n}{d(kr)} \Big|_{r=r'+\epsilon}^{r=r'-\epsilon} = Bx \frac{dK_n(x)}{dx} - Ax \frac{dI_n(x)}{dx} \quad (7-319)$$

where  $x = kr'$ . Our equations for  $A$  and  $B$  now read

$$AI_n(x) - BK_n(x) = 0 \quad (7-320)$$

$$AxI'_n(x) - BxK'_n(x) = \frac{1}{4\pi^2} e^{-i(n\theta'+kz')} \quad (7-321)$$

The determinant of the system is

$$\Delta = \begin{vmatrix} I_n(x) & -K_n(x) \\ xI'_n(x) & -xK'_n(x) \end{vmatrix} = -x(I_n K'_n - K_n I'_n) \quad (7-322)$$

which is almost the Wronskian of  $I_n$  and  $K_n$ , i.e.,

$$\Delta = -xW\{I_n, K_n; x\} \quad (7-323)$$

Abel's formula (6-442) for the Wronskian shows that

$$xW\{I_n, K_n; x\} = x_0 W\{I_n, K_n; x_0\} \quad (7-324)$$

Since the left side of Eq. (7-324) is a function of  $x$  alone while the right side is a function of  $x_0$  alone, (7-324) can hold only if each side is equal to one and the same constant; thus

$$W\{I_n, K_n; x\} = \frac{\text{const}}{x} \quad (7-325)$$

The constant in Eq. (7-325) is determined by evaluating the Wronskian at some arbitrary but definite point. The easiest points to use are  $x = 0$  or  $x = \infty$  because at these points the formulas for  $I_n(kr)$  and  $K_n(kr)$  become especially simple. For example, using Eqs. (6-273) and (6-274), we find

$$x\{I_n K'_n - K_n I'_n\} = x \left\{ \frac{\pi e^x}{\sqrt{2\pi x}} \frac{d}{dx} \left( \frac{e^{-x}}{\sqrt{2\pi x}} \right) - \frac{\pi e^{-x}}{\sqrt{2\pi x}} \frac{d}{dx} \left( \frac{e^x}{\sqrt{2\pi x}} \right) \right\} \quad (7-326)$$

or

$$x\{I_n K'_n - K_n I'_n\} = -1 \quad (7-327)$$

which gives the Wronskian of  $I_n$  and  $K_n$  as

$$W\{I_n, K_n; x\} = -\frac{1}{x} \quad (7-328)$$

The system (7-320) and (7-321) can now be solved for  $A$  and  $B$  to yield

$$A = \frac{1}{4\pi^2} K_n(kr') e^{-i(n\theta'+kz')} \quad (7-329)$$

$$B = \frac{1}{4\pi^2} I_n(kr') e^{-i(n\theta'+kz')} \quad (7-330)$$

Therefore the solution for  $R_n$  becomes

$$R_n = \frac{1}{4\pi^2} I_n(kr) K_n(kr') e^{-i(n\theta'+kz')} \quad 0 \leq r \leq r' \quad (7-331)$$

$$R_n = \frac{1}{4\pi^2} I_n(kr') K_n(kr) e^{-i(n\theta'+kz')} \quad r' \leq r \leq \infty \quad (7-332)$$

As usual, we can combine Eqs. (7-331) and (7-332) in a single formula by writing

$$R_n = \frac{1}{4\pi^2} I_n(kr_{<}) K_n(kr_{>}) e^{-i(n\theta'+kz')} \quad (7-333)$$

With the aid of (7-333), Eq. (7-304) is written as

$$\begin{aligned} G_\infty(r, r'; \theta, \theta'; z, z') \\ = \frac{1}{4\pi^2} \sum_{n=-\infty}^{+\infty} \int_{-\infty}^{+\infty} I_n(kr_{<}) K_n(kr_{>}) e^{in(\theta-\theta') + k(z-z')} dk \end{aligned} \quad (7-334)$$

which is the same as either

$$\begin{aligned} G_\infty(r, r'; \theta, \theta'; z, z') \\ = \frac{1}{2\pi^2} \sum_{n=-\infty}^{+\infty} \int_0^{+\infty} I_n(kr_{<}) K_n(kr_{>}) e^{in(\theta-\theta')} \cos k(z-z') dk \end{aligned} \quad (7-335)$$

or

$$\begin{aligned} G_\infty = \frac{1}{\pi^2} \left\{ \int_0^\infty \cos k(z-z') dk \left[ \frac{1}{2} I_0(kr_{<}) K_0(kr_{>}) \right. \right. \\ \left. \left. + \sum_{n=1}^\infty I_n(kr_{<}) K_n(kr_{>}) \cos n(\theta-\theta') \right] \right\} \end{aligned} \quad (7-336)$$

On the other hand, we know that Eq. (7-303) is also satisfied by

$$G_\infty = \frac{1}{4\pi R} = \frac{1}{4\pi[(x-x')^2 + (y-y')^2 + (z-z')^2]^{\frac{1}{2}}} \quad (7-337)$$

since (7-337) represents the potential at a distance  $R$  units away from a unit positive point charge; consequently

$$\begin{aligned} \frac{1}{4\pi R} = \frac{1}{\pi^2} \left\{ \int_0^\infty \cos k(z-z') dk \left[ \frac{I_0(kr_{<}) K_0(kr_{>})}{2} \right. \right. \\ \left. \left. + \sum_{n=1}^\infty I_n(kr_{<}) K_n(kr_{>}) \cos n(\theta-\theta') \right] \right\} \end{aligned} \quad (7-338)$$

must follow. Suppose that we let  $x' = 0$ ,  $y' = 0$ ,  $z' = 0$  in Eq. (7-338); then since

$$I_n(0) = \begin{cases} 1 & n = 0 \\ 0 & n = 1, 2, 3, \dots \end{cases} \quad (7-339)$$

Eq. (7-338) reduces to

$$\frac{1}{4\pi\sqrt{r^2 + z^2}} = \frac{1}{2\pi^2} \int_0^\infty \cos kz K_0(kr) dk \quad (7-340)$$

If  $r^2$  in Eq. (7-340) is replaced by  $R^2 = r^2 + (r')^2 - 2rr' \cos(\theta - \theta')$ , then the left side of (7-340) becomes just  $1/[(x - x')^2 + (y - y')^2 + z^2]^{1/2}$  where  $r^2 = x^2 + y^2$  and  $(r')^2 = (x')^2 + (y')^2$ . We can write

$$\left(\frac{1}{4\pi R}\right)_{z'=0} = \frac{1}{2\pi^2} \int_0^\infty \cos kz K_0(kR) dk \quad (7-341)$$

but from Eq. (7-338) we can also write

$$\left(\frac{1}{4\pi R}\right)_{z'=0} = \frac{1}{2\pi^2} \left\{ \int_0^\infty \cos kz dk \left[ I_0(kr_<) K_0(kr_>) + 2 \sum_{n=1}^\infty I_n(kr_<) K_n(kr_>) \cos n(\theta - \theta') \right] \right\} \quad (7-342)$$

so that on comparison of Eqs. (7-341) and (7-342),

$$K_0(kR) = I_0(kr_<) K_0(kr_>) + 2 \sum_{n=1}^\infty I_n(kr_<) K_n(kr_>) \cos n(\theta - \theta') \quad (7-343)$$

where

$$R = [r^2 + (r')^2 - 2rr' \cos(\theta - \theta')]^{1/2} \quad (7-344)$$

If one takes the limit of Eq. (7-343) as  $k \rightarrow 0$ , then the result of Eq. (7-233),

$$G_\infty = \frac{1}{2\pi} \log \frac{1}{R} = \frac{1}{2\pi} \log \frac{1}{r_>} + \frac{1}{2\pi} \sum_{n=1}^\infty \left(\frac{r_<}{r_>}\right)^n \frac{\cos n(\theta - \theta')}{n} \quad (7-345)$$

is recovered.

## 7-16 AN ALTERNATIVE METHOD OF SOLVING BOUNDARY-VALUE PROBLEMS

Having constructed the infinite-medium Green's function (7-334) to (7-336) for Laplace's equation in cylindrical coordinates, we could now construct the Green's functions which satisfy homogeneous Dirichlet or Neumann boundary conditions, say on a cylinder  $r = a$ , by merely adding to  $G_\infty$  appropriate solutions of  $\nabla^2 G_i = 0$  chosen so that  $G = G_\infty + G_i = 0$  on  $r = a$  or  $\partial G / \partial n = 0$  on  $r = a$ . The solutions of the Dirichlet and Neumann problems are then given by

$$\phi(r', \theta', z') = - \int_{S(V)} \phi \frac{\partial G}{\partial n} dS \quad (7-346)$$

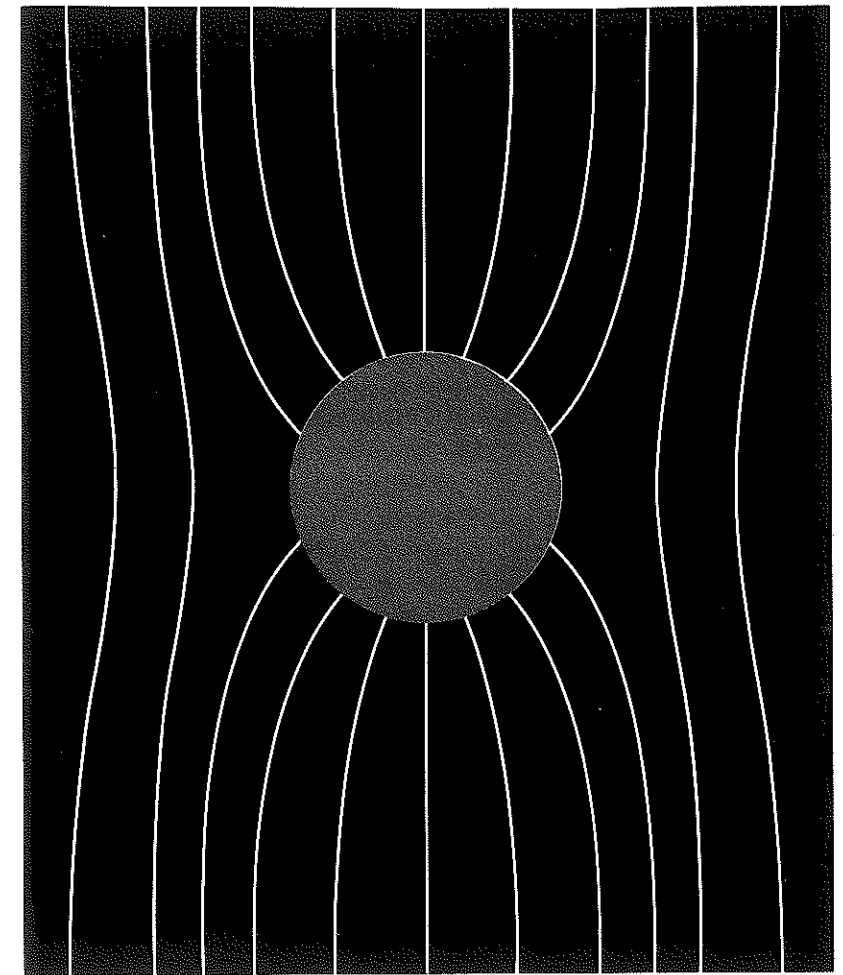


Figure 7-4 Electric field configuration for a grounded conducting cylinder placed in an initially uniform electric field.

and

$$\phi(r', \theta', z') = \int_{S(V)} G \frac{\partial \phi}{\partial n} dS \quad (7-347)$$

respectively.

When there is a great deal of symmetry in the problem, or when the source is at infinity, an alternative procedure is sometimes simpler. For example, we may wish to determine the potential at any point outside of an infinitely long, grounded conducting cylinder when it is in a uniform electric field  $E_0$  whose direction is perpendicular to the axis of the cylinder (Fig. 7-4). By symmetry, the electric field and the potential should depend only on the distance  $r$  from the center of the cylinder and on the angle  $\theta$  relative to the initially

uniform field  $E_0$ . Thus  $\phi$  should be independent of  $z$ , and we must solve

$$\nabla^2 \phi = \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} = 0 \quad (7-348)$$

subject to the boundary condition

$$\phi = 0 \text{ on } r = a \quad (7-349)$$

Let  $\phi_0$  represent the potential of the initially uniform electric field  $E_0$  in which the cylinder is placed. We have  $\phi_0 = E_0 x = E_0 r \cos \theta$  corresponding to a uniform field  $E_0 = -E_0 e_x$  directed along the negative  $x$  axis. The potential  $\phi$  must consist of two parts, an induced portion  $\phi_i$  due to the presence of the cylinder, and the original uniform field  $\phi_0$ . Consequently

$$\phi = \phi_0 + \phi_i \quad (7-350)$$

must satisfy Eqs. (7-348) and (7-349). The potential  $\phi_i$  must represent the effect of the cylinder so that in addition to satisfying  $\nabla^2 \phi_i = 0$ , it must vanish very far away from the cylinder. In other words,

$$\lim_{r \rightarrow \infty} \phi_i(r, \theta) = 0 \quad (7-351)$$

An expression of the form

$$\phi(r, \theta) = E_0 r \cos \theta + \sum_{n=1}^{\infty} r^{-n} (C_n \cos n\theta + D_n \sin n\theta) \quad (7-352)$$

is a good trial solution since it satisfies Eq. (7-348) and behaves correctly at infinity.

The symmetry of the problem can be used to eliminate the sine terms from Eq. (7-352). By symmetry

$$\phi(r, \theta) = \phi(r, -\theta) \quad (7-353)$$

and Eq. (7-353) cannot hold unless  $D_n = 0$  for  $n = 1, 2, 3, \dots$ . This leaves us with

$$\phi(r, \theta) = E_0 r \cos \theta + \sum_{n=1}^{\infty} C_n r^{-n} \cos n\theta \quad (7-354)$$

Applying Eq. (7-349) to (7-354) gives

$$0 = a E_0 \cos \theta + \sum_{n=1}^{\infty} C_n a^{-n} \cos n\theta \quad (7-355)$$

and therefore

$$C_1 = -a^2 E_0, C_2 = 0, C_3 = 0, \dots, C_n = 0, \dots$$

Our result is

$$\phi(r, \theta) = E_0 r \cos \theta - \frac{a^2 E_0}{r} \cos \theta = \left(1 - \frac{a^2}{r^2}\right) E_0 r \cos \theta \quad (7-356)$$

and  $E = -\nabla \phi$  gives us

$$E_r = -\frac{\partial \phi}{\partial r} = -\left(1 + \frac{a^2}{r^2}\right) E_0 \cos \theta \quad (7-357)$$

$$E_\theta = -\frac{1}{r} \frac{\partial \phi}{\partial \theta} = \left(1 - \frac{a^2}{r^2}\right) E_0 \sin \theta \quad (7-358)$$

as the radial and tangential components of the electric field.

## 7.17 LAPLACE'S EQUATION IN SPHERICAL COORDINATES

Laplace's equation in spherical coordinates  $(r, \theta, \phi)$  becomes

$$\nabla^2 \psi = \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2} = 0 \quad (7-359)$$

The Green's function for Eq. (7-359) is the solution of

$$\nabla^2 G = \frac{-\delta(r - r') \delta(\theta - \theta') \delta(\phi - \phi')}{r^2 \sin \theta} \quad (7-360)$$

and  $G(r, r'; \theta, \theta'; \phi, \phi')$  represents the potential at  $(r', \theta', \phi')$  due to a unit positive *point* charge located at  $(r, \theta, \phi)$ . The factor  $1/r^2 \sin \theta$  appearing in Eq. (7-360) is required because an element of volume in spherical coordinates is  $dV = r^2 \sin \theta dr d\theta d\phi$  and the volume integral of the right side of (7-360) must be normalized to unity when the source is inside in the region of integration.

To apply the method of separation of variables, we seek solutions of Eq. (7-359) of the form  $\psi(r, \theta, \phi) = R(r)\Theta(\theta)\Phi(\phi)$ . On substituting  $\psi = R\Theta\Phi$  into (7-359) and dividing the result by  $R\Theta\Phi/r^2 \sin^2 \theta$ , we obtain

$$\frac{\sin^2 \theta}{R} \frac{d}{dr} \left( r^2 \frac{dR}{dr} \right) + \frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left( \sin \theta \frac{d\Theta}{d\theta} \right) = -\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} \quad (7-361)$$

The right side of Eq. (7-361) depends only on  $\phi$  while the left side depends on  $r$  and  $\theta$ . Consequently, the right side of (7-361) must be a constant, and one writes

$$-\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = m^2 \quad (7-362)$$

We are left with

$$\frac{1}{R} \frac{d}{dr} \left( r^2 \frac{dR}{dr} \right) = - \left[ \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left( \sin \theta \frac{d\Theta}{d\theta} \right) - \frac{m^2}{\sin^2 \theta} \right] \quad (7-363)$$

where again the left side of Eq. (7-363) depends only on  $r$  while the right side depends only on  $\theta$ . It follows that both sides must be equal to one and the same constant, say  $k$ . Thus

$$\frac{d}{dr} \left( r^2 \frac{dR}{dr} \right) - kR = 0 \quad (7-364)$$

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left( \sin \theta \frac{d\Theta}{d\theta} \right) + \left( k - \frac{m^2}{\sin^2 \theta} \right) \Theta = 0 \quad (7-365)$$

and by separating variables, we have reduced Eq. (7-359) to three ordinary differential equations,

$$\frac{d^2\Phi}{d\phi^2} + m^2\Phi = 0 \quad (7-366)$$

$$\frac{1}{\sin\theta} \frac{d}{d\theta} \left( \sin\theta \frac{d\Theta}{d\theta} \right) + \left( k - \frac{m^2}{\sin^2\theta} \right) \Theta = 0 \quad (7-367)$$

$$\frac{d}{dr} \left( r^2 \frac{dR}{dr} \right) - kR = 0 \quad (7-368)$$

containing two separation parameters  $m$  and  $k$ . The allowed values of these separation parameters depend on the boundary conditions which  $\psi$  must satisfy. For example, if there are no boundary planes for  $\phi = \text{constant}$ , and  $\phi$  is allowed to run from 0 to  $2\pi$ , then  $\psi(r, \theta, \phi)$  should be periodic in  $\phi$  with period  $2\pi$ . This requires  $m$  to be a positive or a negative integer.

Suppose we let  $R(r) = r^\lambda$  in Eq. (7-368). The result is that  $R = r^\lambda$  satisfies (7-368) provided that

$$\lambda^2 + \lambda - k = 0 \quad (7-369)$$

If we choose  $\lambda = n$  in Eq. (7-369), then  $k = n(n+1)$ . However, if  $\lambda = -(n+1)$ , Eq. (7-369) still gives  $k = n(n+1)$ . Therefore, when  $k = n(n+1)$ , Eq. (7-368) has  $r^n$  and  $r^{-(n+1)}$  as linearly independent solutions. So far,  $n$  is still an arbitrary parameter. To further restrict  $n$  we must examine Eq. (7-367) more closely. Let  $x = \cos\theta$  and  $\Theta(\theta) = y(x)$  in (7-367). Then since  $\frac{d}{d\theta} = \frac{dx}{d\theta} \frac{d}{dx}$ , we find

$$\frac{1}{\sin\theta} \frac{d}{d\theta} \left( \sin\theta \frac{d\Theta}{d\theta} \right) = \frac{d}{dx} \left[ (1-x^2) \frac{dy}{dx} \right] \quad (7-370)$$

With the aid of Eq. (7-370), the system (7-364) to (7-368) can be reduced to

$$\frac{d^2\Phi}{d\phi^2} + m^2\Phi = 0 \quad (7-371)$$

$$\frac{d}{dx} \left[ (1-x^2) \frac{dy}{dx} \right] + \left[ n(n+1) - \frac{m^2}{1-x^2} \right] y(x) = 0 \quad (7-372)$$

$$\frac{d}{dr} \left( r^2 \frac{dR}{dr} \right) - n(n+1)R = 0 \quad (7-373)$$

Equation (7-371) has either trigonometric or hyperbolic solutions, depending on whether  $m$  is real or imaginary, and the solutions of Eq. (7-373) are already known to be  $r^n$  and  $r^{-(n+1)}$ . Now Eq. (7-372) is recognized as Legendre's associated differential equation. Its solutions are  $P_n^m(x)$  and  $Q_n^m(x)$ . Consequently, the complete solutions of Eqs. (7-371) to (7-373) become

$$R(r) = Ar^n + Br^{-(n+1)} \quad (7-374)$$

$$\Theta(\theta) = CP_n^m(\cos\theta) + DQ_n^m(\cos\theta) \quad (7-375)$$

$$\Phi(\phi) = E \cos m\phi + F \sin m\phi \quad (7-376)$$

We have already seen that  $m$  must be an integer if  $\psi(r, \theta, \phi)$  is to be periodic in  $\phi$ . In most problems  $\psi(r, \theta, \phi)$  must also remain finite over any sphere of finite radius. Therefore Eq. (7-375) must remain finite for  $0 \leq \theta \leq \pi$ , i.e., for  $-1 \leq \cos\theta \leq 1$ . This requires  $D$  to vanish and  $n$  to take only integer values in (7-375). The reason is that  $Q_n^m(x)$  is singular at  $\pm 1$  and  $P_n^m(x)$  remains finite for  $-1 \leq x \leq 1$  only when  $n$  and  $m$  are both integers and  $n \geq m$ . For the singular case  $m > 0$ ,  $n = 0$ , Eqs. (7-374) to (7-376) reduce to

$$R(r) = A + Br^{-1} \quad (7-377)$$

$$\Theta(\theta) = C \cot^m \frac{\theta}{2} + D \tan^m \frac{\theta}{2} \quad (7-378)$$

$$\Phi(\phi) = E \cos m\phi + F \sin m\phi \quad (7-379)$$

and for  $n = 0$ ,  $m = 0$ ,

$$R(r) = A + Br^{-1} \quad (7-380)$$

$$\Theta(\theta) = C \log \tan \frac{\theta}{2} + D \quad (7-381)$$

$$\Theta(\phi) = D\phi + E \quad (7-382)$$

## 7-18 CONSTRUCTION OF THE GREEN'S FUNCTION

The Green's function satisfies

$$\nabla^2 G = \frac{-\delta(r-r') \delta(\theta-\theta') \delta(\phi-\phi')}{r^2 \sin\theta} \quad (7-383)$$

and is constructed by making an expansion in terms of the separated solutions of Eq. (7-359) and then substituting the expansion into Eq. (7-383) to determine the expansion coefficients. Thus let

$$G = \sum_{n=0}^{\infty} \sum_{m=-n}^{+n} R_n^{|m|}(r) P_n^{|m|}(\cos\theta) e^{im\phi} \quad (7-384)$$

and corresponding to Eq. (7-384), choose

$$\frac{\delta(r-r') \delta(\theta-\theta') \delta(\phi-\phi')}{r^2 \sin\theta} = \sum_{n=0}^{\infty} \sum_{m=-n}^{+n} C_n^{|m|} P_n^{|m|}(\cos\theta) e^{im\phi} \quad (7-385)$$

To compute the expansion coefficients  $C_n^{|m|}$ , multiply both sides of Eq. (7-385) by  $P_{\lambda}^{|\mu|}(\cos\theta) e^{-i\mu\phi} \sin\theta d\theta d\phi$ , and then integrate  $\theta$  and  $\phi$  over the surface of a unit sphere, using

$$\begin{aligned} \int_0^{2\pi} \int_0^{\pi} P_n^{|m|}(\cos\theta) P_{\lambda}^{|\mu|}(\cos\theta) e^{i(m-\mu)\phi} \sin\theta d\theta d\phi \\ = \frac{4\pi}{2n+1} \frac{(n+|m|)!}{(n-|m|)!} \delta_{n\lambda} \delta_{m\mu} \end{aligned} \quad (7-386)$$

This gives  $C_n^{|m|}$  as

$$C_n^{|m|} = \frac{\delta(r-r')}{4\pi r^2} \frac{(n-|m|)!}{(n+|m|)!} (2n+1) P_n^{|m|}(\cos\theta') e^{-im\phi'} \quad (7-387)$$

and consequently

$$\frac{\delta(r-r') \delta(\theta-\theta') \delta(\phi-\phi')}{r^2 \sin \theta} = \frac{1}{4\pi} \sum_{n=0}^{\infty} \sum_{m=-n}^{+n} \frac{\delta(r-r')}{r^2} \frac{(n-|m|)!}{(n+|m|)!} (2n+1) P_n^{|m|}(\cos \theta) P_n^{|m|}(\cos \theta') e^{im(\phi-\phi')} \quad (7-388)$$

Substituting Eqs. (7-384) and (7-388) into (7-383) gives us

$$\sum_{n=0}^{\infty} \sum_{m=-n}^{+n} \left[ \frac{1}{r^2} \frac{d}{dr} \left( r^2 \frac{dR_n^{|m|}}{dr} \right) - \frac{n(n+1)}{r^2} R_n^{|m|} \right] P_n^{|m|}(\cos \theta) e^{im\phi} = -\frac{1}{4\pi} \sum_{n=0}^{\infty} \sum_{m=-n}^{+n} \frac{\delta(r-r')}{r^2} \frac{(n-|m|)!}{(n+|m|)!} (2n+1) P_n^{|m|}(\cos \theta) P_n^{|m|}(\cos \theta') e^{im(\phi-\phi')} \quad (7-389)$$

and on equating the coefficients of  $P_n^{|m|}(\cos \theta) e^{im\phi}$  on both sides of Eq. (7-389), we obtain the equation determining  $R_n^{|m|}(r)$ :

$$\frac{1}{r^2} \frac{d}{dr} \left( r^2 \frac{dR_n^{|m|}}{dr} \right) - \frac{n(n+1)}{r^2} R_n^{|m|} = \frac{-\delta(r-r')}{4\pi r^2} \frac{(n-|m|)!}{(n+|m|)!} (2n+1) P_n^{|m|}(\cos \theta') e^{-im\phi'} \quad (7-390)$$

To solve Eq. (7-390) one notes that when  $r \neq r'$ ,

$$R_n^{|m|} = \begin{cases} Ar^n \\ Br^{-(n+1)} \end{cases} \quad (7-391)$$

are linearly independent solutions of (7-390). When the source is neither at the origin  $r=0$  nor at infinity, the Green's function should remain finite at  $r=0$  and  $r=\infty$ . Thus the pair of solutions in Eq. (7-391) are assigned so that

$$R_n^{|m|}(r) = Ar^n \quad 0 \leq r < r' \quad (7-392)$$

$$R_n^{|m|}(r) = Br^{-(n+1)} \quad r' < r < \infty \quad (7-393)$$

Continuity of Eqs. (7-392) and (7-393) at  $r=r'$  requires that

$$A(r')^n = B(r')^{-(n+1)} \quad (7-394)$$

giving us one equation for  $A$  and  $B$ . A second equation is obtained by integrating Eq. (7-390) with respect to  $r$  from  $r=r'-\epsilon$  to  $r=r'+\epsilon$  where  $\epsilon$  is a small positive number. We can write (7-390) as

$$\frac{1}{r^2} \frac{d}{dr} \left( r^2 \frac{dR_n^{|m|}}{dr} \right) - \frac{n(n+1)}{r^2} R_n^{|m|} = -Q \frac{\delta(r-r')}{r^2} \quad (7-395)$$

where

$$Q = \frac{2n+1}{4\pi} \frac{(n-|m|)!}{(n+|m|)!} P_n^{|m|}(\cos \theta') e^{-im\phi} \quad (7-396)$$

and therefore

$$\int_{r'-\epsilon}^{r'+\epsilon} \frac{d}{dr} \left( r^2 \frac{dR_n^{|m|}}{dr} \right) dr - n(n+1) \int_{r'-\epsilon}^{r'+\epsilon} R_n^{|m|} dr = -Q \quad (7-397)$$

Taking the limit of Eq. (7-397) as  $\epsilon \rightarrow 0$  and remembering that the second term vanishes as  $\epsilon \rightarrow 0$  because of continuity, it follows that

$$\lim_{\epsilon \rightarrow 0} r^2 \frac{dR_n^{|m|}}{dr} \Big|_{r=r'-\epsilon}^{r=r'+\epsilon} = -Q \quad (7-398)$$

Equations (7-398), (7-392), and (7-393) give us the second equation for  $A$  and  $B$ :

$$An(r')^{n+1} + B(n+1)(r')^{-n} = Q \quad (7-399)$$

The system of equations determining  $A$  and  $B$  is therefore

$$A(r')^n - B(r')^{-(n+1)} = 0 \quad (7-400)$$

$$An(r')^{n+1} + B(n+1)(r')^{-n} = Q \quad (7-401)$$

Solving this system for  $A$  and  $B$ , one finds

$$A = \frac{Q}{2n+1} (r')^{-(n+1)} \quad (7-402)$$

$$B = \frac{Q}{2n+1} (r')^n \quad (7-403)$$

and therefore Eqs. (7-392) and (7-393) reduce to

$$R_n^{|m|}(r) = \frac{Q}{2n+1} \frac{r^n}{(r')^{n+1}} \quad 0 \leq r \leq r' \quad (7-404)$$

$$R_n^{|m|}(r) = \frac{Q}{2n+1} \frac{(r')^n}{(r)^{n+1}} \quad r' \leq r < \infty \quad (7-405)$$

In terms of a single formula, Eqs. (7-404) and (7-405) combine to give

$$R_n^{|m|}(r) = \frac{Q}{2n+1} \frac{(r_{<})^n}{(r_{>})^{n+1}} \quad (7-406)$$

and using Eqs. (7-406), (7-396), and (7-384), the infinite-medium Green's function is found to be

$$G_{\infty} = \frac{1}{4\pi} \sum_{n=0}^{\infty} \sum_{m=-n}^{+n} \frac{(r_{<})^n}{(r_{>})^{n+1}} \frac{(n-|m|)!}{(n+|m|)!} P_n^{|m|}(\cos \theta) P_n^{|m|}(\cos \theta') e^{im(\phi-\phi')} \quad (7-407)$$

Equation (7-407) is the same as

$$G_{\infty}(r, r'; \theta, \theta'; \phi, \phi') = \frac{1}{4\pi} \sum_{n=0}^{\infty} \sum_{m=-n}^{+n} \frac{(r_{<})^n}{(r_{>})^{n+1}} \left[ P_n(\cos \theta) P_n(\cos \theta') + \frac{2(n-|m|)!}{(n+|m|)!} P_n^{|m|}(\cos \theta) P_n^{|m|}(\cos \theta') \cos m(\phi-\phi') \right] \quad (7-408)$$

However, we also know that

$$G_{\infty} = \frac{1}{4\pi R} = \frac{1}{4\pi |\mathbf{r} - \mathbf{r}'|} = \frac{1}{4\pi \sqrt{r^2 + (r')^2 - 2rr' \cos \gamma}} \quad (7-409)$$

where  $r = |\mathbf{r}| = (x^2 + y^2 + z^2)^{1/2}$ ,  $r' = |\mathbf{r}'| = [(x')^2 + (y')^2 + (z')^2]^{1/2}$ , and  $\cos \gamma = \mathbf{r} \cdot \mathbf{r}' / |\mathbf{r}| |\mathbf{r}'|$  is the cosine of the angle between  $\mathbf{r}$  and  $\mathbf{r}'$ .

Equation (7-409) can be expanded as a power series,

$$G_{\infty} = \frac{1}{4\pi R} = \frac{1}{4\pi} \sum_{n=0}^{\infty} \frac{(r_{<})^n}{(r_{>})^{n+1}} P_n(\cos \gamma) \quad (7-410)$$

by noticing that Eq. (7-409) is a generating formula for Legendre polynomials. A comparison of Eqs. (7-408) and (7-410) then yields the addition formula for Legendre polynomials,

$$P_n(\cos \gamma) = P_n(\cos \theta)P_n(\cos \theta') \\ + 2 \sum_{m=1}^n \frac{(n-m)!}{(n+m)!} P_n^m(\cos \theta)P_n^m(\cos \theta') \cos m(\phi - \phi') \quad (7-411)$$

where the spherical coordinates of  $\mathbf{r}$  and  $\mathbf{r}'$  are  $(r, \theta, \phi)$  and  $(r', \theta', \phi')$ , respectively.

### 7-19 SOLUTION OF THE INTERIOR AND EXTERIOR DIRICHLET PROBLEMS FOR A GROUNDED CONDUCTING SPHERE

Let us construct the Green's function for the interior problem, of a unit charge inside a grounded conducting sphere of radius  $r = a$ . We set

$$G = G_{\infty} - G_i \quad (7-412)$$

where  $G$  satisfies Eq. (7-383), together with the boundary condition

$$G = 0 \text{ on } r = a \quad (7-413)$$

The induced potential  $G_i$  is a solution of Laplace's equation,

$$\nabla^2 G_i = 0 \quad (7-414)$$

and, for the interior problem,  $G_i$  must be regular throughout the interior of the sphere  $r = a$ . Therefore we choose

$$G = \frac{1}{4\pi R} - \frac{1}{4\pi} \sum_{n=0}^{\infty} \sum_{m=0}^n A r^n \epsilon_m \frac{(n-m)!}{(n+m)!} \\ P_n^m(\cos \theta)P_n^m(\cos \theta') \cos m(\phi - \phi') \quad (7-415)$$

where

$$\epsilon_m = \begin{cases} 1 & m = 0 \\ 2 & m = 1, 2, 3 \end{cases} \quad (7-416)$$

and where  $A$  is to be determined through the boundary condition (7-413). With the aid of (7-408), Eq. (7-415) becomes

$$G = \frac{1}{4\pi} \sum_{n=0}^{\infty} \sum_{m=0}^n \left( \frac{r_{<}^n}{r_{>}^{n+1}} - A r^n \right) \\ \epsilon_m \frac{(n-m)!}{(n+m)!} P_n^m(\cos \theta)P_n^m(\cos \theta') \cos m(\phi - \phi') \quad (7-417)$$

and on applying Eq. (7-413), keeping in mind that both  $r'$  and  $r$  must be less than or equal to  $a$  in the interior problem, we obtain

$$\frac{(r')^n}{a^{n+1}} = A a^n \quad (7-418)$$

or

$$A = \frac{(r')^n}{a^{2n+1}} \quad (7-419)$$

Thus the Green's function for the interior Dirichlet problem is

$$G = \frac{1}{4\pi R} - \frac{1}{4\pi} \sum_{n=0}^{\infty} \sum_{m=0}^n \epsilon_m \frac{(r r')^n}{a^{2n+1}} \\ P_n^m(\cos \theta)P_n^m(\cos \theta') \cos m(\phi - \phi') \quad (7-420)$$

The exterior Green's function is obtained in the same way except that now the point charge is located outside of the grounded conducting sphere and the induced potential  $G_i$  must remain finite as  $r \rightarrow \infty$  instead of as  $r \rightarrow 0$ .

Therefore we choose

$$G = \frac{1}{4\pi R} - \frac{1}{4\pi} \sum_{n=0}^{\infty} \sum_{m=0}^n \frac{A \epsilon_m (n-m)!}{r^{n+1} (n+m)!} \\ P_n^m(\cos \theta)P_n^m(\cos \theta') \cos m(\phi - \phi') \quad (7-421)$$

and with the aid of Eq. (7-408), this becomes

$$G = \frac{1}{4\pi} \sum_{n=0}^{\infty} \sum_{m=0}^n \left( \frac{r_{<}^n}{r_{>}^{n+1}} - \frac{A}{r^{n+1}} \right) \\ \epsilon_m \frac{(n-m)!}{(n+m)!} P_n^m(\cos \theta)P_n^m(\cos \theta') \cos m(\phi - \phi') \quad (7-422)$$

On equating  $G$  to zero on  $r = a$ , remembering that both  $r'$  and  $r$  are greater than or equal to  $a$  in the exterior problem, Eq. (7-422) yields

$$\frac{a^n}{(r')^{n+1}} = \frac{A}{a^{n+1}} \quad (7-423)$$

Hence

$$A = \frac{a^{2n+1}}{(r')^{n+1}} \quad (7-424)$$

and the Green's function for the exterior Dirichlet problem must be

$$G = \frac{1}{4\pi R} - \frac{1}{4\pi} \sum_{n=0}^{\infty} \sum_{m=0}^n \frac{\epsilon_m a^{2n+1}}{(r r')^{n+1}} \frac{(n-m)!}{(n+m)!} \\ P_n^m(\cos \theta)P_n^m(\cos \theta') \cos m(\phi - \phi') \quad (7-425)$$

The additional term in expression (7-425), which causes  $G$  to vanish on  $r = a$ , is identical with that which would arise from an image charge of magnitude  $a/r'$  at the point  $r = a^2/r'$ ,  $\theta = \theta'$ ,  $\phi = \phi'$ .

If we compare Eqs. (7-425) and (7-420), we see that the image of a point charge  $q$  at point  $(r', \theta', \phi')$  in a grounded spherical surface

at  $r = a$  (with  $a$  either larger or smaller than  $r'$ ) is a charge  $q(a/r')$  at the point  $(a^2/r', \theta', \phi')$ . When the charge is outside the surface ( $r' > a$ ), the image is inside the surface and is smaller than the original; if the charge is inside, the image is outside and larger than the original.

According to this discussion, both the interior and exterior Green's functions can be represented by

$$G = \frac{1}{4\pi R} - \frac{a/r'}{4\pi R'} \quad (7-426)$$

where

$$R = |\mathbf{r} - \mathbf{r}'| = [r^2 + (r')^2 - 2rr' \cos \gamma]^{1/2} \quad (7-427)$$

$$R' = \left| \mathbf{r} - \left( \frac{a}{r'} \right)^2 \mathbf{r}' \right| = \left[ r^2 + \frac{a^4}{(r')^2} - \frac{2a^2 r}{r'} \cos \gamma \right]^{1/2} \quad (7-428)$$

and

$$\frac{r'R'}{a} = \left[ a^2 + \frac{(rr')^2}{a^2} - 2rr' \cos \gamma \right]^{1/2} \quad (7-429)$$

With the aid of these expressions  $G$  reduces to

$$G = \frac{1}{4\pi[r^2 + (r')^2 - 2rr' \cos \gamma]^{1/2}} - \frac{1}{4\pi[a^2 + (rr')^2/a^2 - 2rr' \cos \gamma]^{1/2}} \quad (7-430)$$

The interior Dirichlet problem for Laplace's equation in a sphere requires us to determine a function  $\psi(r, \theta, \phi)$ , satisfying the conditions

$$\nabla^2 \psi = 0 \quad r < a \quad (7-431)$$

$$\psi = \psi(a, \theta, \phi) \text{ on } r = a \quad (7-432)$$

The solution is easily obtained if we insert Eqs. (7-430) and (7-431) into Green's symmetric identity:

$$\int_V \{G \nabla^2 \psi - \psi \nabla^2 G\} dV = \int_{S(V)} \left\{ G \frac{\partial \psi}{\partial n} - \psi \frac{\partial G}{\partial n} \right\} dS \quad (7-433)$$

In Eq. (7-431)  $V$  is the region enclosed by a sphere of radius  $r = a$  and boundary surface  $S(V)$ . If we note that the outward normals to  $S(V)$  point radially away from the center of the sphere  $r = a$ , it follows that  $(\partial G / \partial n)_{r=a} = (\partial G / \partial r)_{r=a}$  on  $S(V)$ . Since  $\nabla^2 \psi = 0$  in  $V$  and  $G = 0$  on  $S(V)$ , Eq. (7-433) gives us

$$\psi(r', \theta', \phi') = - \int_0^{2\pi} \int_0^\pi \psi(a, \theta, \phi) \left( \frac{\partial G}{\partial r} \right)_{r=a} a^2 \sin \theta d\theta d\phi \quad (7-434)$$

and differentiating Eq. (7-430) partially with respect to  $r$ ,

$$\left( \frac{\partial G}{\partial r} \right)_{r=a} = \frac{1}{4\pi a} \frac{(r')^2 - a^2}{[a^2 + (r')^2 - 2ar' \cos \gamma]^{3/2}} \quad (7-435)$$

Therefore

$$\psi(r', \theta', \phi') = \frac{a[a^2 - (r')^2]}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\psi(a, \theta, \phi) \sin \theta d\theta d\phi}{[a^2 + (r')^2 - 2ar' \cos \gamma]^{3/2}} \quad (7-436)$$

is the required solution of Eqs. (7-431) and (7-432).

The exterior Dirichlet problem asks us to construct a function which is harmonic *outside* of the sphere  $r = a$  and which assumes prescribed values on the spherical boundary. The problem is exactly the same as (7-431) and (7-432) except that the inequality in (7-431) is reversed. The solutions of the interior and exterior Dirichlet problems are identical apart from a minus sign which enters because the outward normal for the exterior region points toward instead of away from the center of the sphere  $r = a$ . Thus  $(\partial G / \partial n)_{r=a} = -(\partial G / \partial r)_{r=a}$ , and consequently

$$\psi(r', \theta', \phi') = \frac{a[(r')^2 - a^2]}{4\pi} \int_0^{2\pi} \int_0^\pi \frac{\psi(a, \theta, \phi) \sin \theta d\theta d\phi}{[a^2 + (r')^2 - 2ar' \cos \gamma]^{3/2}} \quad (7-437)$$

replaces Eq. (7-436). Solutions (7-436) and (7-437) are known as "Poisson's solution" of the Dirichlet problem. Double integrals like those appearing in (7-436) and (7-437) are called "Poisson's integrals," and they play an important role in potential theory.

## 7-20 THE ONE-DIMENSIONAL WAVE EQUATION

Suppose that a flexible, elastic string is stretched between two supports on the  $x$  axis (Fig. 7-5). Let  $u(x, t)$  be the vertical displacement of the string as measured from the  $x$  axis. The mass of a length  $ds$  of string located between  $x$  and  $x + dx$  is

$$dm = \rho ds \quad (7-438)$$

where  $\rho$  represents the mass per unit length of the string. The vertical component of force acting on  $dm$  can be resolved into two parts:

$$F_V = F_T + F_E \quad (7-439)$$

In Eq. (7-439),  $F_T$  represents the vertical component of force exerted on  $dm$  due to tension in the string, and  $F_E$  represents the vertical component of all other external forces acting on  $dm$ , such as gravity and so on. The vertical component of acceleration of  $dm$  is  $\partial^2 u / \partial t^2$ , and therefore

$$F_V = \rho ds \frac{\partial^2 u}{\partial t^2} \quad (7-440)$$

To compute  $F_T$ , let  $T$  be the tension in the string, and let  $\theta$  be the angle between the tension vector and the  $x$  axis; then

$$F_T = (T \sin \theta)_{x+dx} - (T \sin \theta)_x \quad (7-441)$$

If the external force per unit length acting vertically on the string is

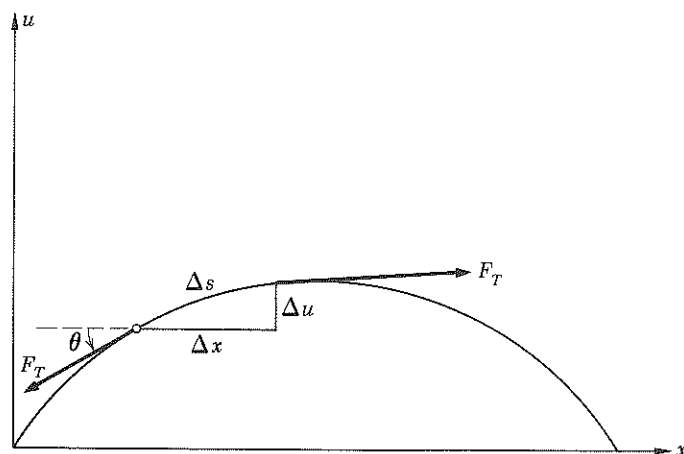


Figure 7-5 The vibrating string.

given by  $F(x, t)$ , then the amount of external force acting on  $dm$  is

$$F_E = F(x, t) ds \quad (7-442)$$

and we have obtained

$$F_V = [(T \sin \theta)_{x+dx} - (T \sin \theta)_x] + F(x, t) ds \quad (7-443)$$

A further simplification can be effected by assuming that the displacement  $u(x, t)$  is always small, for then  $\theta$  is never a large angle, and the small-angle approximation

$$\sin \theta \cong \tan \theta = \frac{\partial u}{\partial x} \quad (7-444)$$

is applicable. Moreover, since

$$ds = \left[ 1 + \left( \frac{\partial u}{\partial x} \right)^2 \right]^{1/2} dx \quad (7-445)$$

the small-angle approximation  $\left| \frac{\partial u}{\partial x} \right| \ll 1$  permits us to use  $dx$  in place of  $ds$ . As a result, Eqs. (7-440) and (7-443) reduce to

$$\rho dx \frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial x} \left( T \frac{\partial u}{\partial x} \right) dx + F(x, t) dx \quad (7-446)$$

The tension  $T$  and linear density  $\rho$  are constants. Therefore, on setting

$$c^2 = \frac{T}{\rho} \quad (7-447)$$

the one-dimensional inhomogeneous wave equation,

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} + \frac{F(x, t)}{\rho} \quad (7-448)$$

follows from Eq. (7-446). Since  $T$  has the dimensions of force and  $\rho$  the dimensions of mass per unit length, the quantity  $c$  defined in

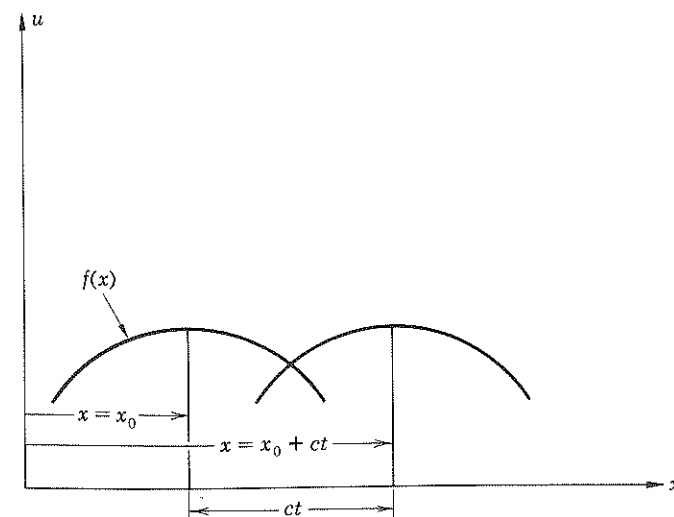


Figure 7-6 The curve  $u = f(x)$  is caused to propagate to the right at speed  $c$  by substituting  $x - ct$  for  $x$ .

Eq. (7-447) must have the dimensions of a velocity. Indeed, we shall see presently that  $c$  is just the speed with which the displacement  $u(x, t)$  propagates along the string.

Consider a curve  $u = f(x)$  in the  $(u, x)$  plane. So far this curve is static, but if  $x$  is now replaced by  $x - ct$ , then the possibility of motion exists. The new function,  $u = f(x - ct)$ , has the property that when  $t$  increases, the argument  $x - ct$  may be kept constant by increasing  $x$  proportionately. This corresponds to translating the entire curve  $u = f(x)$  to the right with a constant velocity equal to  $c$  (Fig. 7-6). In other words, a point which was at position  $x_0$  at time  $t = 0$  and is at position  $x = x_0 + ct$  at time  $t$  will always be assigned the same fixed amplitude  $u = f(x_0)$ .

Had we replaced  $x$  in  $f(x)$  with  $x + ct$  instead of  $x - ct$ , the same argument would still prevail except that  $u = f(x)$  would now translate to the left instead of to the right. This happens because as  $t$  increases,  $x$  must decrease proportionately in order to keep  $x + ct$  constant. A point at  $x = x_0$  at time  $t = 0$  and at  $x = x_0 - ct$  at time  $t$  would always be assigned the same fixed amplitude  $u = f(x_0)$ .

This discussion suggests that in the absence of external forces  $F(x, t)$ , we should expect Eq. (7-448) to have a general solution of the form

$$u(x, t) = f(x - ct) + g(x + ct) \quad (7-449)$$

wherein  $f$  and  $g$  are arbitrary functions of their arguments. It is easy to verify that this is actually the case. Let  $r = x - ct$  and

$s = x + ct$ ; then

$$\frac{\partial u}{\partial x} = \frac{\partial f}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial g}{\partial s} \frac{\partial s}{\partial x} = \frac{\partial f}{\partial r} + \frac{\partial g}{\partial s}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 f}{\partial r^2} + \frac{\partial^2 g}{\partial s^2}$$

$$\frac{\partial u}{\partial t} = \frac{\partial f}{\partial r} \frac{\partial r}{\partial t} + \frac{\partial g}{\partial s} \frac{\partial s}{\partial t} = -c \frac{\partial f}{\partial r} + c \frac{\partial g}{\partial s}$$

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left( \frac{\partial^2 f}{\partial r^2} + \frac{\partial^2 g}{\partial s^2} \right)$$

so that

$$c^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2} \quad (7-450)$$

is indeed satisfied by Eq. (7-449).

### The initial-value problem

The initial-value problem for the one-dimensional wave equation is to solve Eq. (7-450) for a string of unlimited length, given that the initial shape of the string is

$$u(x, 0) = F(x) \quad (7-451)$$

and that the initial velocity of each point on the string is

$$\frac{\partial u}{\partial t}(x, 0) = G(x) \quad (7-452)$$

From Eqs. (7-449), (7-451), and (7-452) it follows that

$$F(x) = f(x) + g(x) \quad (7-453)$$

$$G(x) = -cf'(x) + cg'(x) \quad (7-454)$$

and therefore

$$2cg'(x) = G(x) + cF'(x) \quad (7-455)$$

$$-2cf'(x) = G(x) - cF'(x) \quad (7-456)$$

Integrating Eqs. (7-455) and (7-456) directly, one finds

$$f(x) = \frac{1}{2c} F(x) - \frac{1}{2c} \int^x G(s) ds + \text{const} \quad (7-457)$$

$$g(x) = \frac{1}{2c} F(x) + \frac{1}{2c} \int^x G(s) ds + \text{const} \quad (7-458)$$

On replacing  $x$  by  $x - ct$  in Eq. (7-457) and by  $x + ct$  in Eq. (7-458), and then adding the results, there follows:

$$u(x, t) = f(x - ct) + g(x + ct) = \frac{1}{2} F(x - ct) + \frac{1}{2} F(x + ct) + \frac{1}{2c} \int_{x-ct}^{x+ct} G(s) ds + \text{const} \quad (7-459)$$

The constant in Eq. (7-459) is evaluated by setting  $t = 0$  and noting that because of Eq. (7-451), the constant must be zero. Thus

$$u(x, t) = \frac{1}{2} F(x - ct) + \frac{1}{2} F(x + ct) + \frac{1}{2c} \int_{x-ct}^{x+ct} G(s) ds \quad (7-460)$$

which is d'Alembert's solution of the initial-value problem, gives the motion of a string subjected to arbitrary initial displacements and velocities.

### Mixed boundary- and initial-value problems

In general, strings are not infinitely long, and external forces are not always zero. Hence we frequently want to find a solution of

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} + \frac{1}{\rho} f(x, t) \quad (7-461)$$

which satisfies not only the initial conditions,

$$u(x, 0) = F(x) \quad (7-462)$$

$$\frac{\partial u}{\partial t}(x, 0) = G(x) \quad (7-463)$$

but the boundary conditions,

$$u(0, t) = 0 \quad (7-464)$$

$$u(a, t) = 0 \quad (7-465)$$

at  $x = 0$  and  $x = a$  as well. This mixed initial- and boundary-value problem deals with the forced vibrations of a flexible elastic string supported between rigid walls at  $x = 0$  and  $x = a$ .

One of the easiest ways to solve Eq. (7-461) is to employ the finite sine transform introduced in Sec. 5-10. This transform is particularly well suited to the present problem because, as one notes from Eq. (5-176), the boundary conditions (7-464) and (7-465) are exactly the ones useful for the finite sine transform.

Equation (7-461) is multiplied through by  $\sin n\pi x/a$  and then integrated by parts with respect to  $x$  from  $x = 0$  to  $x = a$ , using Eqs. (5-176), (7-464), and (7-465). The result is that

$$\frac{d^2 \bar{u}_s}{dt^2}(n, t) + \left( \frac{n\pi c}{a} \right)^2 \bar{u}_s(n, t) = \frac{1}{\rho} \bar{f}_s(n, t) \quad (7-466)$$

where

$$\bar{u}_s(n, t) = \int_0^a u(x, t) \sin \frac{n\pi x}{a} dx \quad (7-467)$$

$$\bar{f}_s(n, t) = \int_0^a f(x, t) \sin \frac{n\pi x}{a} dx \quad (7-468)$$

are finite sine transforms of  $u$  and  $f$ . Equation (7-466) must be solved subject to the transformed initial conditions,

$$\bar{u}_s(n, 0) = \bar{F}_s(n) \quad (7-469)$$

$$\frac{\partial \bar{u}_s}{\partial t}(n, 0) = \bar{G}_s(n) \quad (7-470)$$

obtained by multiplying Eqs. (7-462) and (7-463) by  $\sin n\pi x/a$  and then integrating with respect to  $x$  from  $x = 0$  to  $x = a$ .

The solution of Eq. (7-466) subject to (7-469) and (7-470) is given in Sec. 5-24 in the form

$$\begin{aligned} \bar{u}_s(n,t) = & \bar{F}_s(n) \cos \omega_n t + \bar{G}_s(n) \frac{\sin \omega_n t}{\omega_n} \\ & + \frac{1}{\rho \omega_n} \int_0^t \bar{f}_s(n,\tau) \sin \omega_n(t - \tau) d\tau \end{aligned} \quad (7-471)$$

where  $\omega_n = n\pi c/a$ . To obtain  $u(x,t)$ , the inversion theorem (5-151) must be applied to Eq. (7-471). When this is done, we obtain

$$\begin{aligned} u(x,t) = & \frac{2}{a} \sum_{n=1}^{\infty} \left\{ \bar{F}_s(n) \cos \omega_n t + \bar{G}_s(n) \frac{\sin \omega_n t}{\omega_n} \right\} \sin k_n x \\ & + \frac{2}{\rho a} \sum_{n=1}^{\infty} \frac{\sin k_n x}{\omega_n} \int_0^t \bar{f}_s(n,\tau) \sin \omega_n(t - \tau) d\tau \end{aligned} \quad (7-472)$$

where  $k_n = n\pi/a$ . Equation (7-472) expresses the solution of the mixed initial- and boundary-value problem for the one-dimensional wave equation in terms of the Fourier coefficients of the external force and the Fourier coefficients of the initial conditions.

It is instructive to examine the form of solution (7-472). It can be written as the sum of a complementary function  $u_c(x,t)$  and a particular integral  $u_P(x,t)$ :

$$u(x,t) = u_c(x,t) + u_P(x,t) \quad (7-473)$$

The complementary function  $u_c$  satisfies the homogeneous wave equation,

$$\frac{\partial^2 u_c}{\partial t^2} = c^2 \frac{\partial^2 u_c}{\partial x^2} \quad (7-474)$$

the initial conditions,

$$u_c(x,0) = F(x) \quad (7-475)$$

$$\frac{\partial u_c}{\partial t}(x,0) = G(x) \quad (7-476)$$

and the boundary conditions,

$$u_c(0,t) = 0 \quad (7-477)$$

$$u_c(a,t) = 0 \quad (7-478)$$

The particular integral  $u_P$  satisfies the inhomogeneous wave equation,

$$\frac{\partial^2 u_P}{\partial t^2} = c^2 \frac{\partial^2 u_P}{\partial x^2} + \frac{1}{\rho} f(x,t) \quad (7-479)$$

the initial conditions,

$$u_P(x,0) = 0 \quad (7-480)$$

$$\frac{\partial u_P}{\partial t}(x,0) = 0 \quad (7-481)$$

and the boundary conditions,

$$u_P(0,t) = 0 \quad (7-482)$$

$$u_P(a,t) = 0 \quad (7-483)$$

Notice that  $u_P$  and  $u_c$  satisfy the boundary conditions at  $x = 0$  and  $x = a$  independently of each other. The reason is that the initial conditions and the force term have both been expanded in terms of the same eigenfunctions, namely  $\sin n\pi x/a$ , and these eigenfunctions all vanish at  $x = 0$  and  $x = a$ .

## 7-21 THE TWO-DIMENSIONAL WAVE EQUATION

The inhomogeneous scalar wave equation appears most frequently in the form

$$\left( \nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) u(\mathbf{r},t) = -F(\mathbf{r},t) \quad (7-484)$$

where  $\nabla^2$  is the two-dimensional Laplacian operator. One basic procedure employed in solving Eq. (7-484) depends on the nature of the source term  $F(\mathbf{r},t)$ . If  $F(\mathbf{r},t)$  is defined for all positive and negative time, then it is possible to represent  $F(\mathbf{r},t)$  as a Fourier integral,

$$F(\mathbf{r},t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \bar{F}(\mathbf{r},\omega) e^{i\omega t} d\omega \quad (7-485)$$

and because Eq. (7-484) is a linear equation, it is reasonable to expect that the solution of (7-484) will also be a superposition of the same form:

$$u(\mathbf{r},t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \bar{u}(\mathbf{r},\omega) e^{i\omega t} d\omega \quad (7-486)$$

Inserting Eqs. (7-485) and (7-486) into (7-484) gives as a partial differential equation for the Fourier transform of  $u$ :

$$(\nabla^2 + k^2) \bar{u}(\mathbf{r},\omega) = -\bar{F}(\mathbf{r},\omega) \quad k = \frac{\omega}{c} \quad (7-487)$$

Equation (7-487) is called the "inhomogeneous Helmholtz equation."

A more realistic situation is one in which  $u(\mathbf{r},t)$ , together with all of its partial derivatives, is identically zero up to a certain instant, say  $t = 0$ . At  $t = 0$ , the source term  $F(\mathbf{r},t)$  is suddenly "turned on," and the initial values of  $u$  and  $\partial u/\partial t$  are prescribed. One then determines the later evolution of the solution  $u(\mathbf{r},t)$ . This type of problem is readily attacked with the aid of a Laplace transform. One multiplies Eq. (7-484) by  $e^{-st}$ , integrates twice by parts with respect to  $t$  from  $t = 0$  to  $t = \infty$ , and obtains thereby

$$(\nabla^2 - k^2) \bar{u}(\mathbf{r},s) = -\bar{F}(\mathbf{r},s) - \frac{1}{c^2} \left( \frac{\partial u}{\partial t} + su \right)_{t=0} \quad (7-488)$$

which is a Helmholtz type of equation for the Laplace transform

of  $u$ . In Eq. (7-487) it is assumed that

$$\bar{u}(\mathbf{r}, s) = \int_0^\infty u(\mathbf{r}, t) e^{-st} dt \quad (7-489)$$

$$\bar{F}(\mathbf{r}, s) = \int_0^\infty F(\mathbf{r}, t) e^{-st} dt \quad (7-490)$$

$$\lim_{t \rightarrow \infty} \left[ \frac{\partial u}{\partial t} + su(\mathbf{r}, t) \right] e^{-st} = 0 \quad (7-491)$$

and that  $k = s/c$ .

If  $u$  and  $\partial u/\partial t$  evolve continuously from the quiescent state  $u = 0$ , then initially we must have

$$u(\mathbf{r}, 0) = 0 \quad (7-492)$$

$$\left( \frac{\partial u}{\partial t} \right)_{t=0} = 0 \quad (7-493)$$

and Eq. (7-487) reduces to

$$(\nabla^2 - k^2)\bar{u}(\mathbf{r}, s) = -\bar{F}(\mathbf{r}, s) \quad (7-494)$$

We see that the Fourier or Laplace transform of the time-dependent wave equation leads, under appropriate circumstances, to a Helmholtz equation which must be solved.

The required solution can be obtained with the aid of the by now familiar Green's-function technique. Consider the problem of solving

$$(\nabla^2 + k^2)\bar{u}(\mathbf{r}, \omega) = -\bar{F}(\mathbf{r}, \omega) \quad (7-495)$$

We first introduce a Green's function, defined as the solution of

$$(\nabla^2 + k^2)\bar{G}(\mathbf{r}|\mathbf{r}') = -\delta(\mathbf{r} - \mathbf{r}') \quad (7-496)$$

where the Dirac delta function has the property that

$$\int_V \bar{F}(\mathbf{r}) \delta(\mathbf{r} - \mathbf{r}') dV = \begin{cases} \bar{F}(\mathbf{r}') & \text{if } \mathbf{r}' \text{ is the position vector} \\ & \text{of a point in } V \\ 0 & \text{if } \mathbf{r}' \text{ lies outside of } V \end{cases} \quad (7-497)$$

Next we observe that Green's symmetric identity can always be written as

$$\begin{aligned} \int_V \{ \bar{G}(\nabla^2 + k^2)\bar{u} - \bar{u}(\nabla^2 + k^2)\bar{G} \} dV \\ = \int_{S(V)} \left\{ \bar{G} \frac{\partial \bar{u}}{\partial n} - \bar{u} \frac{\partial \bar{G}}{\partial n} \right\} dS \end{aligned} \quad (7-498)$$

which is useful in connection with Eqs. (7-495) and (7-496). With the aid of these equations, (7-498) reduces to

$$\bar{u}(\mathbf{r}', \omega) = \int_V \bar{G}(\mathbf{r}|\mathbf{r}') \bar{F}(\mathbf{r}, \omega) dV + \int_{S(V)} \left\{ \bar{G} \frac{\partial \bar{u}}{\partial n} - \bar{u} \frac{\partial \bar{G}}{\partial n} \right\} dS \quad (7-499)$$

Here  $V$  represents the volume of a closed region of space,  $\mathbf{r}'$  is the position vector of a point in  $V$ , and  $S(V)$  is the surface enclosing  $V$ .

When we are concerned with a two-dimensional problem, Eq. (7-499) is replaced by

$$\bar{u}(\mathbf{r}', \omega) = \int_S \bar{G}(\mathbf{r}|\mathbf{r}') \bar{F}(\mathbf{r}, \omega) dS + \int_{C(S)} \left\{ \bar{G} \frac{\partial \bar{u}}{\partial n} - \bar{u} \frac{\partial \bar{G}}{\partial n} \right\} ds \quad (7-500)$$

Here  $S$  represents the area of a bordered surface,  $\mathbf{r}'$  is a point of  $S$ , and  $C(S)$  is the complete boundary curve bordering  $S$ . For example, if  $S$  is the surface of an annulus or "washer," then  $\mathbf{r}'$  is a point on the washer,  $S$  is the surface of the washer, and  $C(S)$  consists of the inner and outer circumference of the washer.

If we replace  $k^2$  by  $-k^2$  in Eqs. (7-496) and (7-498), then we can write the three-dimensional solution of (7-494) as

$$\bar{u}(\mathbf{r}', s) = \int_S \bar{G}(\mathbf{r}|\mathbf{r}') \bar{F}(\mathbf{r}, s) dV + \int_{S(V)} \left\{ \bar{G} \frac{\partial \bar{u}}{\partial n} - \bar{u} \frac{\partial \bar{G}}{\partial n} \right\} dS \quad (7-501)$$

and the corresponding two-dimensional solution becomes

$$\bar{u}(\mathbf{r}', s) = \int_S \bar{G}(\mathbf{r}|\mathbf{r}') \bar{F}(\mathbf{r}, s) dS + \int_{C(S)} \left\{ \bar{G} \frac{\partial \bar{u}}{\partial n} - \bar{u} \frac{\partial \bar{G}}{\partial n} \right\} d\sigma \quad (7-502)$$

In order to avoid confusion with the Laplace transform parameter  $s$ , the element of arc length in Eq. (7-502) has been changed from  $ds$  to  $d\sigma$ . Equations (7-499) to (7-502) are formal solutions of Eqs. (7-487) and (7-494). In order to make them useful, one must know, not only the Green's function  $\bar{G}$ , but also the boundary values of  $\bar{u}$  and  $\partial \bar{u}/\partial n$ . The solution in this form is overspecified, since one cannot arbitrarily assign both  $\bar{u}$  and  $\partial \bar{u}/\partial n$  on  $C(S)$ . When  $\bar{u}$  is a known function on  $C(S)$ , it is convenient to eliminate  $\partial \bar{u}/\partial n$  from the solution by making the Green's function satisfy a homogeneous Dirichlet condition,  $\bar{G} = 0$  on  $C(S)$ . Similarly, if it is  $\partial \bar{u}/\partial n$  that is to be assigned on  $C(S)$ , the boundary values of  $\bar{u}$  can be eliminated by making  $\partial \bar{G}/\partial n$  satisfy a homogeneous Neumann condition,  $\partial \bar{G}/\partial n = 0$  on  $C(S)$ . Occasionally, one wishes to specify a linear combination such as  $\partial \bar{u}/\partial n + \alpha \bar{u} = f$  on  $C(S)$ . This can be done by making  $\bar{G}$  and  $\partial \bar{G}/\partial n$  satisfy a so-called "impedance" boundary condition  $\partial \bar{G}/\partial n + \alpha \bar{G} = 0$  on  $C(S)$ .

### Properties of the Green's function

The Green's function satisfying Eq. (7-496) must have the following properties:

1.  $(\nabla^2 + k^2)\bar{G}(\mathbf{r}|\mathbf{r}') = 0$  except at  $\mathbf{r} = \mathbf{r}'$ .
2.  $\bar{G}(\mathbf{r}|\mathbf{r}')$  is continuous except at  $\mathbf{r} = \mathbf{r}'$ .
3.  $\bar{G}(\mathbf{r}|\mathbf{r}')$  satisfies homogeneous boundary conditions on  $C(S)$ .
4. The Green's function satisfies the reciprocity relation  $\bar{G}(\mathbf{r}|\mathbf{r}') = \bar{G}(\mathbf{r}'|\mathbf{r})$  obtained by interchanging  $\mathbf{r}$  and  $\mathbf{r}'$ .

Let us consider the reciprocity property of the Green's function. It says, for example, that if a point source of sound is placed in the center of a room and a point recorder is placed in one corner of

the room, the record obtained is exactly the same as that which would be obtained if the source and recorder were interchanged. This is not particularly obvious. One might expect the corner to affect the source and recorder differently. Reciprocity guarantees that there will be absolutely no difference on interchanging the source and recorder.

To prove the reciprocity property, consider  $\tilde{G}(\mathbf{r}|\mathbf{r}')$ , the solution at the point  $\mathbf{r}$  for an applied impulse at  $\mathbf{r}'$ , and also  $\tilde{G}(\mathbf{r}|\mathbf{r}'')$ , the solution at  $\mathbf{r}$  when the impulse is located at another point  $\mathbf{r}''$ , subject to the same *homogeneous* boundary condition. Then

$$\int_S \{ \tilde{G}(\mathbf{r}|\mathbf{r}') \nabla^2 \tilde{G}(\mathbf{r}|\mathbf{r}'') - \tilde{G}(\mathbf{r}|\mathbf{r}'') \nabla^2 \tilde{G}(\mathbf{r}|\mathbf{r}') \} dS \\ = \int_{C(S)} \left\{ \tilde{G}(\mathbf{r}|\mathbf{r}') \frac{\partial \tilde{G}(\mathbf{r}|\mathbf{r}'')}{\partial n} - \tilde{G}(\mathbf{r}|\mathbf{r}'') \frac{\partial \tilde{G}(\mathbf{r}|\mathbf{r}')}{\partial n} \right\} ds \quad (7-503)$$

Now if  $\tilde{G}(\mathbf{r}|\mathbf{r}')$  vanishes on  $C(S)$ , then so does  $\tilde{G}(\mathbf{r}|\mathbf{r}'')$ , by hypothesis. Similarly, if  $\partial \tilde{G}(\mathbf{r}|\mathbf{r}')/\partial n$  vanishes on  $C'(S)$ , then so does  $\partial \tilde{G}(\mathbf{r}|\mathbf{r}'')/\partial n$ , again by hypothesis. In either case,

$$\int_S \{ \tilde{G}(\mathbf{r}|\mathbf{r}') \nabla^2 \tilde{G}(\mathbf{r}|\mathbf{r}'') - \tilde{G}(\mathbf{r}|\mathbf{r}'') \nabla^2 \tilde{G}(\mathbf{r}|\mathbf{r}') \} dS = 0 \quad (7-504)$$

However,

$$\nabla^2 \tilde{G}(\mathbf{r}|\mathbf{r}') = -k^2 \tilde{G}(\mathbf{r}|\mathbf{r}') - \delta(\mathbf{r} - \mathbf{r}') \quad (7-505)$$

and

$$\nabla^2 \tilde{G}(\mathbf{r}|\mathbf{r}'') = -k^2 \tilde{G}(\mathbf{r}|\mathbf{r}'') - \delta(\mathbf{r} - \mathbf{r}'') \quad (7-506)$$

yield

$$\int_S \{ \tilde{G}(\mathbf{r}|\mathbf{r}') \delta(\mathbf{r} - \mathbf{r}'') - \tilde{G}(\mathbf{r}|\mathbf{r}'') \delta(\mathbf{r} - \mathbf{r}') \} dS = 0 \quad (7-507)$$

from which we obtain

$$\tilde{G}(\mathbf{r}''|\mathbf{r}') = \tilde{G}(\mathbf{r}'|\mathbf{r}'') \quad (7-508)$$

proving the reciprocity principle.

### The two-dimensional Green's function for an infinite medium

The two-dimensional infinite-medium Green's function for a line source is defined as the solution of

$$(\nabla^2 + k^2) \tilde{G}(\mathbf{r}|\mathbf{r}') = -\delta(\mathbf{r} - \mathbf{r}') \quad (7-509)$$

in polar coordinates. The Green's function  $\tilde{G}(\mathbf{r}|\mathbf{r}')$  represents the disturbance at  $\mathbf{r}$  due to a line source passing through the point  $\mathbf{r}'$  or vice versa. In a medium of unlimited extent,  $\tilde{G}$  should depend *only* on the distance between the source and receiver. Therefore

$$\tilde{G}(\mathbf{r}|\mathbf{r}') = \tilde{G}(|\mathbf{r} - \mathbf{r}'|) = \tilde{G}(R) \quad (7-510)$$

and (7-509) can be written as

$$(\nabla^2 + k^2) \tilde{G}(R) = -\delta(\mathbf{r} - \mathbf{r}') \quad (7-511)$$

It is convenient to choose a polar coordinate system centered at the source; then  $r = R$  and  $r' = 0$ . In this coordinate system

$dS = R dR d\theta$ , and

$$\delta(\mathbf{r} - \mathbf{r}') = \frac{\delta(R)}{R} \delta(\theta) \quad (7-512)$$

in order that

$$\int_0^\infty \int_{-\pi}^\pi \delta(\mathbf{r} - \mathbf{r}') R dR d\theta = 1 \quad (7-513)$$

Consequently Eq. (7-511) may be written as

$$\frac{1}{R} \frac{\partial}{\partial R} \left( R \frac{\partial \tilde{G}}{\partial R} \right) + k^2 \tilde{G}(R) = \frac{-\delta(R)}{R} \delta(\theta) \quad (7-514)$$

Equation (7-514) is now multiplied through by  $d\theta$ , and both sides are integrated from  $\theta = -\pi$  to  $\theta = \pi$  to give

$$\frac{1}{R} \frac{\partial}{\partial R} \left( R \frac{\partial \tilde{G}}{\partial R} \right) + k^2 \tilde{G}(R) = \frac{-\delta(R)}{2\pi R} \quad (7-515)$$

Integrating each side of Eq. (7-515) with respect to  $dR$  from  $R = -\epsilon$  to  $R = \epsilon$  and letting  $\epsilon \rightarrow 0$  then yields

$$\frac{\partial \tilde{G}}{\partial R} = -\frac{1}{2\pi R} \quad (7-516)$$

Equation (7-515) is recognized as Bessel's equation of order zero, and thus its solutions must be found among  $J_0(kR)$ ,  $N_0(kR)$ ,  $H_0^{(1)}(kR)$ , or  $H_0^{(2)}(kR)$ . To see which one is appropriate, we must recall from Eq. (7-486) that

$$G(R, t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \tilde{G}(R, \omega) e^{i\omega t} d\omega \quad (7-517)$$

is supposed to represent the disturbance generated by a *source* at  $R = 0$ . Thus as time increases, the disturbance must move *away* from the source. The asymptotic expansions in Sec. 6-11 lead to the conclusion that  $H_0^{(2)}(kR)$  is the correct function to use, since for very large  $kR$ ,

$$H_0^{(2)}(kR) e^{i\omega t} \sim \left( \frac{2}{\pi kR} \right)^{1/2} e^{i\pi/4} e^{-i(kR - \omega t)} \quad (7-518)$$

In other words, for very large  $kR$ ,  $H_0^{(2)}(kR) e^{i\omega t}$  reduces to a plane wave moving *away* from the source. Thus  $\tilde{G}$  is chosen as

$$\tilde{G}(kR) = A H_0^{(2)}(kR) \quad (7-519)$$

where the constant  $A$  is determined from Eq. (7-516) and the results in Sec. 6-11. One finds

$$\frac{d}{dR} [\lim_{R \rightarrow 0} A H_0^{(2)}(kR)] = \frac{2iA}{\pi} \frac{d}{dR} (\log R) = \frac{-1}{2\pi R} \quad (7-520)$$

or

$$A = \frac{i}{4} \quad (7-521)$$

and therefore

$$\tilde{G}(kR) = \frac{i}{4} H_0^{(2)}(kR) \quad (7-522)$$

In the same way we can show that the infinite-medium Green's function for

$$(\nabla^2 - k^2)\tilde{G}(\mathbf{r}|\mathbf{r}') = -\delta(\mathbf{r} - \mathbf{r}') \quad (7-523)$$

is given by

$$\tilde{G}(kR) = \frac{1}{2\pi} K_0(kR) \quad (7-524)$$

where as before

$$R = |\mathbf{r} - \mathbf{r}'| = [(x - x')^2 + (y - y')^2]^{1/2} \quad (7-525)$$

## 7-22 THE HELMHOLTZ EQUATION IN CYLINDRICAL COORDINATES

In this section we shall examine the problem of solving

$$(\nabla^2 + k^2)\tilde{G} = -\delta(\mathbf{r} - \mathbf{r}') \quad (7-526)$$

in cylindrical coordinates  $(r, \theta, z)$ . If we recall that  $dV = r dr d\theta dz$ , the delta function becomes

$$\delta(\mathbf{r} - \mathbf{r}') = \frac{\delta(r - r') \delta(\theta - \theta') \delta(z - z')}{r} \quad (7-527)$$

If we set  $\phi = \theta - \theta'$ , then Eq. (7-526) may be written as

$$\begin{aligned} \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial \tilde{G}}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \tilde{G}}{\partial \phi^2} + \frac{\partial^2 \tilde{G}}{\partial z^2} + k^2 \tilde{G} \\ = \frac{-\delta(r - r') \delta(\phi) \delta(z - z')}{r} \end{aligned} \quad (7-528)$$

and a solution of Eq. (7-528) may be sought in the form

$$\tilde{G} = \frac{1}{2\pi} \sum_{n=-\infty}^{+\infty} \tilde{G}_n e^{in\phi} \quad (7-529)$$

Let Eq. (7-528) be multiplied through by  $e^{-in\phi}$  and both sides integrated with respect to  $\phi$  from  $\phi = -\pi$  to  $\phi = \pi$ . First of all,

$$\int_{-\pi}^{\pi} \frac{\partial^2 \tilde{G}}{\partial \phi^2} e^{-in\phi} d\phi = e^{-in\phi} \left( \frac{\partial \tilde{G}}{\partial \phi} + in\tilde{G} \right) \Big|_{-\pi}^{\pi} - n^2 \int_{-\pi}^{\pi} \tilde{G} e^{-in\phi} d\phi \quad (7-530)$$

and in an unbounded medium,  $\tilde{G}$  should be periodic in  $\phi = \theta - \theta'$ , so that the integrated part in Eq. (7-530) vanishes. Because of the orthogonality properties of  $e^{-in\phi}$ ,

$$\tilde{G}_n = \int_{-\pi}^{\pi} \tilde{G} e^{-in\phi} d\phi \quad (7-531)$$

and thus

$$\begin{aligned} \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial \tilde{G}_n}{\partial r} \right) + \left( k^2 - \frac{n^2}{r^2} \right) \tilde{G}_n + \frac{\partial^2 \tilde{G}_n}{\partial z^2} \\ = \frac{-\delta(r - r') \delta(z - z')}{r} \end{aligned} \quad (7-532)$$

where  $n = 0, \pm 1, \pm 2, \dots$ . The  $z$  dependence in Eq. (7-532) is now eliminated by taking a Fourier transform of (7-532) with respect to  $z$ . Let (7-532) be multiplied through by  $e^{-ihz}$  and integrated with respect to  $z$  from  $z = -\infty$  to  $z = \infty$ . The result is

$$\frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial \tilde{F}_n}{\partial r} \right) - \frac{n^2}{r^2} \tilde{F}_n + (k^2 - h^2) \tilde{F}_n = \frac{-\delta(r - r')}{r} e^{-ihz'} \quad (7-533)$$

where

$$\tilde{F}_n = \int_{-\infty}^{+\infty} \tilde{G}_n e^{-ihz} dz \quad (7-534)$$

and

$$\tilde{G}_n = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \tilde{F}_n e^{ihz} dh \quad (7-535)$$

In obtaining Eq. (7-533) we have assumed that

$$\left( \frac{\partial \tilde{G}_n}{\partial z} + ih\tilde{G}_n \right) e^{-ihz} \Big|_{z=-\infty}^{z=+\infty} = 0 \quad (7-536)$$

and this will have to be justified later on.

### Application of the Hankel transform

In order to solve Eq. (7-533) it is convenient to introduce a new transform based on the orthogonality and completeness properties of Bessel functions. Just as trigonometric functions serve to generate the Fourier integral theorem,

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \cos xu du \int_0^{\infty} f(y) \cos uy dy \quad (7-537)$$

so Bessel functions may be expected to provide a similar result.

In fact, if  $f(x)$  is an arbitrary function subject only to mild restrictions, it can be shown that for  $\nu \geq -1/2$ ,

$$f(x) = \int_0^{\infty} J_{\nu}(xu) (xu)^{1/2} du \int_0^{\infty} f(y) J_{\nu}(uy) (uy)^{1/2} dy \quad (7-538)$$

In order for Eq. (7-538) to hold, it is sufficient that  $\int_0^{\infty} |f(y)| dy$  converge and that  $f(y)$  be of bounded variation in the neighborhood of  $x$ . If  $f(x)$  is discontinuous at  $x$ , then the left side of (7-538) is replaced by  $1/2[f(x+0) + f(x-0)]$ .

It is customary to use  $x^{1/2}f(x)$  instead of  $f(x)$  in (7-538) and to write the result in terms of a transform and an inversion theorem. Thus

$$H_{\nu}(u) = \int_0^{\infty} y f(y) J_{\nu}(uy) dy \quad (7-539)$$

is referred to as a Hankel transform of order  $\nu$  of  $f(y)$ , and

$$f(x) = \int_0^{\infty} u H_{\nu}(u) J_{\nu}(ux) du \quad (7-540)$$

is the corresponding inversion theorem.

Returning to Eq. (7-533), let

$$\bar{H}_n(\lambda, r', h) = \int_0^\infty \bar{F}_n(r, r', h) J_n(\lambda r) r dr \quad (7-541)$$

be a Hankel transform of order  $n$  of the solution of Eq. (7-533), and let

$$\bar{F}_n(r, r', h) = \int_0^\infty \bar{H}_n(\lambda, r', h) J_n(\lambda r) \lambda d\lambda \quad (7-542)$$

be the corresponding inversion formula. The usual transform method calls for us to multiply (7-533) on both sides by  $J_n(\lambda r) r dr$  and then integrate both sides from  $r = 0$  to  $r = \infty$ . This task is facilitated by taking note of Bessel's equation (Sec. 6-8), which yields

$$\frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial J_n}{\partial r} \right) - \frac{n^2}{r^2} J_n(\lambda r) = -\lambda^2 J_n(\lambda r) \quad (7-543)$$

and therefore

$$\begin{aligned} \int_0^\infty \left[ \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial \bar{F}_n}{\partial r} \right) - \left( \frac{n^2}{r^2} + h^2 - k^2 \right) \bar{F}_n \right] J_n(\lambda r) r dr \\ = r \left[ J_n(\lambda r) \frac{\partial \bar{F}_n}{\partial r} - \lambda \bar{F}_n J_n'(\lambda r) \right] \Big|_{r=0}^{r=\infty} - (\lambda^2 + h^2 - k^2) \bar{H}_n \\ = -J_n(\lambda r') e^{-ihz'} \end{aligned} \quad (7-544)$$

The integrated term

$$r \left[ J_n(\lambda r) \frac{\partial \bar{F}_n}{\partial r} - \lambda \bar{F}_n J_n'(\lambda r) \right] \Big|_{r=0}^{r=\infty} \quad (7-545)$$

vanishes automatically at the lower limit  $r = 0$ . At the upper limit  $r = \infty$ , we recall that we have assumed a priori that  $\int_0^\infty |\bar{F}_n| \sqrt{r} dr$  exists in order to use the Hankel transform. Therefore,  $\bar{F}_n$  must vanish faster than  $r^{-1/2}$ . However, for large  $r$ ,

$$J_n(\lambda r) \sim \left( \frac{2}{\pi r \lambda} \right)^{1/2} \cos \left( \lambda r - \frac{n\pi}{2} - \frac{\pi}{4} \right) \quad (7-546)$$

and as a result, both  $J_n(\lambda r) \partial \bar{F}_n / \partial r$  and  $\bar{F}_n J_n'(\lambda r)$  vanish faster than  $r^{-1}$ ; hence the integrated part vanishes as  $r \rightarrow \infty$ , and Eq. (7-544) reduces to

$$\bar{H}_n(\lambda, r', h, z') = \frac{J_n(\lambda r') e^{-ihz'}}{\lambda^2 + h^2 - k^2} \quad (7-547)$$

The inversion formulas (7-542) and (7-535) are now applied to Eq. (7-547) to obtain

$$\bar{G}_n(r, r'; z - z') = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{ih(z-z')} dh \int_0^\infty \frac{J_n(\lambda r) J_n(\lambda r') \lambda d\lambda}{h^2 + (\lambda^2 - k^2)} \quad (7-548)$$

Interchanging the order of integration in Eq. (7-548) and factoring the denominator permits us to write

$$\begin{aligned} \bar{G}_n = \int_0^\infty J_n(\lambda r) J_n(\lambda r') \lambda d\lambda \\ \cdot \frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{e^{ih(z-z')} dh}{(h + i\sqrt{\lambda^2 - k^2})(h - i\sqrt{\lambda^2 - k^2})} \end{aligned} \quad (7-549)$$

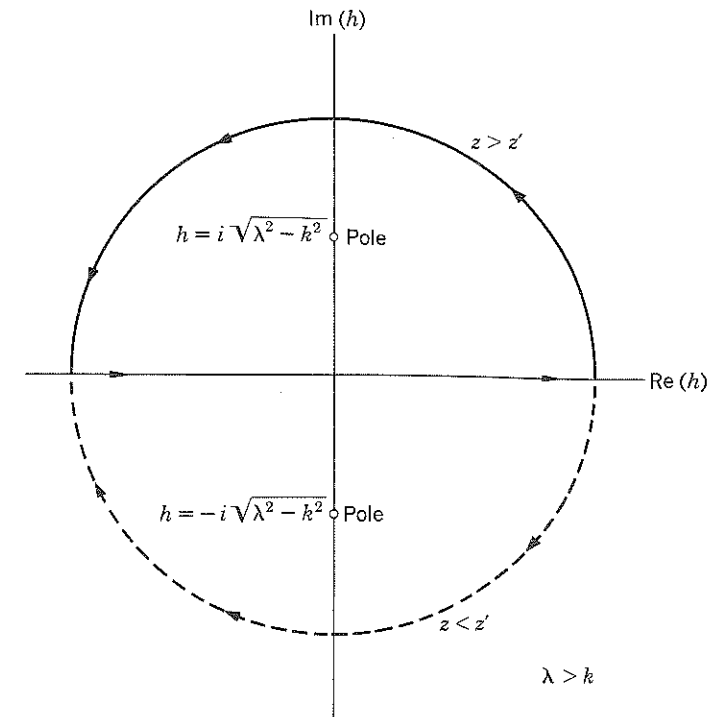


Figure 7-7 The complex  $h$  plane.

We wish to evaluate the integral

$$\frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{e^{ih(z-z')} dh}{(h + i\sqrt{\lambda^2 - k^2})(h - i\sqrt{\lambda^2 - k^2})} \quad (7-550)$$

by the method of residues (Sec. 4-18). Apparently the integrand in (7-550) has first-order poles at

$$h = \pm i\sqrt{\lambda^2 - k^2} \quad (7-551)$$

in the complex  $h$  plane (Fig. 7-7). Two cases,  $z > z'$  and  $z < z'$ , must be considered. Suppose first that  $z > z'$ . In order to use the residue theorem (4-187), we must close the path of integration in (7-550) with a semicircle whose radius approaches infinity. This semicircle must lie in either the upper or lower half of the complex  $h$  plane. The choice is made in such a way that on the semicircular path  $e^{ih(z-z')} \rightarrow 0$  as  $|h| \rightarrow \infty$ . This is accomplished, when  $z > z'$ , by choosing the semicircle to lie in the upper half plane, so that on it  $\text{Im}(h) > 0$ . The pole which contributes to (7-550) is the one which lies in the upper half of the complex  $h$  plane. In order to fix this pole, we choose the branch of  $\sqrt{\lambda^2 - k^2}$  in (7-550) which satisfies

$$\text{Re}(\sqrt{\lambda^2 - k^2}) > 0 \quad (7-552)$$

Then the residue contribution to (7-550) comes from the pole at

$$h = i\sqrt{\lambda^2 - k^2} \quad (7-553)$$

and we have

$$\frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{e^{ih(z-z')} dh}{(h+i\sqrt{\lambda^2-k^2})(h-i\sqrt{\lambda^2-k^2})} = \frac{e^{-\sqrt{\lambda^2-k^2}(z-z')}}{2\sqrt{\lambda^2-k^2}} \quad (7-554)$$

Suppose next that  $z < z'$ . For this case, the path of integration in (7-550) should be closed with a semicircle in the lower half plane  $\text{Im}(h) < 0$ . Then the residue contribution to (7-550) comes from the pole at  $h = -i\sqrt{\lambda^2-k^2}$ , provided we keep the branch of the radical selected in inequality (7-552). Keeping in mind that the pole at  $h = -i\sqrt{\lambda^2-k^2}$  is encircled in the negative (clockwise) direction, we have

$$\frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{e^{ih(z-z')} dh}{(h+i\sqrt{\lambda^2-k^2})(h-i\sqrt{\lambda^2-k^2})} = \frac{e^{-\sqrt{\lambda^2-k^2}(z'-z)}}{2\sqrt{\lambda^2-k^2}} \quad (7-555)$$

Both Eqs. (7-554) and (7-555) can be combined into a single formula by taking the absolute value of  $z - z'$ . The result is

$$\frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{e^{ih(z-z')} dh}{h^2 + \lambda^2 - k^2} = \frac{e^{-\sqrt{\lambda^2-k^2}|z-z'|}}{2\sqrt{\lambda^2-k^2}} \quad (7-556)$$

and thus Eq. (7-548) reduces to

$$\tilde{G}_n(r, r', z, z') = \frac{1}{2} \int_0^\infty \frac{J_n(\lambda r) J_n(\lambda r') e^{-\sqrt{\lambda^2-k^2}|z-z'|}}{\sqrt{\lambda^2-k^2}} \lambda d\lambda \quad (7-557)$$

where  $\text{Re}(\sqrt{\lambda^2-k^2}) > 0$ . Finally, using Eq. (7-529), the solution of Eq. (7-528) becomes

$$\begin{aligned} \tilde{G}(r, r', \theta - \theta', |z - z'|) \\ = \frac{1}{4\pi} \sum_{n=-\infty}^{+\infty} e^{in(\theta-\theta')} \int_0^\infty \frac{J_n(\lambda r) J_n(\lambda r') e^{-\sqrt{\lambda^2-k^2}|z-z'|}}{\sqrt{\lambda^2-k^2}} \lambda d\lambda \end{aligned} \quad (7-558)$$

The Green's function (7-558) can be regarded as the result of superposing separated solutions of the Helmholtz equation:

$$(\nabla^2 + k^2)\psi = 0 \quad (7-559)$$

Let

$$\psi_{n>} = J_n(\lambda r_{>}) e^{in\theta_{>}} e^{-\sqrt{\lambda^2-k^2} z_{>}} \quad (7-560)$$

and

$$\psi_{n<} = J_n(\lambda r_{<}) e^{-in\theta_{<}} e^{+\sqrt{\lambda^2-k^2} z_{<}} \quad (7-561)$$

where the notation  $>$  is as in Eq. (6-485). Equations (7-560) and (7-561) are separated solutions of Eq. (7-559) with separation parameters  $n$  and  $\lambda$ . Using (7-560) and (7-561), the Green's function (7-558) assumes the form

$$\tilde{G} = \frac{1}{4\pi} \sum_{n=-\infty}^{+\infty} \int_0^\infty \psi_{n>} \psi_{n<} \frac{\lambda d\lambda}{\sqrt{\lambda^2-k^2}} \quad (7-562)$$

If one considers a plane passing through the source perpendicular to the  $z$  axis, then Eq. (7-562) corresponds to expanding the Green's function  $\tilde{G}$  in exponentially attenuated standing-wave solutions of Helmholtz's equation. The exponential attenuation takes place along the  $z$  axis as one recedes from the source plane.

#### Independent representation of the Green's function for a point source

In spherical coordinates the Green's function for a point source should be spherically symmetric and thus should depend only on the radial coordinate  $R$ . We have

$$(\nabla^2 + k^2)\tilde{G}(R) = -\delta(R) \quad (7-563)$$

or

$$\frac{1}{R} \frac{\partial^2}{\partial R^2} [R\tilde{G}(R)] + k^2\tilde{G}(R) = -\delta(R) \quad (7-564)$$

When  $R \neq 0$ , Eq. (7-564) is solved by

$$R\tilde{G}(R) = Ae^{ikR} + Be^{-ikR} \quad (7-565)$$

where  $k = \omega/c$ .

Since

$$G(R, t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \tilde{G}(R) e^{i\omega t} d\omega \quad (7-566)$$

only the  $Be^{-ikR}$  term corresponds to a disturbance traveling away from the source at  $R = 0$  as time increases. Consequently,  $A$  is equated to zero, and there remains only

$$\tilde{G}(R) = \frac{Be^{-ikR}}{R} \quad (7-567)$$

To determine  $B$ , let Eq. (7-563) be integrated over a sphere of volume  $V$  centered at  $R = 0$ , and then let  $V$  shrink to zero. Since

$$\int_V \delta(R) dV = \iiint_{\text{Sphere}} \delta(x) \delta(y) \delta(z) dx dy dz = 1 \quad (7-568)$$

we have

$$\int_V (\nabla^2 + k^2)\tilde{G}(R) dV = -1 \quad (7-569)$$

and using the divergence theorem, Eq. (7-569) becomes

$$\int_{S(V)} \frac{\partial \tilde{G}}{\partial R} dS + k^2 \int \tilde{G}(R) dV = -1 \quad (7-570)$$

In terms of the element of solid angle  $d\Omega$  we have  $dS = R^2 d\Omega$  and  $dV = R^2 d\Omega dR$  so that Eqs. (7-567) and (7-570) become

$$B \int_{S(V)} \frac{\partial}{\partial R} \left( \frac{e^{-ikR}}{R} \right) R^2 d\Omega + k^2 B \int_V e^{-ikR} R d\Omega dR = -1 \quad (7-571)$$

The volume of the sphere is now permitted to shrink to zero by letting  $R \rightarrow 0$ . Since

$$\lim_{R \rightarrow 0} k^2 B \int_V e^{-ikR} R d\Omega dR = 0 \quad (7-572)$$

and

$$\lim_{R \rightarrow 0} B \int_{S(V)} \frac{\partial}{\partial R} \left( \frac{e^{-ikR}}{R} \right) R^2 d\Omega = -4\pi B \quad (7-573)$$

Eq. (7-571) yields

$$B = \frac{1}{4\pi} \quad (7-574)$$

and thus

$$\bar{G}(R) = \frac{e^{-ikR}}{4\pi R} \quad (7-575)$$

is another important expression for the Green's function for a point source. On comparison of Eqs. (7-575) and (7-558), which both represent the Green's function for a point source, the expansion

$$\frac{e^{-ikR}}{4\pi R} = \frac{1}{4\pi} \sum_{n=-\infty}^{+\infty} e^{in(\theta-\theta')} \int_0^\infty \frac{J_n(\lambda r) J_n(\lambda r') e^{-\sqrt{\lambda^2 - k^2}|z-z'|}}{\sqrt{\lambda^2 - k^2}} \lambda d\lambda \quad (7-576)$$

is obtained. In Eq. (7-576)  $(r, \theta, z)$  are cylindrical coordinates, and  $R$  is the distance  $R = [(x-x')^2 + (y-y')^2 + (z-z')^2]^{1/2}$  between the source point  $(r', \theta', z')$  and the point of observation  $(r, \theta, z)$ . A number of interesting results can be derived from (7-576). For example, in the static limit  $k \rightarrow 0$ , and (7-576) becomes

$$\frac{1}{R} = \sum_{n=-\infty}^{+\infty} e^{in(\theta-\theta')} \int_0^\infty e^{-\lambda|z-z'|} J_n(\lambda r) J_n(\lambda r') d\lambda \quad (7-577)$$

for a static source at the origin  $r' = 0$  and  $z' = 0$ . Since  $J_n(0) = 0$  when  $n \neq 0$ , Eq. (7-577) reduces to

$$\frac{1}{\sqrt{r^2 + z^2}} = \int_0^\infty e^{-\lambda|z|} J_0(\lambda r) d\lambda \quad (7-578)$$

which is called the "Lipshitz integral." If  $r = 1$  in Eq. (7-578), then the equation takes the form of the Laplace transform of  $J_0(\lambda)$ . The inversion theorem (5-473) for the Laplace transform can then be applied to (7-578) to give an integral representation for  $J_0$ : i.e.,

$$J_0(x) = \frac{1}{2\pi i} \int_{\sigma_0 - i\infty}^{\sigma_0 + i\infty} \frac{e^{sx}}{\sqrt{1+s^2}} ds \quad (7-579)$$

For a nonstatic source ( $k \neq 0$ ) at the origin, the previous remarks show that Eq. (7-576) yields

$$\frac{e^{-ikR}}{4\pi R} = \frac{1}{4\pi} \int_0^\infty e^{-\sqrt{\lambda^2 - k^2}|z|} \frac{J_0(\lambda r) \lambda d\lambda}{\sqrt{\lambda^2 - k^2}} \quad (7-580)$$

where  $R = (x^2 + y^2 + z^2)^{1/2} = (r^2 + z^2)^{1/2}$ . Equation (7-580) has the form of a Hankel transform of order zero of  $e^{-\sqrt{\lambda^2 - k^2}|z|}/\sqrt{\lambda^2 - k^2}$ . Therefore, using the Hankel-transform inversion theorem (7-542) we

obtain Sommerfeld's integral,

$$\frac{e^{-\sqrt{\lambda^2 - k^2}|z|}}{\sqrt{\lambda^2 - k^2}} = \int_0^\infty \frac{e^{-ikR}}{R} J_0(\lambda r) r dr \quad (7-581)$$

where, as in all of these formulas,  $\text{Re}(\sqrt{\lambda^2 - k^2}) > 0$ .

We are now able to return to the matter of justifying Eq. (7-536):

$$\left( \frac{\partial \bar{G}_n}{\partial z} + ik\bar{G}_n \right) e^{-ihz} \Big|_{z=-\infty}^{z=+\infty} = 0 \quad (7-582)$$

With the aid of Eq. (7-531),

$$\begin{aligned} \left( \frac{\partial \bar{G}_n}{\partial z} + ik\bar{G}_n \right) e^{-ihz} \Big|_{z=-\infty}^{z=+\infty} \\ = \int_{-\pi}^{\pi} e^{-in\phi} \left( \frac{\partial \bar{G}}{\partial z} + ik\bar{G} \right) e^{-ihz} \Big|_{z=-\infty}^{z=+\infty} d\phi \end{aligned} \quad (7-583)$$

However

$$\bar{G}(R) = \frac{e^{-ikR}}{4\pi R} \quad (7-584)$$

and thus

$$\begin{aligned} \left( \frac{\partial \bar{G}_n}{\partial z} + ik\bar{G}_n \right) e^{-ihz} \Big|_{z=-\infty}^{z=+\infty} \\ = - \int_{-\pi}^{\pi} e^{-in\phi} \frac{z(1+ikR)e^{-i(kR+h z)}}{4\pi R^3} \Big|_{z=-\infty}^{z=+\infty} d\phi \end{aligned} \quad (7-585)$$

Since for very large  $z$  the radial distance  $R \sim z$ , it follows that

$$\lim_{z \rightarrow \infty} \frac{z(1+ikR)}{4\pi R^3} = 0 \quad (7-586)$$

and therefore Eq. (7-582) must hold.

According to the remarks following Eq. (7-84), we should expect that there are two alternative ways to divide up the space around the source. One way, which we have just used, is to divide space up with a plane passing through the source perpendicular to the  $z$  axis. The other way, which we shall examine now, is to divide up space with a cylinder passing through the source and having generators parallel to the  $z$  axis. For this purpose, let us go back to Eq. (7-533),

$$\frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial \bar{F}_n}{\partial r} \right) - \frac{n^2}{r^2} \bar{F}_n + (k^2 - h^2) \bar{F}_n = \frac{-\delta(r-r')}{r} e^{-ihz'} \quad (7-587)$$

which is just Bessel's equation. Instead of using a Hankel transform on Eq. (7-587) we can solve it directly as in Eq. (7-311). When  $r \neq r'$ ,

$$\bar{F}_n = \begin{cases} A J_n(\sqrt{k^2 - h^2} r) \\ B H_n^{(2)}(\sqrt{k^2 - h^2} r) \end{cases} \quad (7-588)$$

provides linearly independent solutions of (7-587). When the source is neither at the origin nor at infinity, the Green's function should

remain finite at  $r = 0$  and  $r = \infty$ . One therefore chooses

$$\bar{F}_n = AJ_n(\sqrt{k^2 - h^2} r) \quad 0 \leq r < r' \quad (7-589)$$

$$\bar{F}_n = BH_n^{(2)}(\sqrt{k^2 - h^2} r) \quad r' < r < \infty \quad (7-590)$$

The choice of  $H_n^{(2)}$  is made instead of  $H_n^{(1)}$  for the reason mentioned in connection with Eq. (7-560). The constants  $A$  and  $B$  are determined as in Sec. 7-15, and the result is that

$$\bar{F}_n = -\frac{J_n(\sqrt{k^2 - h^2} r_<)H_n^{(2)}(\sqrt{k^2 - h^2} r_>)e^{-ihz'}}{\sqrt{k^2 - h^2} r' W\{J_n, H_n^{(2)}; r'\}} \quad (7-591)$$

where

$$W\{J_n, H_n^{(2)}; r'\}$$

denotes the Wronskian of  $J_n$  and  $H_n^{(2)}$ . The Wronskian can be evaluated as in Sec. 7-15 to give

$$W\{J_n, H_n^{(2)}; r'\} = \frac{-2i}{\pi \sqrt{k^2 - h^2} r'} \quad (7-592)$$

so that Eq. (7-591) becomes

$$\bar{F}_n = \frac{-i\pi}{2} J_n(\sqrt{k^2 - h^2} r_<)H_n^{(2)}(\sqrt{k^2 - h^2} r_>)e^{-ihz'} \quad (7-593)$$

Consequently, Eqs. (7-529), (7-535), and (7-593) yield

$$\bar{G}(R) = \frac{e^{-ikR}}{4\pi R} = \frac{-i}{8\pi} \sum_{n=-\infty}^{+\infty} e^{in(\theta-\theta')} \int_{-\infty}^{+\infty} J_n(\sqrt{k^2 - h^2} r_<)H_n^{(2)}(\sqrt{k^2 - h^2} r_>)e^{ih(z-z')} dh \quad (7-594)$$

for the infinite-medium Green's function.

The Green's function (7-594) can be regarded as the result of superposing separated solutions of Helmholtz's equation:

$$(\nabla^2 + k^2)\psi = 0 \quad (7-595)$$

Let

$$\psi_{n<} = J_n(\sqrt{k^2 - h^2} r_<)e^{i(n\theta + hz_<)} \quad (7-596)$$

and

$$\psi_{n>} = H_n^{(2)}(\sqrt{k^2 - h^2} r_>)e^{-i(n\theta + hz_>)} \quad (7-597)$$

These are separated solutions of Eq. (7-595) with separation parameters  $n$  and  $h$ . Using (7-596) and (7-597), the Green's function (7-594) becomes

$$\bar{G} = \frac{-i}{8\pi} \sum_{n=-\infty}^{+\infty} \int_{-\infty}^{+\infty} \psi_{n>} \psi_{n<} dh \quad (7-598)$$

If one considers a cylinder passing through the source at  $(r', \theta', z')$  and having generators parallel to the  $z$  axis, then Eq. (7-598) divides space into two regions, one inside and one outside the cylinder. The

expansion (7-598) corresponds to expanding the Green's function in a set of waves which propagate along  $z$  instead of decaying exponentially along  $z$ .

Just as for Eq. (7-576), a number of interesting formulas can be obtained from (7-594). For example, in the static limit  $k \rightarrow 0$ , (7-594) reduces to

$$\frac{1}{R} = \frac{-i}{2} \sum_{n=-\infty}^{+\infty} e^{in(\theta-\theta')} \int_{-\infty}^{+\infty} J_n(-i|h|r_<)H_n^{(2)}(-i|h|r_>)e^{ih(z-z')} dh \quad (7-599)$$

The limiting form (7-599) arises because the branch of  $\sqrt{k^2 - h^2}$  in (7-594) is determined by the requirement that

$$\text{Im}(\sqrt{k^2 - h^2}) < 0 \quad (7-600)$$

To see why inequality (7-600) must hold, notice that

$$\lim_{x \rightarrow \infty} \frac{1}{R} = \lim_{x \rightarrow \infty} \frac{1}{[(x-x')^2 + (y-y')^2 + (z-z')^2]^{1/2}} = 0 \quad (7-601)$$

must apply to both sides of Eq. (7-599). As  $x \rightarrow \infty$ ,

$$\lim_{x \rightarrow \infty} H_n^{(2)}(-i|h|r_>) \rightarrow \sqrt{\frac{-2}{\pi|h|x}} e^{-i\{ -i|h|x - \frac{n\pi}{2} - \frac{\pi}{4} \}} \rightarrow 0 \quad (7-602)$$

has the correct behavior; however, if

$$\text{Im}(\sqrt{k^2 - h^2}) > 0 \quad (7-603)$$

then for  $k \rightarrow 0$ ,  $\sqrt{k^2 - h^2} \rightarrow i|h|$ , and the right side of Eq. (7-594) will diverge when  $x$  is allowed to become infinite. It is left as an exercise for the reader to show that

$$\frac{e^{ikR}}{4\pi R} = \frac{i}{8\pi} \sum_{n=-\infty}^{+\infty} e^{in(\theta-\theta')} \int_{-\infty}^{+\infty} J_n(\sqrt{k^2 - h^2} r_<)H_n^{(1)}(\sqrt{k^2 - h^2} r_>)e^{ih(z-z')} dh \quad (7-604)$$

In the static limit  $k = 0$ , the asymptotic form of  $H_n^{(1)}(\sqrt{k^2 - h^2} r_>)$  as  $x \rightarrow \infty$  requires that

$$\text{Im}(\sqrt{k^2 - h^2}) > 0 \quad (7-605)$$

and consequently, Eq. (7-604) gives

$$\frac{1}{R} = \frac{i}{2} \sum_{n=-\infty}^{+\infty} e^{in(\theta-\theta')} \int_{-\infty}^{+\infty} J_n(i|h|r_<)H_n^{(1)}(i|h|r_>)e^{ih(z-z')} dh \quad (7-606)$$

Since

$$J_n(i|h|r_<) = e^{in\pi/2} I_n(|h|r_<) \quad (7-607)$$

and

$$H_n^{(1)}(i|h|r_>) = \frac{2}{\pi i} e^{-in\pi/2} K_n(|h|r_>) \quad (7-608)$$

we can write Eq. (7-606) as

$$\frac{1}{R} = \frac{1}{\pi} \sum_{n=-\infty}^{+\infty} e^{in(\theta-\theta')} \int_{-\infty}^{+\infty} I_n(|h|r_<) K_n(|h|r_>) e^{ih(z-z')} dh \quad (7-609)$$

which is the same as Eq. (7-334). The static limit,  $k = 0$ , with the source at the origin,  $r' = 0$ ,  $z' = 0$ , yields

$$\frac{1}{R} = \frac{1}{\sqrt{r^2 + z^2}} = \frac{1}{\pi} \int_{-\infty}^{+\infty} e^{ihz} K_0(hr) dh \quad (7-610)$$

or

$$\frac{1}{\sqrt{r^2 + z^2}} = \frac{2}{\pi} \int_0^{+\infty} K_0(hr) \cos hz dh \quad (7-611)$$

This corresponds to Eq. (7-340). When there is a source of spherical waves at the origin, then  $k \neq 0$ ,  $r' = 0$ ,  $z' = 0$ ,  $R = (r^2 + z^2)^{1/2}$ , and

$$\frac{e^{ikR}}{R} = \frac{i}{2} \int_{-\infty}^{+\infty} H_0^{(1)}(\sqrt{k^2 - h^2} r) e^{ihz} dh \quad (7-612)$$

This last result is called "Weyrich's formula."

## 7-23 THE HELMHOLTZ EQUATION IN RECTANGULAR CARTESIAN COORDINATES

Consider the problem of solving

$$(\nabla^2 + k^2)\bar{G} = -\delta(r - r') = -\delta(x - x') \delta(y - y') \delta(z - z') \quad (7-613)$$

in rectangular Cartesian coordinates. We already know that one solution is

$$\bar{G}(R) = \frac{e^{-ikR}}{4\pi R} \quad (7-614)$$

However, it is of some interest to derive this result on its own merits by applying a triple Fourier transform. Beginning with

$$\left[ \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) + k^2 \right] \bar{G} = -\delta(x - x') \delta(y - y') \delta(z - z') \quad (7-615)$$

we multiply this by  $e^{-i\xi x} dx e^{-i\eta y} dy e^{-i\zeta z} dz$  and integrate both sides from  $x = -\infty$  to  $x = \infty$ ,  $y = -\infty$  to  $y = \infty$ ,  $z = -\infty$  to  $z = \infty$  by parts to obtain

$$[(\xi^2 + \eta^2 + \zeta^2) - k^2] \bar{F}(\xi, \eta, \zeta) = e^{-i(\xi x' + \eta y' + \zeta z')} \quad (7-616)$$

on the assumption that

$$\left( \frac{\partial \bar{G}}{\partial x} + i\xi \bar{G} \right) e^{-i\xi x} \Big|_{x=-\infty}^{x=+\infty} = 0 \quad (7-617)$$

$$\left( \frac{\partial \bar{G}}{\partial y} + i\eta \bar{G} \right) e^{-i\eta y} \Big|_{y=-\infty}^{y=+\infty} = 0 \quad (7-618)$$

$$\left( \frac{\partial \bar{G}}{\partial z} + i\zeta \bar{G} \right) e^{-i\zeta z} \Big|_{z=-\infty}^{z=+\infty} = 0 \quad (7-619)$$

Equations (7-617) to (7-619) are justified in the same way as Eq. (7-582). The triple Fourier transform of  $\bar{G}$  is therefore given by

$$\bar{F}(\xi, \eta, \zeta) = \frac{e^{-i(\xi x' + \eta y' + \zeta z')}}{(\xi^2 + \eta^2 + \zeta^2) - k^2} \quad (7-620)$$

where

$$\bar{F}(\xi, \eta, \zeta) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \bar{G} e^{-i(\xi x + \eta y + \zeta z)} dx dy dz \quad (7-621)$$

The corresponding inversion theorem (5-373) is

$$\bar{G} = \frac{1}{(2\pi)^3} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \bar{F}(\xi, \eta, \zeta) e^{i(\xi x + \eta y + \zeta z)} d\xi d\eta d\zeta \quad (7-622)$$

and thus

$$\bar{G} = \frac{1}{(2\pi)^3} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{e^{i[\xi(x-x') + \eta(y-y') + \zeta(z-z')]}}{(\xi^2 + \eta^2 + \zeta^2) - k^2} d\xi d\eta d\zeta \quad (7-623)$$

represents the Green's-function solution of Eq. (7-615). In Eq. (7-623) we regard  $(\xi, \eta, \zeta)$  as forming a right-handed rectangular Cartesian coordinate system in  $\mathbf{K}$  space.

Equation (7-623) can be written more compactly by observing that

$$\mathbf{K} = (\xi, \eta, \zeta) \quad (7-624)$$

$$K^2 = |\mathbf{K}|^2 = \xi^2 + \eta^2 + \zeta^2 \quad (7-625)$$

$$\mathbf{R} = [(x - x'), (y - y'), (z - z')] \quad (7-626)$$

$$R^2 = |\mathbf{R}|^2 = (x - x')^2 + (y - y')^2 + (z - z')^2 \quad (7-627)$$

$$dV_K = d\xi d\eta d\zeta \quad (7-628)$$

$$\mathbf{K} \cdot \mathbf{R} = \xi(x - x') + \eta(y - y') + \zeta(z - z') \quad (7-629)$$

so that Eq. (7-623) reduces to

$$\bar{G}(R) = \frac{1}{(2\pi)^3} \int_{\text{All } \mathbf{K} \text{ space}} \frac{e^{i\mathbf{K} \cdot \mathbf{R}}}{K^2 - k^2} dV_K \quad (7-630)$$

The evaluation of Eq. (7-630) is simplified by the following tricks. First of all, we imagine  $\mathbf{R}$  to be a fixed vector in space. Then we rotate the  $(\xi, \eta, \zeta)$  coordinate system until the  $\zeta$  axis coincides with the direction of  $\mathbf{R}$ . Next, we introduce spherical coordinates  $(K, \alpha, \beta)$  into  $(\xi, \eta, \zeta)$  space (Fig. 7-8). In these coordinates  $\alpha$  is the colatitude measured with respect to the  $\zeta$  axis,  $\beta$  is the colongitude measured from the  $\xi$  axis, and  $K$  is the radial distance from the origin of coordinates. The transformation from  $(\xi, \eta, \zeta)$  to  $(K, \alpha, \beta)$  is therefore

$$\xi = K \sin \alpha \cos \beta \quad (7-631)$$

$$\eta = K \sin \alpha \sin \beta \quad (7-632)$$

$$\zeta = K \cos \alpha \quad (7-633)$$

The volume element  $dV_K = d\xi d\eta d\zeta$  is replaced by

$$dV_K = K^2 \sin \alpha dK d\alpha d\beta \quad (7-634)$$

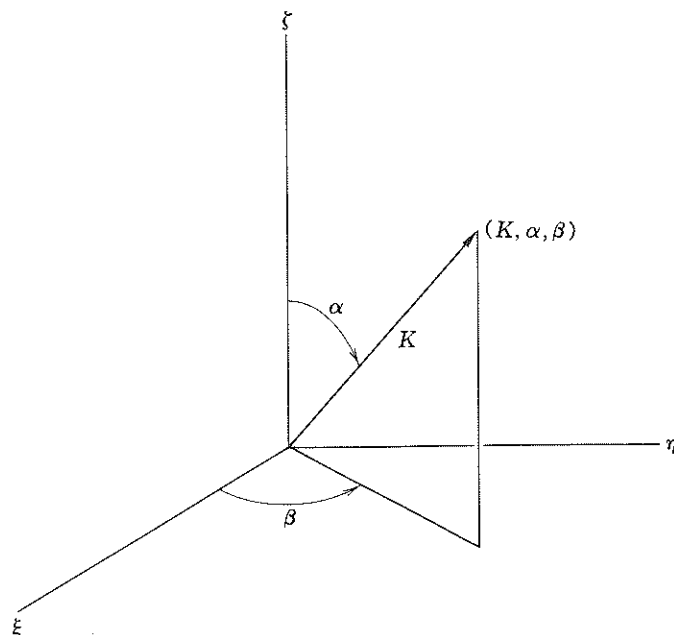


Figure 7-8 Spherical coordinates in  $(\xi, \eta, \zeta)$  space.

and since we have chosen the  $\zeta$  axis to coincide with the direction of  $\mathbf{R}$ , the colatitude angle  $\alpha$  of  $\mathbf{K}$  is just the angle between  $\mathbf{K}$  and  $\mathbf{R}$ ; thus

$$\mathbf{K} \cdot \mathbf{R} = KR \cos \alpha \quad (7-635)$$

With these remarks Eq. (7-630) reduces to

$$\bar{G}(R) = \frac{1}{(2\pi)^3} \int_0^\infty \int_0^\pi \int_0^{2\pi} \frac{e^{iKR \cos \alpha}}{K^2 - k^2} K^2 \sin \alpha \, dK \, d\alpha \, d\beta \quad (7-636)$$

The integrations over  $\alpha$  and  $\beta$  can be performed immediately using

$$\int_0^\pi e^{ikR \cos \alpha} \sin \alpha \, d\alpha = \frac{e^{iKR} - e^{-iKR}}{iKR} \quad (7-637)$$

and

$$\int_0^{2\pi} d\beta = 2\pi \quad (7-638)$$

The result is

$$\bar{G}(R) = \frac{1}{(2\pi)^2} \int_0^\infty \frac{e^{iKR} - e^{-iKR}}{iR(K^2 - k^2)} K \, dK \quad (7-639)$$

or

$$\bar{G}(R) = \frac{1}{(2\pi)^2} \int_0^\infty \frac{e^{iKR}}{iR(K^2 - k^2)} K \, dK - \frac{1}{(2\pi)^2} \int_0^\infty \frac{e^{-iKR}}{iR(K^2 - k^2)} K \, dK \quad (7-640)$$

Suppose that we replace  $K$  by  $-K$  in the first term of Eq. (7-640) and then reverse the limits of integration. We obtain

$$\bar{G}(R) = \frac{-1}{(2\pi)^2} \int_{-\infty}^{+\infty} \frac{e^{-iKR}}{iR(K^2 - k^2)} K \, dK \quad (7-641)$$

which must be evaluated by the method of residues (Sec. 4-18). The integrand in Eq. (7-641) has simple first-order poles at  $K = k$  and  $K = -k$ . When  $k$  is real, these poles lie on the path of integration, which complicates the evaluation of (7-641). One way to resolve this difficulty is to replace  $k$  by  $k + i\epsilon$  in (7-641). Then the poles are at  $K = k + i\epsilon$  and  $K = -k - i\epsilon$ , and no longer fall on the path of integration. The evaluation of (7-641) by residues gives a definite result which depends on  $\epsilon$  and can be evaluated for  $\epsilon \rightarrow 0$ . A similar procedure based on replacing  $k$  by  $k - i\epsilon$  in (7-641) also leads to a definite but different result as  $\epsilon \rightarrow 0$ . One of these two results is then eliminated on the basis of physical considerations.

Thus, consider

$$\bar{G}(R) = -\frac{1}{(2\pi)^2} \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{+\infty} \frac{e^{-iKR}}{iR[K^2 - (k + i\epsilon)^2]} K \, dK \quad (7-642)$$

where  $\epsilon$  is a small positive number. Notice that  $R$  is a distance and is always non-negative; also,  $K = \text{Re}(K) + i \text{Im}(K)$ , so that

$$|e^{-iKR}| = e^{R \text{Im}(K)} \quad (7-643)$$

In order to integrate (7-643) by residues, the path of integration must be closed with a semicircle of infinite radius. Equation (7-643) shows that the semicircle must lie in the lower half of the complex  $K$  plane (Fig. 7-9). Then  $\text{Im}(K) < 0$  on the semicircle, and its contribution to Eq. (7-642) vanishes exponentially. The pole in the lower half plane occurs at  $K = -k - i\epsilon$  and is encircled in the negative (clockwise) direction. Using Eqs. (4-191) and (4-195), we find

$$\bar{G}(R) = \lim_{\epsilon \rightarrow 0} \frac{e^{i(k+i\epsilon)R}}{4\pi R} = \frac{e^{ikR}}{4\pi R} \quad (7-644)$$

Next, let  $k$  be replaced by  $k - i\epsilon$ ; then Eq. (7-641) becomes

$$\bar{G}(R) = \frac{1}{(2\pi)^2} \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{+\infty} \frac{e^{-iKR}}{iR[K^2 - (k - i\epsilon)^2]} K \, dK \quad (7-645)$$

However, Eq. (7-643) still applies, and the path of integration must be completed with a semicircle in the lower half plane. The pole in the lower half plane now occurs at  $K = k - i\epsilon$ , and the residue theorem gives

$$\bar{G}(R) = \lim_{\epsilon \rightarrow 0} \frac{e^{-i(k-i\epsilon)R}}{4\pi R} = \frac{e^{-ikR}}{4\pi R} \quad (7-646)$$

To see whether (7-644) or (7-646) is correct, let us evaluate  $G(R, t)$  itself. We have

$$G(R, t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \bar{G}(R, \omega) e^{i\omega t} \, d\omega \quad (7-647)$$

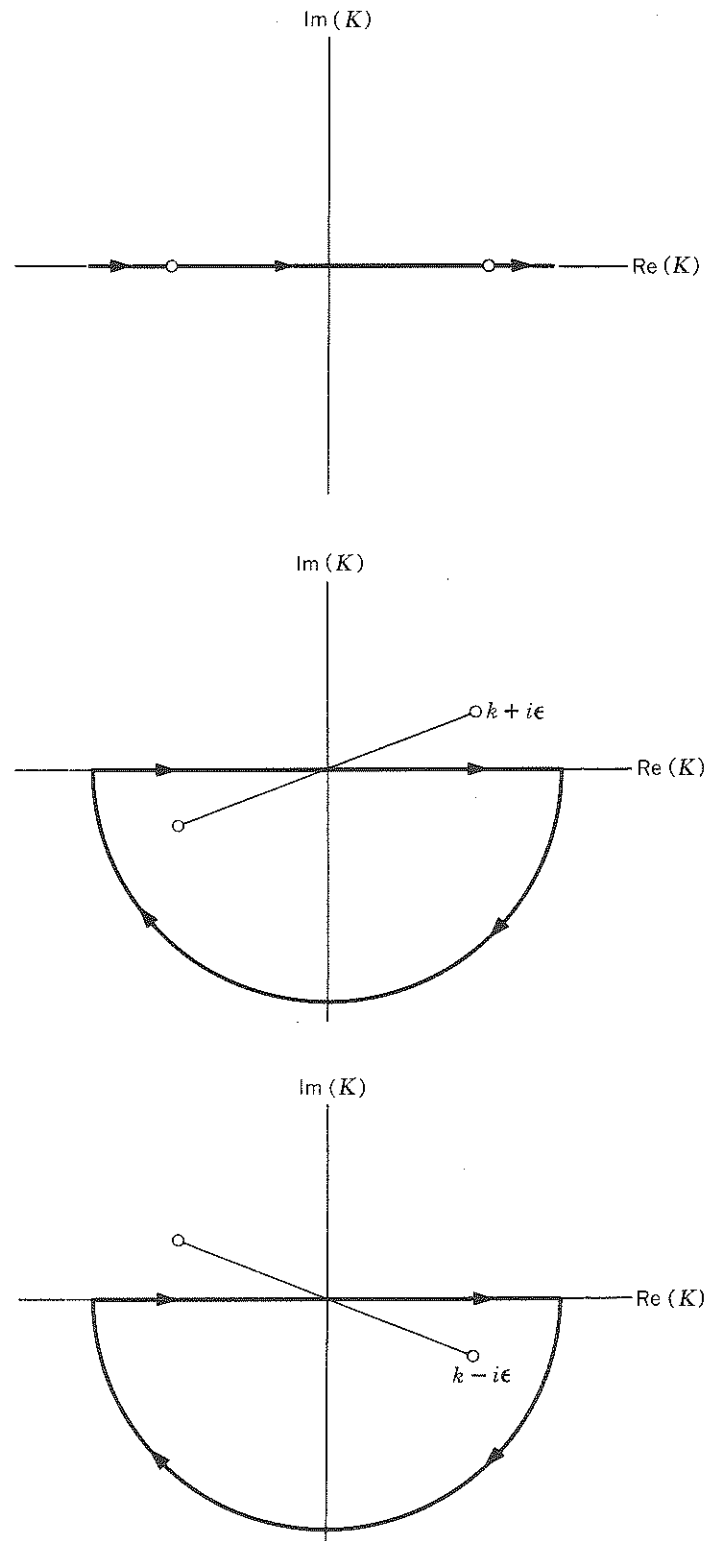


Figure 7-9 Integration contours in the complex  $K$  plane.

and therefore Eq. (7-644) gives

$$G(R, t) = \frac{1}{4\pi R} \cdot \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{i\omega(R/c+t)} d\omega \quad (7-648)$$

where  $k = \omega/c$ . Recalling (5-331), Eq. (7-648) becomes

$$G(R, t) = \frac{\delta(t + R/c)}{4\pi R} \quad (7-649)$$

while Eq. (7-646) leads to

$$G(R, t) = \frac{\delta(t - R/c)}{4\pi R} \quad (7-650)$$

We see that Eq. (7-649) corresponds to a spherical pulse which contracts toward the origin  $R = 0$  as time increases, while (7-650) corresponds to a spherical pulse which expands away from the origin  $R = 0$  as time increases. The physical behavior of a point source at  $R = 0$  fired off at  $t = 0$  corresponds to (7-650) and not to (7-649). Therefore,

$$\bar{G}(R) = \frac{e^{-ikR}}{4\pi R} \quad (7-651)$$

is the correct result provided that the inverse Fourier transform of  $\bar{G}(R)$  is defined as in Eq. (7-647).

Instead of using spherical coordinates, we can deal directly with Eq. (7-623):

$$\bar{G}(R) = \frac{1}{(2\pi)^3} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{e^{i[\xi(x-x') + \eta(y-y') + \zeta(z-z')]} d\xi d\eta d\zeta}{(\xi^2 + \eta^2 + \zeta^2) - k^2} \quad (7-652)$$

Let us perform the integration over  $d\zeta$  by residues; we write Eq. (7-652) as

$$\bar{G} = \frac{1}{(2\pi)^3} \int_{-\infty}^{+\infty} \frac{e^{i\xi(x-x')}}{\{\gamma^2 + \zeta^2\}} d\zeta \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} e^{i[\xi(x-x') + \eta(y-y')]} d\xi d\eta \quad (7-653)$$

where

$$\gamma^2 = (\xi^2 + \eta^2) - k^2 \quad (7-654)$$

The integrand of the  $d\zeta$  integral in Eq. (7-653) has first-order poles at  $\zeta = \pm i\gamma$ . Suppose that  $z > z'$ , and let  $\zeta = \text{Re}(\zeta) + i \text{Im}(\zeta)$ ; then  $|e^{i\zeta(z-z')}| = e^{-(z-z') \text{Im}(\zeta)}$

shows that in order to obtain exponential decay of the semicircular contribution to (7-653), the semicircle must lie in the upper half,  $\text{Im}(\zeta) > 0$ , of the complex  $\zeta$  plane (Fig. 7-10). The path of integration in

$$\int_{-\infty}^{+\infty} \frac{e^{i\xi(x-x')}}{\gamma^2 + \zeta^2} d\zeta \quad (7-656)$$

is closed with a semicircle on which  $\text{Im}(\zeta) > 0$ , and the branch of

$$\gamma = \sqrt{(\xi^2 + \eta^2) - k^2} \quad (7-657)$$

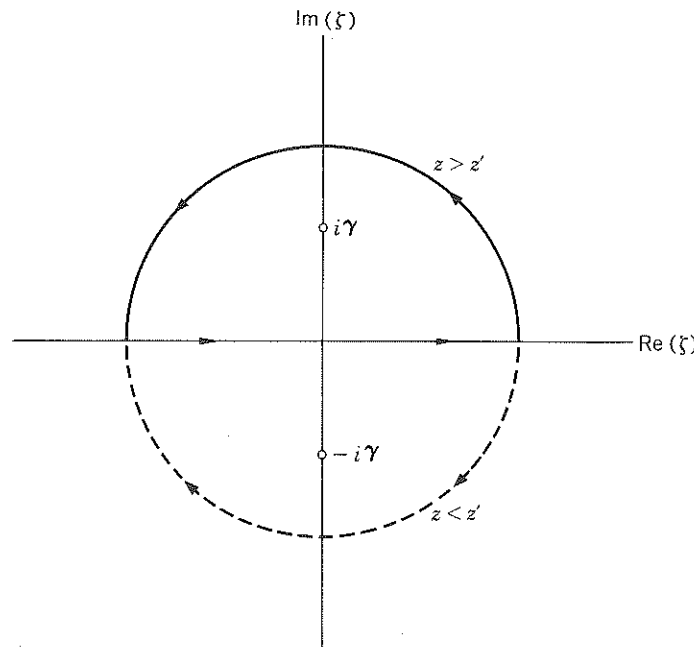


Figure 7-10 The complex  $\zeta$  plane.

in (7-656) is fixed by choosing it so that

$$\operatorname{Re}(\gamma) > 0 \quad (7-658)$$

With these conventions, the pole in the upper half of the complex  $\zeta$  plane occurs at  $\zeta = i\gamma$ , provided that the radicand in Eq. (7-657) is real and positive. When  $z > z'$ , the residue theorem (4-191) and (4-195) applied to (7-656) yields

$$\int_{-\infty}^{+\infty} \frac{e^{i\zeta(z-z')}}{\gamma^2 + \zeta^2} d\zeta = \frac{\pi e^{-\gamma(z-z')}}{\gamma} \quad (7-659)$$

with  $\operatorname{Re}(\gamma) > 0$ .

If  $z < z'$ , then the path of integration in (7-656) must be closed with a semicircle in the lower half plane  $\operatorname{Im}(\zeta) < 0$ . For this case the pole  $\zeta = -i\gamma$  is encircled in the negative (clockwise) direction, and

$$\int_{-\infty}^{+\infty} \frac{e^{i\zeta(z-z')}}{\gamma^2 + \zeta^2} d\zeta = \frac{\pi e^{-\gamma(z'-z)}}{\gamma} \quad (7-660)$$

where again  $\operatorname{Re}(\gamma) > 0$ . Combining Eqs. (7-659) and (7-660) into one expression, we write

$$\int_{-\infty}^{+\infty} \frac{e^{i\zeta(z-z')}}{\gamma^2 + \zeta^2} d\zeta = \pi \frac{e^{-\gamma|z-z'|}}{\gamma} \quad (7-661)$$

With the aid of (7-661), Eq. (7-653) reduces to

$$\bar{G} = \frac{1}{8\pi^2} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{e^{-\gamma|z-z'|}}{\gamma} e^{i[\xi(x-x') + \eta(y-y')]} d\xi d\eta \quad (7-662)$$

and this represents  $\bar{G}e^{i\omega t}$  as a superposition of plane waves, propagating along  $x$  and  $y$  and exponentially attenuated along  $z$ . To further evaluate Eq. (7-662), we transform it to plane polar coordinates, using

$$x - x' = r \cos \theta \quad (7-663)$$

$$y - y' = r \sin \theta \quad (7-664)$$

$$\xi = \lambda \cos \phi \quad (7-665)$$

$$\eta = \lambda \sin \phi \quad (7-666)$$

$$d\xi d\eta = \lambda d\lambda d\phi \quad (7-667)$$

$$\gamma = \sqrt{\lambda^2 - k^2} \quad (7-668)$$

with the result that

$$\bar{G} = \frac{1}{8\pi^2} \int_0^\infty \int_{-\pi}^\pi \frac{e^{-\sqrt{\lambda^2 - k^2}|z-z'|}}{\sqrt{\lambda^2 - k^2}} e^{i\lambda \cos(\theta-\phi)} \lambda d\lambda d\theta \quad (7-669)$$

Since

$$J_0(\lambda r) = \frac{1}{2\pi} \int_{-\pi}^\pi e^{i\lambda r \cos \Phi} d\Phi \quad (7-670)$$

Eq. (7-669) is the same as the expression

$$\bar{G} = \frac{1}{4\pi} \int_0^\infty \frac{e^{-\sqrt{\lambda^2 - k^2}|z-z'|}}{\sqrt{\lambda^2 - k^2}} J_0(\lambda r) \lambda d\lambda \quad (7-671)$$

obtained in Eq. (7-580). Our analysis in Eqs. (7-644) and (7-646) led us to the result that

$$\bar{G}(R) = \begin{cases} \frac{e^{ikR}}{4\pi R} & \operatorname{Im}(k) \rightarrow 0^+ \\ \frac{e^{-ikR}}{4\pi R} & \operatorname{Im}(k) \rightarrow 0^- \end{cases} \quad (7-672)$$

According to Eq. (7-672), when  $k$  is regarded as a complex variable, analytic continuation (see Sec. 4-5) demands that

$$\frac{1}{4\pi} \int_0^\infty \frac{e^{-\sqrt{\lambda^2 - k^2}|z-z'|}}{\sqrt{\lambda^2 - k^2}} J_0(\lambda r) \lambda d\lambda = \begin{cases} \frac{e^{ikR}}{4\pi R} & \operatorname{Im}(k) \geq 0 \\ \frac{e^{-ikR}}{4\pi R} & \operatorname{Im}(k) \leq 0 \end{cases} \quad (7-673)$$

where  $R = \sqrt{r^2 + (z - z')^2}$ . In Eq. (7-673)  $r$  and  $z - z'$  are real, and  $\operatorname{Re}(\sqrt{\lambda^2 - k^2}) > 0$ . It must follow that

$$\begin{aligned} \bar{G} &= \frac{1}{8\pi^2} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{e^{-\gamma|z-z'|}}{\gamma} e^{i[\xi(x-x') + \eta(y-y')]} d\xi d\eta \\ &= \begin{cases} \frac{e^{ikR}}{4\pi R} & \operatorname{Im}(k) \geq 0 \\ \frac{e^{-ikR}}{4\pi R} & \operatorname{Im}(k) \leq 0 \end{cases} \end{aligned} \quad (7-674)$$

where  $\gamma = \sqrt{(\xi^2 + \eta^2) - k^2}$  and  $\operatorname{Re}(\gamma) > 0$ .

## 7-24 THE HELMHOLTZ EQUATION IN SPHERICAL COORDINATES

In spherical coordinates  $(r, \theta, \phi)$  the inhomogeneous Helmholtz equation,

$$(\nabla^2 + k^2)\bar{G} = -\delta(r - r') \quad (7-675)$$

takes the form

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial \bar{G}}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial \bar{G}}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \bar{G}}{\partial \phi^2} + k^2 \bar{G} = \frac{-\delta(r - r') \delta(\theta - \theta') \delta(\phi - \phi')}{r^2 \sin \theta} \quad (7-676)$$

It is convenient to separate the  $r$  portion of the Laplacian from the  $\theta$  and  $\phi$  portions, and so we define  $\nabla_r^2$  and  $\nabla_{\theta, \phi}^2$  by

$$r^2 \nabla^2 = \nabla_r^2 + \nabla_{\theta, \phi}^2 \quad (7-677)$$

where

$$\nabla_r^2 = \frac{\partial}{\partial r} \left( r^2 \frac{\partial}{\partial r} \right) \quad (7-678)$$

and

$$\nabla_{\theta, \phi}^2 = \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \quad (7-679)$$

The results obtained in Sec. 7-18 suggest that we assume a solution for  $\bar{G}$  of the form

$$\bar{G}(r, r'; \theta, \theta'; \phi - \phi') = \sum_{n=0}^{\infty} \sum_{m=-n}^{+n} \bar{F}_n^m(r, r'; \theta', \phi') P_n^{|m|}(\cos \theta) e^{im\phi} \quad (7-680)$$

Since

$$\nabla_{\theta, \phi}^2 [P_n^{|m|}(\cos \theta) e^{im\phi}] = -n(n+1) P_n^{|m|}(\cos \theta) e^{im\phi} \quad (7-681)$$

it follows that on substituting Eq. (7-680) into

$$(\nabla_r^2 + \nabla_{\theta, \phi}^2 + k^2 r^2) \bar{G} = \frac{-\delta(r - r') \delta(\theta - \theta') \delta(\phi - \phi')}{\sin \theta} \quad (7-682)$$

we obtain

$$\begin{aligned} \sum_{n=0}^{\infty} \sum_{m=-n}^n [\nabla_r^2 + k^2 r^2 - n(n+1)] \bar{F}_n^m(r, r'; \theta', \phi') P_n^{|m|}(\cos \theta) e^{im\phi} \\ = \frac{-\delta(r - r') \delta(\theta - \theta') \delta(\phi - \phi')}{\sin \theta} \end{aligned} \quad (7-683)$$

for the determination of  $\bar{F}_n^m(r, r'; \theta', \phi')$ . To compute  $\bar{F}_n^m$ , let both sides of Eq. (7-683) be multiplied by  $P_{\lambda}^{|m|}(\cos \theta) e^{-im\phi} \sin \theta d\theta d\phi$  and then integrated with respect to  $\theta$  and  $\phi$  over the surface of a unit sphere. With the aid of the orthogonality relation (7-386) we find

$$\begin{aligned} [\nabla_r^2 + k^2 r^2 - n(n+1)] \bar{F}_n^{|m|} \frac{(n+|m|)!}{(n-|m|)!} \frac{4\pi}{2n+1} \\ = -\delta(r - r') P_n^{|m|}(\cos \theta') e^{-im\phi'} \end{aligned} \quad (7-684)$$

For convenience, let us define

$$\bar{F}_n^{|m|}(r, r', \theta', \phi') = \frac{2n+1}{4\pi} \frac{(n-|m|)!}{(n+|m|)!} P_n^{|m|}(\cos \theta') e^{-im\phi'} K_n(r, r') \quad (7-685)$$

Then Eq. (7-684) becomes

$$[\nabla_r^2 + k^2 r^2 - n(n+1)] K_n(r, r') = -\delta(r - r') \quad (7-686)$$

or

$$\frac{d}{dr} \left( r^2 \frac{dK_n}{dr} \right) + [k^2 r^2 - n(n+1)] K_n(r, r') = -\delta(r - r') \quad (7-687)$$

Equation (7-687) can be put into a recognizable form by making the substitution

$$K_n(r, r') = \frac{v_n(r, r')}{\sqrt{kr}} \quad (7-688)$$

Using Eq. (7-688),

$$\frac{d}{dr} \left( r^2 \frac{dK_n}{dr} \right) = \frac{1}{\sqrt{kr}} \left( r^2 \frac{d^2 v_n}{dr^2} + r \frac{dv_n}{dr} - \frac{1}{4} v_n \right) \quad (7-689)$$

so that when  $r \neq r'$ , Eq. (7-687) becomes

$$r^2 \frac{d^2 v_n}{dr^2} + r \frac{dv_n}{dr} - \frac{v_n}{4} + [k^2 r^2 - n(n+1)] v_n = 0 \quad (7-690)$$

However,

$$-n(n+1) - \frac{1}{4} = -(n + \frac{1}{2})^2 \quad (7-691)$$

and therefore

$$r^2 \frac{d^2 v_n}{dr^2} + r \frac{dv_n}{dr} + [k^2 r^2 - (n + \frac{1}{2})^2] v_n = 0 \quad (7-692)$$

Equation (7-692) was discussed in Sec. 6-16 and has Bessel functions of order  $n + \frac{1}{2}$  as solutions. Consequently, when  $r \neq r'$ , the solutions of Eq. (7-687) are *spherical* Bessel functions. For example,

$$K_n(r, r') = \begin{cases} A j_n(kr) \\ B h_n^{(1)}(kr) \end{cases} \quad (7-693)$$

is one linearly independent set of solutions of (7-687) when  $r \neq r'$ . Since the source is neither at the origin  $r = 0$  nor at infinity,  $K_n$  should remain finite at  $r = 0$  and  $r = \infty$ . The exact behavior of  $h_n^{(1)}(kr)$  is given by Eq. (6-345) as

$$h_n^{(1)}(kr) = (-i)^{n+1} \frac{e^{ikr}}{r} \quad (7-694)$$

We see from Eq. (7-694) that  $K_n$  will remain finite as  $r \rightarrow \infty$ , provided that

$$\text{Im}(k) \geq 0 \quad (7-695)$$

Thus the pair of solutions (7-693) is assigned so that

$$K_n = A j_n(kr) \quad 0 \leq r < r' \quad (7-696)$$

$$K_n = B h_n^{(1)}(kr) \quad r' < r < \infty \quad (7-697)$$

Since  $K_n$  must be continuous at  $r = r'$ , we find

$$A j_n(kr') - B h_n^{(1)}(kr') = 0 \quad (7-698)$$

and integrating both sides of Eq. (7-687) over an interval  $2\epsilon$  centered at  $r = r'$  then gives

$$\lim_{\epsilon \rightarrow 0} r^2 \frac{dK_n}{dr} \Big|_{r=r'-\epsilon}^{r=r'+\epsilon} = -1 \quad (7-699)$$

or

$$A k r^2 \frac{d j_n}{d(kr)} - B k r^2 \frac{d h_n^{(1)}}{d(kr)} = 1 \quad (7-700)$$

Solving Eqs. (7-698) and (7-700) simultaneously, we find

$$A = \frac{h_n^{(1)}(kr')}{\Delta} \quad (7-701)$$

$$B = \frac{j_n(kr')}{\Delta} \quad (7-702)$$

where

$$\Delta = -kr^2 \begin{vmatrix} j_n & h_n^{(1)} \\ j_n' & h_n^{(1)'} \end{vmatrix} \quad (7-703)$$

The determinant in Eq. (7-703) is the Wronskian of  $j_n$  and  $h_n^{(1)}$  and is presumably constant. Therefore, we can evaluate it by using the limiting forms (6-338), (6-340), and (6-341) for  $j_n$  and  $h_n^{(1)}$  as  $kr \rightarrow 0$ . The result is

$$\Delta = -kr^2 \begin{vmatrix} \frac{(kr)^n}{(2n+1)!!} \left[ \frac{(kr)^n}{(2n+1)!!} - \frac{i(2n-1)!!}{(kr)^{n+1}} \right] \\ \frac{n(kr)^{n-1}}{(2n+1)!!} \left[ \frac{n(kr)^{n-1}}{(2n+1)!!} + \frac{i(n+1)(2n-1)!!}{(kr)^{n+2}} \right] \end{vmatrix} \quad (7-704)$$

which reduces to

$$\begin{aligned} \Delta &= -kr^2 \left[ \frac{i(n+1)(2n-1)!!}{(kr)^2(2n+1)!!} + \frac{i(2n-1)!!}{(2n+1)!!} \frac{n}{(kr)^2} \right] \\ &= -kr^2 \left[ \frac{i(2n+1)(2n-1)!!}{(kr)^2(2n+1)!!} \right] = \frac{1}{ik} \end{aligned} \quad (7-705)$$

and consequently Eqs. (7-696), (7-697), (7-701), (7-702), and (7-705) combine to give

$$K_n(r, r') = ik j_n(kr_{<}) h_n^{(1)}(kr_{>}) \quad (7-706)$$

where

$$\text{Im}(k) \geq 0 \quad (7-707)$$

By means of Eq. (7-685),  $F_n^{lm}(r, r', \theta', \phi')$  is recovered and inserted into Eq. (7-680) to yield

$$\bar{G}(r, r'; \theta, \theta'; \phi - \phi')$$

$$\begin{aligned} &= \frac{e^{ikR}}{4\pi R} = \frac{ik}{4\pi} \sum_{n=0}^{\infty} (2n+1) j_n(kr_{<}) h_n^{(1)}(kr_{>}) \\ &\sum_{m=-n}^n \frac{(n-|m|)!}{(n+|m|)!} P_n^{|m|}(\cos \theta) P_n^{|m|}(\cos \theta') e^{im(\phi-\phi')} \end{aligned} \quad (7-708)$$

Equation (7-708) is the desired expansion in spherical coordinates of Green's function defined by Eq. (7-675).

Suppose that the source is placed on the polar axis of the coordinate system. Then  $\theta' = 0$ , and  $\theta$  becomes the angle between the source and the observer. Let us distinguish this angle by calling it  $\theta = \gamma$ . It follows from Eqs. (6-296) and (6-293) that

$$P_n^{|m|}(1) = 0 \quad m > 0 \quad (7-709)$$

and

$$P_n^{|m|}(1) = 1 \quad m = 0 \quad (7-710)$$

Thus for  $\theta' = 0$  and  $\theta = \gamma$ ,

$$\sum_{m=-n}^n \frac{(n-|m|)!}{(n+|m|)!} P_n^{|m|}(\cos \theta) P_n^{|m|}(\cos \theta') e^{im(\phi-\phi')} = P_n(\cos \gamma) \quad (7-711)$$

This shows that the complicated sum on the left side of Eq. (7-711) is nothing more than an  $n$ th-order Legendre polynomial whose argument is the cosine of the angle between the source and the observer. With this remark Eq. (7-708) reduces to

$$\bar{G} = \frac{e^{ikR}}{4\pi R} = \frac{ik}{4\pi} \sum_{n=0}^{\infty} (2n+1) j_n(kr_{<}) h_n^{(1)}(kr_{>}) P_n(\cos \gamma) \quad (7-712)$$

ending our discussion of the expansion of  $\bar{G}$  in spherical coordinates.

## 7-25 INTERPRETATION OF THE INTEGRAL SOLUTION OF HELMHOLTZ'S EQUATION

Let us examine the integral representation of the solution of

$$(\nabla^2 + k^2) \bar{u}(\mathbf{r}, \omega) = -\bar{F}(\mathbf{r}, \omega) \quad (7-713)$$

a little more closely. It is given by Eq. (7-499) as

$$\bar{u}(\mathbf{r}', \omega) = \int_V \bar{G}(\mathbf{r}|\mathbf{r}') \bar{F}(\mathbf{r}, \omega) dV + \int_{S(V)} \left\{ \bar{G} \frac{\partial \bar{u}}{\partial n} - \bar{u} \frac{\partial \bar{G}}{\partial n} \right\} dS \quad (7-714)$$

We know from our previous discussions that if

$$\bar{G} = \int_{-\infty}^{+\infty} G e^{-i\omega t} dt \quad (7-715)$$

then

$$\bar{G} = \frac{e^{-ikR}}{4\pi R} \quad (7-716)$$

and if

$$\bar{G} = \int_{-\infty}^{+\infty} G e^{i\omega t} dt \quad (7-717)$$

then

$$\bar{G} = \frac{e^{ikR}}{4\pi R} \quad (7-718)$$

in order that  $G(R, t)$  represent the radiation from a point source. Although we normally use Eq. (7-715) as the definition of  $G$ , Eq. (7-717) is equally valid and probably occurs more frequently in the literature. If (7-717) is chosen as the definition of  $\tilde{G}$ , then Eq. (7-714) reduces to

$$\bar{u}(\mathbf{r}', \omega) = \frac{1}{4\pi} \int_V \bar{F}(\mathbf{r}, \omega) \frac{e^{ikR}}{R} dV + \frac{1}{4\pi} \int_{S(V)} \left[ \frac{e^{ikR}}{R} \frac{\partial \bar{u}}{\partial n} - \bar{u} \frac{\partial}{\partial n} \left( \frac{e^{ikR}}{R} \right) \right] dS \quad (7-719)$$

where  $R = |\mathbf{r} - \mathbf{r}'|$ .

Equation (7-719) represents the disturbance  $\bar{u}$  at  $\mathbf{r}'$  in  $V$  as the result of superposing contributions from volume sources distributed inside  $V$  with a density  $\bar{F}(\mathbf{r}, \omega)$ , and surface sources distributed over the surface  $S(V)$  enclosing  $V$ . The term

$$\int_{S(V)} \frac{e^{ikR}}{4\pi R} \frac{\partial \bar{u}}{\partial n} dS \quad (7-720)$$

represents the contribution to  $\bar{u}$  at  $\mathbf{r}'$  due to surface point sources distributed on  $S(V)$  with a surface density  $\partial \bar{u} / \partial n$ . The remaining term

$$\int_{S(V)} \frac{\partial}{\partial n} \left( \frac{e^{ikR}}{4\pi R} \right) (-\bar{u} dS) \quad (7-721)$$

represents the contribution to  $\bar{u}$  at  $\mathbf{r}'$  due to a surface distribution of dipole radiators (double layer) of moment  $-\bar{u} dS$ . This is most easily seen from Fig. 7-11. Suppose that a pair of periodically oscillating charges of opposite sign are located a small distance  $\Delta n$  apart along the normal  $\mathbf{n}$  to  $S(V)$ . The dipole field generated by adding together the Green's function for each charge is

$$u_{\text{dipole}} = \frac{qe^{ikR_1}}{4\pi R_1} - \frac{qe^{ikR}}{4\pi R} \quad (7-722)$$

The magnitude of the charge is now allowed to increase without bound as  $\Delta n$  shrinks to zero. It is assumed that this happens in such a way that the product of  $q$  and  $\Delta n$  remains fixed: i.e.,

$$q \Delta n = m \text{ (const)} \quad (7-723)$$

Then

$$\bar{u}_{\text{dipole}} = q\Delta \left( \frac{e^{ikR}}{4\pi R} \right) = m \frac{\Delta(e^{ikR}/4\pi R)}{\Delta n} \quad (7-724)$$

and as the two charges approach each other along the normal to  $S$ ,

$$\lim_{\Delta n \rightarrow 0} \bar{u}_{\text{dipole}} = m \frac{\partial}{\partial n} \left( \frac{e^{ikR}}{4\pi R} \right) \quad (7-725)$$

The effect of the radiation sources outside of  $V$  are accounted for by introducing suitable fictitious monopole and dipole distributions on  $S(V)$ . As the surface  $S(V)$  recedes to infinity, the limiting value of the surface integral in Eq. (7-719) represents the effect of

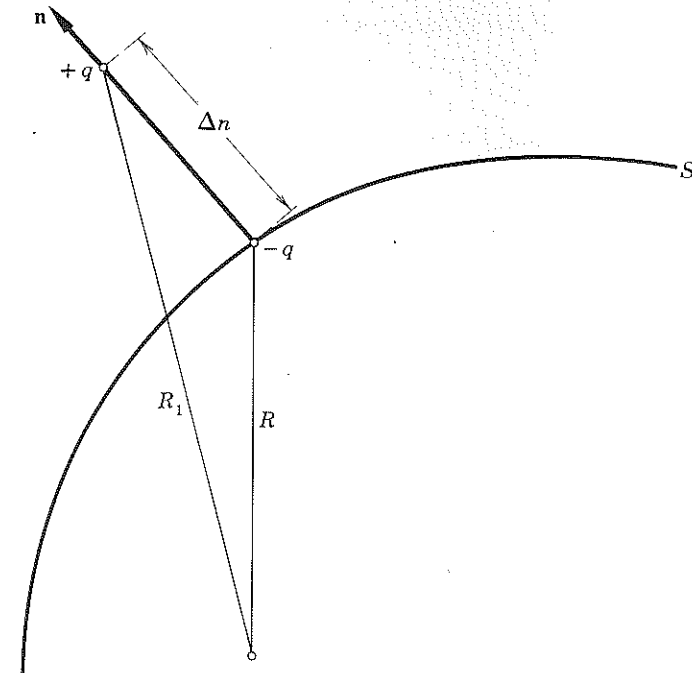


Figure 7-11 Interpretation of a double layer.

sources at infinity. When there are no sources at infinity, the limiting value of the surface integral must be zero, and this places limitations on the behavior of  $\bar{u}$ . These limitations are summarized in the Sommerfeld radiation condition, a sharp form of which we shall now derive.

## 7-26 THE SOMMERFELD RADIATION CONDITION†

**THEOREM:** A sufficient condition for the surface integral

$$\bar{u} = \frac{1}{4\pi} \int_{S(V)} \left[ \frac{e^{ikR}}{R} \left( \frac{\partial \bar{u}}{\partial n} \right) - \bar{u} \frac{\partial}{\partial n} \left( \frac{e^{ikR}}{R} \right) \right] dS \quad (7-726)$$

to vanish as  $S(V)$  recedes to infinity is that  $\bar{u}(r, \theta, \phi)$  should satisfy

$$\lim_{r \rightarrow \infty} r \left( \frac{\partial \bar{u}}{\partial r} - ik\bar{u} + \frac{\bar{u}}{r} \right) = 0 \quad (7-727)$$

and

$$\lim_{r \rightarrow \infty} \frac{\bar{u}}{r} = 0 \quad (7-728)$$

uniformly in  $\theta$  and  $\phi$ .

†I. S. Sokolnikoff and R. M. Redheffer, "Mathematics of Physics and Modern Engineering," p. 501, McGraw-Hill, New York, 1958.

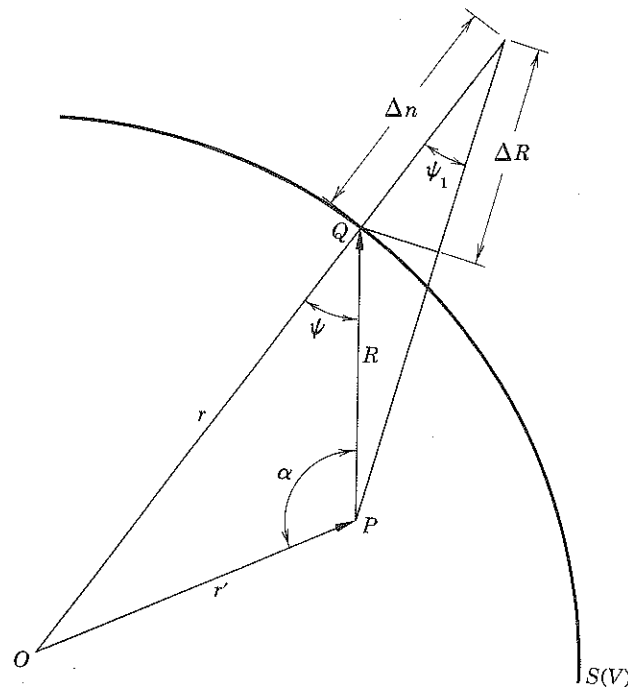


Figure 7-12 Geometry associated with the Sommerfeld radiation condition.

**proof:** In order to prove this theorem, we shall need several simple geometric results obtainable from Fig. 7-12. It is apparent from the figure that since

$$\frac{\Delta R}{\Delta n} = \cos \psi_1 \quad (7-729)$$

$$\frac{\partial R}{\partial n} = \lim_{\Delta n \rightarrow 0} \frac{\Delta R}{\Delta n} = \cos \psi \quad (7-730)$$

we must have

$$\frac{\partial}{\partial n} \left( \frac{e^{ikR}}{R} \right) = \frac{ikRe^{ikR} - e^{ikR}}{R^2} \cos \psi$$

and therefore Eq. (7-726) may be written as

$$\bar{u} = \frac{1}{4\pi} \int_{S(V)} \frac{e^{ikR}}{R} \left[ \frac{\partial \bar{u}}{\partial n} - ik\bar{u} + ik\bar{u}(1 - \cos \psi) + \frac{\bar{u}}{R} \cos \psi \right] dS \quad (7-731)$$

The further reduction of Eq. (7-731) depends on three geometric results,

$$1 - \cos \psi = O(\epsilon^2) \quad (7-732)$$

$$\frac{\cos \psi}{R} = \frac{1}{r} + O(\epsilon^2) \quad (7-733)$$

$$\frac{r}{R} = 1 + O(\epsilon) \quad (7-734)$$

obtained from Fig. 7-12. The quantity  $\epsilon$  defined by

$$\epsilon = \frac{r'}{r} \quad (7-735)$$

eventually becomes an arbitrarily small positive number. This happens as  $r$  recedes to infinity with  $r'$  held fixed. Equations (7-732) and (7-733) are obtained by applying the law of sines to Fig. 7-12, and Eq. (7-734) comes from an application of the triangle inequality.

We see that

$$\frac{\sin \psi}{r'} = \frac{\sin \alpha}{r} \quad (7-736)$$

and therefore

$$1 - \cos^2 \psi = \left( \frac{r'}{r} \right)^2 \sin^2 \alpha \quad (7-737)$$

$$\cos \psi = \left[ 1 - \left( \frac{r'}{r} \right)^2 \sin^2 \alpha \right]^{1/2} \quad (7-738)$$

or

$$\cos \psi = (1 - \epsilon^2 \sin^2 \alpha)^{1/2} \quad (7-739)$$

On expanding Eq. (7-739), we obtain

$$\cos \psi = 1 - \frac{1}{2} \epsilon^2 \sin^2 \alpha - \frac{1}{8} \epsilon^4 \sin^4 \alpha - \dots \quad (7-740)$$

so that for small  $\epsilon$ ,

$$1 - \cos \psi = O(\epsilon^2) \quad (7-741)$$

Also, from Eq. (7-738),

$$\frac{\cos \psi}{R} = \frac{1}{r} \left[ 1 - \left( \frac{r'}{r} \right)^2 \sin^2 \alpha \right]^{1/2} \quad (7-742)$$

Applying the triangle inequality to Fig. 7-12 gives

$$r \leq r' + R \quad (7-743)$$

$$1 \leq \frac{r'}{r} + \frac{R}{r} \quad (7-744)$$

$$\frac{R}{r} \geq 1 - \epsilon \quad (7-745)$$

and therefore

$$0 \leq \frac{r}{R} \leq \frac{1}{1 - \epsilon} \quad (7-746)$$

However

$$\frac{1}{1 - \epsilon} = 1 + \epsilon + \dots \quad (7-747)$$

So we can write

$$\frac{r}{R} = 1 + O(\epsilon) \quad (7-748)$$

for small  $\epsilon$ .

Let  $d\Omega$  be the element of solid angle subtended by  $dS$  at 0 in Fig. 7-12. Then, using  $dS = r^2 d\Omega$ , Eq. (7-731) becomes

$$\bar{u} = \frac{1}{4\pi} \int_{S(V)} r d\Omega \left( \frac{r}{R} \right) e^{ikR} \left[ \left( \frac{\partial \bar{u}}{\partial n} - ik\bar{u} + \frac{\bar{u}}{r} \right) + (1 + ik)\bar{u}O(\epsilon^2) \right] \quad (7-749)$$

or equivalently

$$\begin{aligned} \bar{u} &= \frac{1}{4\pi} \int_{S(V)} d\Omega e^{ikR} [1 + O(\epsilon)] \left[ r \left( \frac{\partial \bar{u}}{\partial n} - ik\bar{u} + \frac{\bar{u}}{r} \right) \right. \\ &\quad \left. + (1 + ik) \frac{\bar{u}}{r} O(1) \right] = \frac{1}{4\pi} \int_{S(V)} d\Omega e^{ikR} \left[ r \left( \frac{\partial \bar{u}}{\partial n} - ik\bar{u} + \frac{\bar{u}}{r} \right) \right. \\ &\quad \left. + (1 + ik) \frac{\bar{u}}{r} O(1) \right] + \frac{1}{4\pi} \int_{S(V)} d\Omega O(\epsilon) e^{ikR} \left[ r \left( \frac{\partial \bar{u}}{\partial n} - ik\bar{u} + \frac{\bar{u}}{r} \right) \right. \\ &\quad \left. + (1 + ik) \frac{\bar{u}}{r} O(1) \right] \end{aligned} \quad (7-750)$$

We see that  $\bar{u}$  is the sum of a pair of integrals,

$$\bar{u} = I_1 + I_2 \quad (7-751)$$

and therefore

$$|\bar{u}| \leq |I_1| + |I_2| \quad (7-752)$$

However,

$$|I_2| \leq |I_1| \quad (7-753)$$

because of the additional factor  $O(\epsilon)$  in the integrand of  $I_2$ , and consequently

$$|\bar{u}| \leq 2|I_1| \quad (7-754)$$

An application of the mean-value theorem to  $I_1$  shows that

$$\begin{aligned} |I_1| &\leq 4\pi \max_{S(V)} \left| r \left( \frac{\partial \bar{u}}{\partial n} - ik\bar{u} + \frac{\bar{u}}{r} \right) + (1 + ik) \frac{\bar{u}}{r} O(1) \right| \\ &\leq 4\pi \max_{S(V)} \left| r \left( \frac{\partial \bar{u}}{\partial n} - ik\bar{u} + \frac{\bar{u}}{r} \right) \right| \\ &\quad + 4\pi \max_{S(V)} O(1) |1 + ik| \frac{|\bar{u}|}{r} \end{aligned} \quad (7-755)$$

Since  $r \rightarrow \infty$  as  $S(V)$  recedes to infinity, it follows from Eqs. (7-726), (7-754), and (7-755) that

$$\lim_{r \rightarrow \infty} \frac{1}{4\pi} \int_{S(V)} \left[ \frac{e^{ikR}}{R} \left( \frac{\partial \bar{u}}{\partial n} \right) - \bar{u} \frac{\partial}{\partial n} \left( \frac{e^{ikR}}{R} \right) \right] dS = 0 \quad (7-756)$$

provided

$$\lim_{r \rightarrow \infty} r \left( \frac{\partial \bar{u}}{\partial n} - ik\bar{u} + \frac{\bar{u}}{r} \right) = 0 \quad (7-757)$$

and

$$\lim_{r \rightarrow \infty} \frac{\bar{u}}{r} = 0 \quad (7-758)$$

uniformly in  $\theta$  and  $\phi$ .

## 7-27 TIME-DEPENDENT PROBLEMS

In Sec. 7-21 it was suggested that one means of solving

$$\left( \nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) u(\mathbf{r}, t) = -F(\mathbf{r}, t) \quad (7-759)$$

is to Fourier-transform it, solve the resulting Helmholtz equation, and then invert the result back into the time domain. An alternative method is to deal directly with Eq. (7-759) by introducing a time-dependent Green's function.

We wish to solve (7-759) subject to specified *initial conditions*,

$$u(\mathbf{r}, 0) = f(\mathbf{r}) \quad (7-760)$$

$$\frac{\partial u}{\partial t}(\mathbf{r}, 0) = g(\mathbf{r}) \quad (7-761)$$

and given *boundary conditions*. These will be of the radiation condition type when the domain is unbounded, and of the Dirichlet or Neumann type when the domain is bounded. The time-dependent Green's function is introduced as the solution of

$$\left( \nabla^2 - \frac{1}{c^2} \frac{\partial}{\partial t^2} \right) G(\mathbf{r}, t; \mathbf{r}', t') = -\delta(\mathbf{r} - \mathbf{r}') \delta(t - t') \quad (7-762)$$

having the following properties:

1.  $G(\mathbf{r}, t; \mathbf{r}', t') = 0 \quad t < t'$ .
2.  $G(\mathbf{r}, t; \mathbf{r}', t')$  satisfies the homogeneous wave equation except at  $t = t'$ ,  $\mathbf{r} = \mathbf{r}'$ .
3.  $G(\mathbf{r}, t; \mathbf{r}', t')$  will satisfy homogeneous boundary conditions on  $S(V)$ , and
4. homogeneous initial conditions  
 $G = 0, \partial G / \partial t = 0$  for  $t < t'$ .
5.  $G(\mathbf{r}, t; \mathbf{r}', t')$  satisfies a reciprocity relation  $G(\mathbf{r}, t; \mathbf{r}', t') = G(\mathbf{r}', -t'; \mathbf{r}, -t)$  for  $t > t'$ .

This Green's function can be interpreted as the disturbance at  $\mathbf{r}$  occurring at time  $t$  due to an earlier disturbance at  $\mathbf{r}'$  which happened at time  $t'$ . Notice that in the time-dependent reciprocity relation one does not merely interchange the primed and unprimed coordinates. The signs of  $t$  and  $t'$  must also be reversed. If this step were not taken, then a disturbance at  $\mathbf{r}$  and  $t$  would be observed at  $\mathbf{r}'$  and  $t'$  when  $t'$  is less than  $t$ . This would imply that the disturbance was detected before it occurred! By reversing the signs, the disturbance occurring at  $\mathbf{r}$  and  $-t$  is observed at  $\mathbf{r}'$  and  $-t'$  where  $-t' > -t$  as it should be.

The proof of the reciprocity property 5 follows along the same lines as the proof of Eq. (7-508). Two Green's functions,  $G_+(\mathbf{r}, t; \mathbf{r}', t')$  and  $G_-(\mathbf{r}, -t; \mathbf{r}', -t')$ , are introduced. Both satisfy identical homogeneous boundary conditions on  $S(V)$  as well as homogeneous initial

conditions, together with the wave equations,

$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) G_+(r, t; r', t') = -\delta(r - r') \delta(t - t') \quad (7-763)$$

$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) G_-(r, -t; r'', -t'') = -\delta(r - r'') \delta(t - t'') \quad (7-764)$$

Applying Green's theorem,

$$\begin{aligned} \int_V \{G_+ \nabla^2 G_- - G_- \nabla^2 G_+\} dV \\ = \int_{S(V)} \left\{ G_+ \frac{\partial G_-}{\partial n} - G_- \frac{\partial G_+}{\partial n} \right\} dS \end{aligned} \quad (7-765)$$

to Eqs. (7-763) and (7-764) yields

$$\begin{aligned} \int_V \{G_+ \nabla^2 G_- - G_- \nabla^2 G_+\} dV = -G_+(r'', t; r', t') \delta(t - t'') \\ + G_-(r', -t; r'', -t'') \delta(t - t'') + \frac{1}{c^2} \int_V \left\{ G_+ \frac{\partial^2 G_-}{\partial t^2} \right. \\ \left. - G_- \frac{\partial^2 G_+}{\partial t^2} \right\} dV = 0 \end{aligned} \quad (7-766)$$

The surface-integral contribution from Eq. (7-765) vanishes because we have assumed that  $G_+$  and  $G_-$  satisfy identical *homogeneous* boundary conditions on  $S(V)$ . The last term in Eq. (7-766) can be rewritten by using

$$\begin{aligned} \int_V \left\{ G_+ \frac{\partial^2 G_-}{\partial t^2} - G_- \frac{\partial^2 G_+}{\partial t^2} \right\} dV \\ = \frac{\partial}{\partial t} \int_V \left\{ G_+ \frac{\partial G_-}{\partial t} - G_- \frac{\partial G_+}{\partial t} \right\} dV \end{aligned} \quad (7-767)$$

Both sides of Eq. (7-766) are now integrated with respect to time from  $t = -\infty$  to  $t = t$  where  $t > t'$  and  $t > t''$ . The result is

$$\begin{aligned} G_+(r'', t'; r', t') - G_-(r', -t'; r'', -t'') \\ = \frac{1}{c^2} \int_V \left\{ G_+ \frac{\partial G_-}{\partial t} - G_- \frac{\partial G_+}{\partial t} \right\} dV \Big|_{t=-\infty}^{t=t} \end{aligned} \quad (7-768)$$

At the lower limit  $t = -\infty$ , the last term vanishes because of property 4 of the Green's function. Also, the Green's function  $G_-(r, -t; r'', -t'')$  is nonzero only when  $-t > -t''$ , and this is true of  $\partial G_-/\partial t$  as well. However,  $-t > -t''$  implies that  $t < t''$ , and in Eq. (7-768) it was assumed that  $t > t'$  and  $t > t''$ . Therefore  $G_-$  and  $\partial G_-/\partial t$  vanish at the upper limit  $t = t$  in (7-768), and the reciprocity relation

$$G_+(r'', t'; r', t') = G_-(r', -t'; r'', -t'') \quad (7-769)$$

is established.

Let us return to the problem of obtaining a formal integral solution of Eqs. (7-759) to (7-761). For this purpose it is convenient to rewrite (7-759) to (7-762) in primed coordinates; thus

$$\left(\nabla'^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t'^2}\right) u(r', t') = -F(r', t') \quad (7-770)$$

and

$$\left(\nabla'^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t'^2}\right) G(r, t; r', t') = -\delta(r - r') \delta(t - t') \quad (7-771)$$

where

$$\nabla'^2 = \frac{\partial^2}{\partial x'^2} + \frac{\partial^2}{\partial y'^2} + \frac{\partial^2}{\partial z'^2} \quad (7-772)$$

By integrating Green's theorem over  $dt'$ , one finds

$$\begin{aligned} \int_0^{t+\epsilon} dt' \int_{V'} \{G \nabla'^2 u - u \nabla'^2 G\} dV' \\ = \int_0^{t+\epsilon} dt' \int_{S(V')} \left\{ G \frac{\partial u}{\partial n'} - u \frac{\partial G}{\partial n'} \right\} dS' \end{aligned} \quad (7-773)$$

where  $\epsilon$  is an arbitrarily small positive number, and  $u$  and  $G$  satisfy Eqs. (7-770) and (7-771). It follows from (7-770), (7-771), and (7-773) that

$$\begin{aligned} - \int_0^{t+\epsilon} dt' \int_{V'} G(r, t; r', t') F(r', t') dV' + u(r, t) \\ + \frac{1}{c^2} \int_{V'} \left\{ G \frac{\partial u}{\partial t'} - u \frac{\partial G}{\partial t'} \right\}_{t'=0}^{t'=t+\epsilon} dV' \\ = \int_0^{t+\epsilon} dt' \int_{S(V')} \left\{ G \frac{\partial u}{\partial n'} - u \frac{\partial G}{\partial n'} \right\} dS' \end{aligned} \quad (7-774)$$

Both  $G(r, t; r', t')$  and  $\partial G/\partial t' (r, t; r', t')$  must vanish when  $t' > t$  according to property 4 of  $G$ . Hence the third term on the left side of Eq. (7-774) must vanish at the upper limit  $t = t' + \epsilon$ . As  $\epsilon \rightarrow 0$ , we obtain

$$\begin{aligned} u(r, t) = \int_0^{t+\epsilon} dt' \int_{V'} G(r, t; r', t') F(r', t') dV' \\ + \int_0^{t+\epsilon} dt' \int_{S(V')} \left\{ G \frac{\partial u}{\partial n'} - u \frac{\partial G}{\partial n'} \right\} dS' \\ + \frac{1}{c^2} \int_{V'} \left\{ G \frac{\partial u}{\partial t'} - u \frac{\partial G}{\partial t'} \right\}_{t'=0} dV' \end{aligned} \quad (7-775)$$

Equation (7-775) provides a formal solution to the mixed initial- and boundary-value problem associated with Eq. (7-759). It is clear from (7-775) that  $u(r, t)$  is made up of contributions from volume sources distributed throughout  $V'$  with a density  $F(r', t')$ , single- and double-layer surface sources distributed over  $S(V')$ , and a term involving the initial conditions at  $t' = 0$ .

The surface  $S(V')$  may now be permitted to recede to infinity. If there are no sources at infinity and the medium is initially at rest so that at  $t' = 0$ ,  $u = 0$  and  $\partial u/\partial t' = 0$  everywhere, then Eq. (7-775) reduces to

$$u(r, t) = \int_0^{t+\epsilon} dt' \int_{\text{All space}} G(r, t; r', t') F(r', t') dV' \quad (7-776)$$

We showed in Eq. (7-650) that when no sources are present at infinity, then

$$G(r, t; r', t') = \frac{\delta[(t - t') - R/c]}{4\pi R} \quad (7-777)$$

where  $t - t'$  represents the time that has elapsed since the source was triggered, and  $R = [(x - x')^2 + (y - y')^2 + (z - z')^2]^{1/2}$ . Thus

$$u(\mathbf{r}, t) = \int_0^{t+\epsilon} dt' \int_{\text{All space}} \frac{\delta[(t - t') - R/c]}{4\pi R} F(\mathbf{r}', t') dV' \quad (7-778)$$

or

$$u(\mathbf{r}, t) = \frac{1}{4\pi} \int_{\text{All space}} \frac{F(\mathbf{r}', t - R/c)}{R} dV' \quad (7-779)$$

Notice that the time  $t$  in the integrand of Eq. (7-779) has been retarded by a factor of  $R/c$ . It is customary to designate retarded functions by enclosing them in square brackets so that

$$[F(\mathbf{r}', t)] = F\left(\mathbf{r}', t - \frac{R}{c}\right) \quad (7-780)$$

With this notation

$$u(\mathbf{r}, t) = \frac{1}{4\pi} \int_{\text{All space}} \frac{[F(\mathbf{r}', t)]}{R} dV' \quad (7-781)$$

The appearance of retarded quantities in the solution of the time-dependent wave equation is a consequence of the finite speed of propagation. In other words, a signal which is here now must have left somewhere else at an earlier time. The time difference or retardation is how long it took the signal to travel here from its source.

For two space dimensions one expects solutions like Eqs. (7-775) and (7-776), except that the volume  $V'$  is replaced by an open surface  $S'$ , and the boundary  $S(V')$  is replaced by the closed curve  $C(S')$  forming the complete boundary of  $S'$ . The time Laplace transform of the two-dimensional Green's function  $G(R, t)$  is given by Eq. (7-524) as

$$\bar{G}\left(\frac{sR}{c}\right) = \frac{1}{2\pi} K_0\left(\frac{sR}{c}\right) \quad (7-782)$$

From a table of Laplace transforms,

$$K_0\left(\frac{sR}{c}\right) = \int_0^\infty \frac{H[t - R/c]}{\sqrt{t^2 - R^2/c^2}} e^{-st} dt \quad (7-783)$$

where  $H[t - R/c]$  is Heaviside's unit step function defined by

$$H(x) = \begin{cases} 1 & x > 0 \\ 0 & x < 0 \end{cases} \quad (7-784)$$

Therefore the inverse Laplace transform of  $\bar{G}(sR/c)$  is

$$G(R, t) = \frac{H[t - R/c]}{2\pi \sqrt{t^2 - R^2/c^2}} \quad (7-785)$$

and the two-dimensional time-dependent Green's function becomes

$$G(\mathbf{r}, t; \mathbf{r}', t') = \frac{H[(t - t') - R/c]}{2\pi \sqrt{(t - t')^2 - R^2/c^2}} \quad (7-786)$$

where  $t - t'$  is the elapsed time measured from the instant of source triggering, and  $R = [(x - x')^2 + (y - y')^2]^{1/2}$ .

## 7-28 POISSON'S SOLUTION OF THE WAVE EQUATION

Suppose we wish to solve

$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) u(\mathbf{r}, t) = 0 \quad (7-787)$$

subject to prescribed initial conditions,

$$u = f(\mathbf{r}) \quad t = 0 \quad (7-788)$$

$$\frac{\partial u}{\partial t} = g(\mathbf{r}) \quad t = 0 \quad (7-789)$$

According to Eq. (7-775), the solution is

$$u(\mathbf{r}, t) = \int_0^{t+\epsilon} dt' \int_{S(V')} \left\{ G \frac{\partial u}{\partial n'} - u \frac{\partial G}{\partial n'} \right\} dS' \quad (7-790)$$

and since

$$G = \frac{\delta[(t - t') - R/c]}{4\pi R} \quad (7-791)$$

Eq. (7-790) is equivalent to

$$u(\mathbf{r}, t) = \frac{1}{4\pi} \int_{S(V')} dS' \int_0^{t+\epsilon} dt' \left( \frac{\delta[(t - t') - R/c]}{R} \frac{\partial u}{\partial n'} - u \frac{\partial}{\partial n'} \left\{ \frac{\delta[(t - t') - R/c]}{R} \right\} \right) \quad (7-792)$$

The integral of the first term in Eq. (7-792) is relatively easy to evaluate; it is simply

$$\begin{aligned} \int_0^{t+\epsilon} dt' \frac{\delta[(t - t') - R/c]}{R} \frac{\partial u(\mathbf{r}', t')}{\partial n'} \\ = \frac{1}{R} \frac{\partial u}{\partial n'}(\mathbf{r}', t') \Big|_{t'=t-R/c} = \frac{1}{R} \left[ \frac{\partial u}{\partial n'} \right] \end{aligned} \quad (7-793)$$

The second term in (7-792) involves a little added difficulty because it is necessary to "differentiate" the Dirac delta function. Proceeding formally, we have

$$\begin{aligned} - \int_0^{t+\epsilon} dt' u(\mathbf{r}', t') \frac{\partial}{\partial n'} \left\{ \frac{\delta[(t - t') - R/c]}{R} \right\} \\ = \int_0^{t+\epsilon} dt' u(\mathbf{r}', t') \\ - \frac{(R/c) \delta'[(t - t') - R/c] + \delta[(t - t') - R/c] \partial R}{R^2} \frac{\partial R}{\partial n'} \end{aligned} \quad (7-794)$$

where from Fig. 7-12,

$$\frac{\partial R}{\partial n'} = \cos(n', R) \quad (7-795)$$

The second term in Eq. (7-794) integrates to

$$[u] \frac{\cos(n', R)}{R^2} \quad (7-796)$$

leaving

$$- \int_0^{t+\epsilon} dt' u(\mathbf{r}', t') \frac{\cos(n', R)}{Rc} \delta' \left[ (t - t') - \frac{R}{c} \right] \quad (7-797)$$

to be evaluated. This can be accomplished by employing integration by parts, and the result

$$\int_{-\epsilon}^{+\epsilon} f(x) \delta'(x) dx = - \int_{-\epsilon}^{+\epsilon} \delta(x) f'(x) dx \quad (7-798)$$

to give

$$\begin{aligned} - \int_0^{t+\epsilon} dt' u(\mathbf{r}', t') \frac{\cos(n', R)}{Rc} \delta' \left[ (t - t') - \frac{R}{c} \right] \\ = \left[ \frac{\partial u}{\partial t'} \right] \frac{\cos(n', R)}{Rc} \end{aligned} \quad (7-799)$$

Thus

$$\begin{aligned} - \int_0^{t+\epsilon} dt' u(\mathbf{r}', t') \frac{\partial}{\partial n'} \left[ \frac{\delta \{ (t - t') - R/c \}}{R} \right] \\ = [u] \frac{\cos(n', R)}{R^2} + \left[ \frac{\partial u}{\partial t} \right] \frac{\cos(n', R)}{Rc} \end{aligned} \quad (7-800)$$

and we have shown that Eq. (7-792) becomes

$$\begin{aligned} u(\mathbf{r}, t) = \frac{1}{4\pi} \int_{S(V')} \left\{ \frac{1}{R} \left[ \frac{\partial u}{\partial n'} \right] + [u] \frac{\cos(n', R)}{R^2} \right. \\ \left. + \left[ \frac{\partial u}{\partial t} \right] \frac{\cos(n', R)}{Rc} \right\} dS' \end{aligned} \quad (7-801)$$

or equivalently

$$\begin{aligned} u(\mathbf{r}, t) = \frac{1}{4\pi} \int_{S(V')} dS' \left[ \frac{1}{R} \frac{\partial u(\mathbf{r}', t')}{\partial n'} + u(\mathbf{r}', t') \frac{\cos(n', R)}{R^2} \right. \\ \left. + \frac{\partial u(\mathbf{r}', t')}{\partial t'} \frac{\cos(n', R)}{Rc} \right]_{t'=t-R/c} \end{aligned} \quad (7-802)$$

The vector form of Eq. (7-802) is

$$\begin{aligned} u(\mathbf{r}, t) = \frac{1}{4\pi} \int_{S(V')} dS \cdot \left[ \frac{1}{R} \nabla' u(\mathbf{r}', t') + \frac{\mathbf{R}}{R^3} u(\mathbf{r}', t') \right. \\ \left. + \frac{\partial u(\mathbf{r}', t')}{\partial t'} \frac{\mathbf{R}}{cR^2} \right]_{t'=t-R/c} \end{aligned} \quad (7-803)$$

where

$$dS = \mathbf{n}' dS' \quad (7-804)$$

and

$$\cos(n', R) = \frac{\mathbf{n}' \cdot \mathbf{R}}{R} \quad (7-805)$$

Poisson's solution of the wave equation is derivable from Eq. (7-802) or (7-803) as follows. Choose  $S(V')$  to be the surface of a sphere of radius  $ct$ . Let  $\mathbf{r}$  be the position vector of the origin of the sphere, and let  $\mathbf{r}'$  be the position vector of any arbitrary point on the surface

of the sphere. Then

$$R = |\mathbf{r} - \mathbf{r}'| = ct \quad (7-806)$$

and therefore

$$t' = t - \frac{R}{c} = 0 \quad (7-807)$$

We also notice that

$$\frac{\partial}{\partial n'} = \frac{\partial}{\partial R} \quad (7-808)$$

since

$$\cos(n', R) = 1 \quad (7-809)$$

on  $S(V')$ .

Using these expressions reduces Eq. (7-802) to

$$\begin{aligned} u(\mathbf{r}, t) = \frac{1}{4\pi} \int_{R=ct} dS' \left\{ \frac{1}{R} \frac{\partial u(\mathbf{r}', 0)}{\partial R} + \frac{u(\mathbf{r}', 0)}{R^2} \right. \\ \left. + \frac{1}{Rc} \left[ \frac{\partial u(\mathbf{r}', t')}{\partial t'} \right]_{t'=0} \right\} \end{aligned} \quad (7-810)$$

The element of surface area  $dS'$  on  $R = ct$  can be expressed in terms of solid angle  $d\Omega'$  by the usual formula,

$$dS' = R^2 d\Omega' \quad (7-811)$$

thus reducing Eq. (7-810) to

$$u(\mathbf{r}, t) = \frac{1}{4\pi} \int_{R=ct} \left\{ \frac{\partial}{\partial R} (Ru(\mathbf{r}', 0)) + \frac{R}{c} \left[ \frac{\partial u(\mathbf{r}', t')}{\partial t'} \right]_{t'=0} \right\} d\Omega' \quad (7-812)$$

One now observes that

$$\mathbf{r}' = \mathbf{R} + \mathbf{r} \quad (7-813)$$

so that for any arbitrary function  $f$ ,

$$f(\mathbf{r}') = f(\mathbf{R} + \mathbf{r}) \quad (7-814)$$

Thus

$$\frac{1}{4\pi} \int_{R=ct} f(\mathbf{r}') d\Omega' = \frac{1}{4\pi} \int_{R=ct} f(\mathbf{R} + \mathbf{r}) d\Omega' \quad (7-815)$$

However,  $\mathbf{r}$  is the position vector of the origin of the sphere  $R = ct$  and thus remains constant in Eq. (7-815). Consequently, only  $\mathbf{R}$  varies over the surface of the sphere. Hence (7-815) gives the mean value of  $f$  over a sphere of radius  $R = ct$  centered at  $\mathbf{r}$ . We indicate this by writing

$$M_{ct}[f(\mathbf{r}')] = \frac{1}{4\pi} \int_{R=ct} f(\mathbf{r}') d\Omega' \quad (7-816)$$

and we use Eq. (7-816) to simplify (7-812). Since  $R = ct$ ,

$$\frac{\partial}{\partial R} [Ru(\mathbf{r}', 0)] = \frac{\partial}{\partial t} [tu(\mathbf{r}', 0)] \quad (7-817)$$

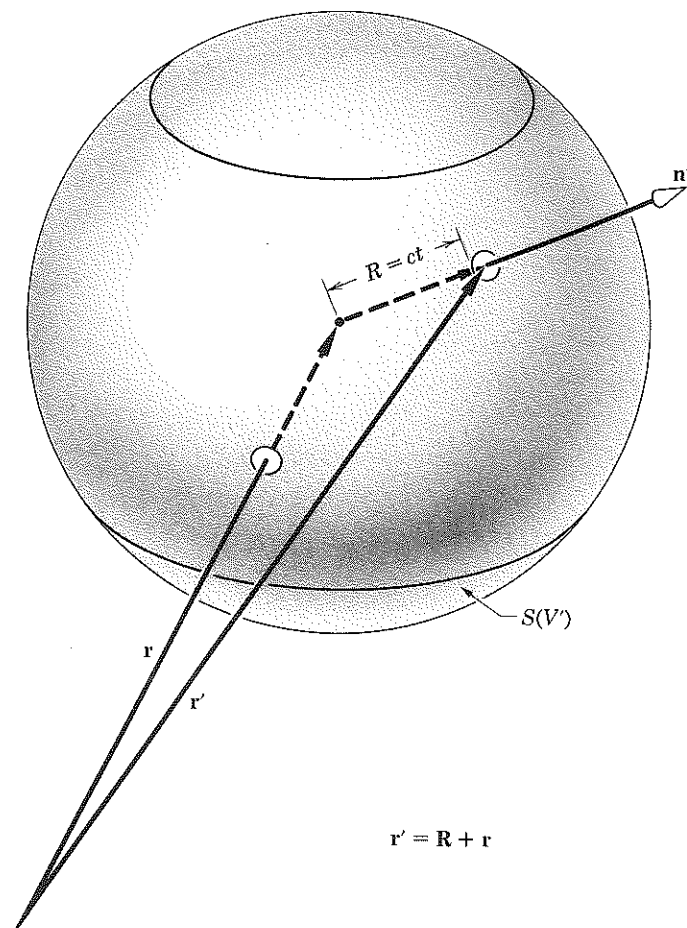


Figure 7-13 Sphere of radius  $R = ct$ .

and

$$\frac{R}{c} \left[ \frac{\partial u(\mathbf{r}, t')}{\partial t'} \right]_{t'=0} = t \frac{\partial u(\mathbf{r}', 0)}{\partial t'} \quad (7-818)$$

Eqs. (7-812) and (7-816) give

$$u(\mathbf{r}, t) = \frac{\partial}{\partial t} M_{ct}[tu(\mathbf{r}', 0)] + tM_{ct} \left[ \frac{\partial u(\mathbf{r}', 0)}{\partial t'} \right] \quad (7-819)$$

or, on using Eqs. (7-788) and (7-789),

$$u(\mathbf{r}, t) = \frac{\partial}{\partial t} M_{ct}[tf(\mathbf{r}')] + tM_{ct}[g(\mathbf{r}')] \quad (7-820)$$

Equation (7-820) is Poisson's solution of Cauchy's initial-value problem (7-787) to (7-789) for the wave equation.

A solution, similar to Eq. (7-820), can be derived for the Cauchy problem in two space dimensions. For this purpose, it is necessary

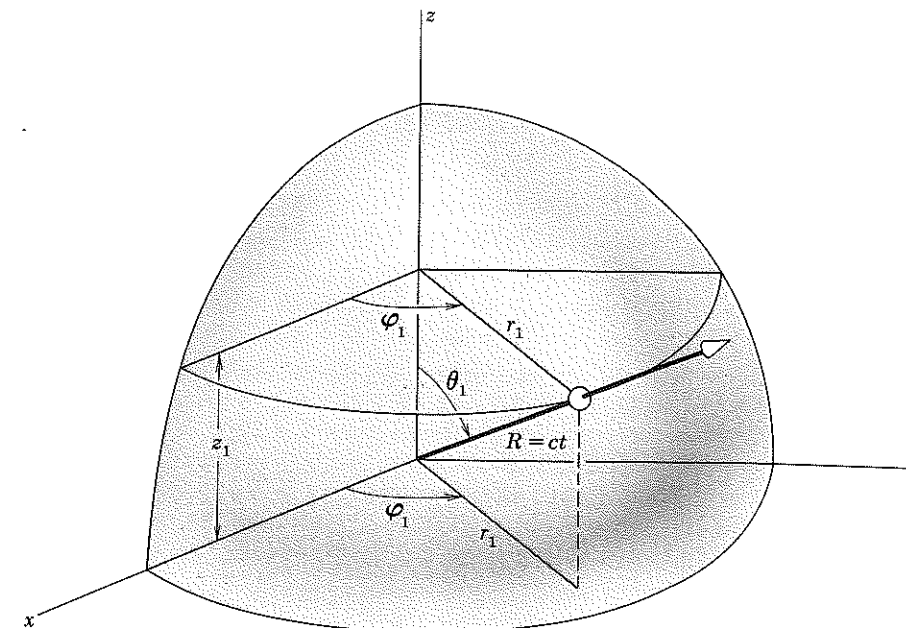


Figure 7-14 The intersection of the sphere  $R = ct$  with the plane  $z = z_1$ .

to reexpress Eq. (7-816) in terms of an integral over a plane. Let the sphere centered at  $\mathbf{r}$  be intersected by a plane at a distance  $z_1$  above the equatorial plane, and let  $\theta_1$  be the colatitude relative to the  $z_1$  axis defined by the normal to the plane. Spherical coordinates  $(R, \theta_1, \phi_1)$ , centered in the sphere, are now introduced. These are related to polar coordinates in the plane of intersection by

$$r_1 = R \sin \theta_1 = ct \sin \theta_1 \quad (7-821)$$

$$z_1 = \sqrt{R^2 - r_1^2} = \sqrt{c^2 t^2 - r_1^2} \quad (7-822)$$

$$\theta_1 = \cot^{-1} \frac{z_1}{r_1} = \cot^{-1} \frac{\sqrt{c^2 t^2 - r_1^2}}{r_1} \quad (7-823)$$

The element of solid angle  $d\Omega'$  relative to the center of the sphere is given by

$$d\Omega' = \sin \theta_1 d\theta_1 d\phi_1 \quad (7-824)$$

However,

$$dr_1 = ct \cos \theta_1 d\theta_1 \quad (7-825)$$

and so we can write

$$d\Omega' = \frac{dr_1 d\phi_1 \sin \theta_1}{dt \cos \theta_1} = \frac{dr_1 d\phi_1}{ct \cot \theta_1} \quad (7-826)$$

or

$$d\Omega' = \frac{dr_1 d\phi_1}{ct \sqrt{c^2 t^2 - r_1^2}} \quad (7-827)$$

The range  $0 \leq \theta_1 \leq \pi/2$ ,  $0 \leq \phi_1 \leq 2\pi$  corresponds to a hemisphere; hence Eq. (7-816) may be written as

$$M_{ct}[f(\mathbf{r}')] = \frac{2}{4\pi} \int_0^{\pi/2} \int_0^{2\pi} f(\mathbf{r} + \mathbf{R}) \sin \theta_1 d\theta_1 d\phi_1 \quad (7-828)$$

As  $\theta_1$  runs from 0 to  $\pi/2$ ,  $r_1$  runs from 0 to  $ct$  because  $z_1$  is fixed. Thus, using (7-824) to (7-827), Eq. (7-828) becomes

$$M_{ct}[f(\mathbf{r}')] = \frac{1}{2\pi} \int_0^{2\pi} \int_0^{ct} \frac{f(\mathbf{r} + \mathbf{R})}{ct \sqrt{c^2 t^2 - r_1^2}} r_1 dr_1 d\phi_1 \quad (7-829)$$

Let the element area  $r_1 dr_1 d\phi_1$  in the plane of intersection with the sphere be designated by

$$dS_1 = r_1 dr_1 d\phi_1 \quad (7-830)$$

Then Eq. (7-829) becomes

$$M_{ct}[f(\mathbf{r}')] = \frac{1}{2\pi c} \int_{r_1 \leq ct} \frac{f(\mathbf{r} + \mathbf{R})}{t \sqrt{c^2 t^2 - r_1^2}} dS_1 \quad (7-831)$$

and Eq. (7-820) is explicitly given as

$$u(\mathbf{r}, t) = \frac{1}{2\pi c} \frac{\partial}{\partial t} \int_{r_1 \leq ct} \frac{f(\mathbf{r} + \mathbf{R})}{\sqrt{c^2 t^2 - r_1^2}} dS_1 + \frac{1}{2\pi c} \int_{r_1 \leq ct} \frac{g(\mathbf{r} + \mathbf{R})}{\sqrt{c^2 t^2 - r_1^2}} dS_1 \quad (7-832)$$

The vector  $\mathbf{r}' = \mathbf{r} + \mathbf{R}$  has components

$$\mathbf{r}' = \mathbf{r} + \mathbf{R} = [x + ct \sin \theta_1 \cos \phi_1, (y + ct \sin \theta_1 \sin \phi_1), (z + ct \cos \theta_1)] \quad (7-833)$$

or equivalently

$$\mathbf{r}' = (x', y', z') = \mathbf{r} + \mathbf{R} = [(x + r_1 \cos \phi_1), (y + r_1 \sin \phi_1), (z + z_1)] \quad (7-834)$$

The wave equation in two space dimensions can always be chosen so that the solution and its initial conditions are independent of the  $z'$  coordinate in some  $(x', y', z')$  coordinate system. For example, let

$$[u(\mathbf{r}', t')]_{t'=0} = u(x', y', 0) = f(x', y') \quad (7-835)$$

$$\left[ \frac{\partial u(\mathbf{r}', t')}{\partial t'} \right]_{t'=0} = \frac{\partial u(x', y', 0)}{\partial t'} = g(x', y') \quad (7-836)$$

Since the initial conditions (7-835) and (7-836) are independent of  $z'$ , it follows from Eq. (7-834) that (7-832) is explicitly given by

$$u(x, y, t) = \frac{1}{2\pi c} \frac{\partial}{\partial t} \iint_{r_1 \leq ct} \frac{f[(x + r_1 \cos \phi_1), (y + r_1 \sin \phi_1)]}{\sqrt{c^2 t^2 - r_1^2}} r_1 dr_1 d\phi_1 + \frac{1}{2\pi c} \iint_{r_1 \leq ct} \frac{g[(x + r_1 \cos \phi_1), (y + r_1 \sin \phi_1)]}{\sqrt{c^2 t^2 - r_1^2}} r_1 dr_1 d\phi_1 \quad (7-837)$$

This solves the initial-value problem for

$$\nabla^2 u = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} \quad (7-838)$$

in two space dimensions, given the initial conditions,

$$u(x, y, 0) = f(x, y) \quad (7-839)$$

$$\frac{\partial u}{\partial t}(x, y, 0) = g(x, y) \quad (7-840)$$

We can read some interesting results from Eqs. (7-820) and (7-837). For this purpose we shall introduce the notion of the support of a function. By definition, the support of a function  $f$  is the closure of the set of points on which  $f \neq 0$ . To say that a function  $f$  has a compact support means that there is some *closed* and *bounded* domain on which  $f$  does not vanish. This domain is called the support of  $f$ . For example, if  $f = f(x, y, z)$  does not vanish anywhere in or on the solid sphere  $x^2 + y^2 + z^2 \leq 1$ , but vanishes everywhere outside of it, then  $f$  is of compact support, and its compact support is the closed sphere  $x^2 + y^2 + z^2 \leq 1$ .

It follows from Eq. (7-820) that if the initial conditions  $u(x', y', z', 0) = f$  and  $\partial u / \partial t'(x', y', z', 0) = g$  are functions of compact support, then at any fixed point  $(x, y, z)$  there exists a *finite* time interval  $t_1 \leq t \leq t_2$ , outside of which  $u(x, y, z, t) = 0$ . To see this, one observes that (7-820) represents  $u$  in terms of averages over an expanding surface of radius  $R = ct$ . Since the surface  $R = ct$  expands with time, its intersections with the supports of  $f$  and  $g$  can each be nonzero only for finite time intervals. Thus there must be a finite time interval outside of which  $u$  vanishes identically.

The situation for the two-dimensional initial-value problem is different. Even if the initial conditions  $u(x', y', 0) = f$  and  $\partial u / \partial t'(x', y', 0) = g$  are of compact support in the  $x', y'$  plane, the disturbance  $u(x, y, t)$ , once it begins, never vanishes at any finite time. That this must be the case can be seen by noticing that Eq. (7-837) represents  $u(x, y, t)$  in terms of integrals over the *interior* of an expanding circle of radius  $r_1 = ct$ . Once the circle passes and encloses the support of  $f$  or  $g$ , this support will always be in the interior of the circle and will always contribute to an integral in (7-837). The two-dimensional initial-value problem is said to lead to a solution having a wake. We can calculate this wake explicitly for the simple case of a line source.

Suppose we let  $x = 0, y = 0$  in (7-837). This places the point at which  $u$  is calculated at the center of the circle  $r_1 = ct$  in the  $r_1, \phi_1$  plane. An infinitely long line source, normal to the  $r_1, \phi_1$  plane, is assumed to pass through the point  $r_1 = r, \phi_1 = \phi$  relative to the center of the circle  $r_1 = ct$ . The corresponding initial conditions on  $u$  are

$$(u)_{t=0} = \delta(r_1 - r) = \frac{\delta(r_1 - r)}{r_1} \delta(\phi_1 - \phi) \quad (7-841)$$

$$\left( \frac{\partial u}{\partial t} \right)_{t=0} = 0 \quad (7-842)$$

and when inserted into Eq. (7-837), these yield

$$u(0,t) = \frac{1}{2\pi c} \frac{\partial}{\partial t} \int_0^{2\pi} \int_0^{ct} \frac{\delta(r_1 - r) \delta(\phi_1 - \phi)}{\sqrt{c^2 t^2 - r_1^2}} dr_1 d\phi_1 \quad (7-843)$$

As a result,

$$u(0,t) = \begin{cases} \frac{1}{2\pi c} \frac{\partial}{\partial t} \left( \frac{1}{\sqrt{c^2 t^2 - r^2}} \right) & r < ct \\ 0 & r > ct \end{cases} \quad (7-844)$$

and therefore

$$u(0,t) = \frac{1}{2\pi c} \frac{\partial}{\partial t} \left[ \frac{H(ct - r)}{\sqrt{c^2 t^2 - r^2}} \right] = \frac{1}{2\pi} \left[ -\frac{ctH(ct - r)}{(c^2 t^2 - r^2)^{3/2}} + \frac{\delta(ct - r)}{\sqrt{c^2 t^2 - r^2}} \right] \quad (7-845)$$

where  $H(x)$  is Heaviside's unit step function,

$$H(x) = \begin{cases} 1 & x > 0 \\ 0 & x < 0 \end{cases} \quad (7-846)$$

and is related to the Dirac delta function by

$$\delta(x) = \frac{dH(x)}{dx} \quad (7-847)$$

We see from Eq. (7-845) that the signal at the origin is zero until  $t = r/c$ , which is the time it takes the signal to propagate from the source at  $r_1 = r$  to the origin at  $r_1 = 0$ . At  $t = r/c$ , the delta-function pulse arrives at the origin, followed by a wake which decays as described by the step-function term in (7-845). When  $t \gg r/c$ , the amplitude of the wake decays as  $1/c^2 t^2$ . The presence of a wake is characteristic of wave propagation in homogeneous two-dimensional media. Wave propagation in one or three space dimensions does not involve a wake. Since the line source extends to infinity in both directions, perpendicular to the plane in which the solution is computed, it is possible to think of the wake as being associated with the successive arrivals of signals from more and more remote portions of the source.

## 7-29 THE DIFFUSION EQUATION

The temperature distribution  $T(\mathbf{r},t)$  in a homogeneous medium in which heat  $H$  is being produced or lost at a rate  $\partial H/\partial t$  per unit volume is governed by the inhomogeneous diffusion equation,

$$\left( \nabla^2 - \frac{1}{\alpha} \frac{\partial}{\partial t} \right) T(\mathbf{r},t) = -\frac{1}{k} \frac{\partial H}{\partial t} = -h(\mathbf{r},t) \quad (7-848)$$

where  $\alpha$  is the diffusivity, and  $k$  is the thermal conductivity of the medium (see Sec. 7-5). All the techniques which we have developed for solving the wave equation and the potential equation are applicable to Eq. (7-848). Thus we can use integral transforms, separation of variables, or a Green's-function method to construct solutions of (7-848).

In harmony with our treatment of wave and potential problems, we shall derive a general solution of (7-848) in terms of a Green's function, thus relegating the real difficulties associated with solving (7-848) to the problem of constructing an appropriate Green's function.

The Green's function for (7-848) is defined as the solution of

$$\left( \nabla^2 - \frac{1}{\alpha} \frac{\partial}{\partial t} \right) G(\mathbf{r},t;\mathbf{r}',t') = -\delta(\mathbf{r} - \mathbf{r}') \delta(t - t') \quad (7-849)$$

having the following properties:

1.  $G(\mathbf{r},t;\mathbf{r}',t') = 0$  for  $t < t'$ . (7-850)
2.  $G(\mathbf{r},t;\mathbf{r}',t')$  satisfies the homogeneous diffusion equation except at  $t = t'$ ,  $\mathbf{r} = \mathbf{r}'$ .
3.  $G(\mathbf{r},t;\mathbf{r}',t')$  will satisfy homogeneous boundary conditions on  $S(V)$ .
4. Homogeneous initial condition  $G = 0$  at  $t = t'$ . (7-851)
5.  $G(\mathbf{r},t;\mathbf{r}',t')$  satisfies the reciprocity relation

$$G(\mathbf{r},t;\mathbf{r}',t') = G(\mathbf{r}',-t';\mathbf{r},-t) \quad (7-852)$$

The Green's function  $G(\mathbf{r},t;\mathbf{r}',t')$  can be interpreted as the temperature at  $\mathbf{r}$  observed at time  $t$  due to the injection of a unit of heat at time  $t' < t$  by an impulsive heat source located at  $\mathbf{r}'$ . Since  $G = 0$  for  $t < t'$ , the Green's function gives the temperature only for future times  $t > t'$  and thus describes how heat diffuses away from its initial position.

The sign reversal in the reciprocity relation (7-852) is required by causality, just as in the case of the wave equation. By reversing signs, the heat injected at  $\mathbf{r}$  and  $-t$  is observed at  $\mathbf{r}'$  and  $-t'$  where  $-t' > -t$  as causality demands. The proof of the reciprocity relation for the diffusion equation differs somewhat from the corresponding proof for the wave equation. The reason is that the wave equation involves a second derivative with respect to time. Therefore, if  $u(\mathbf{r},t)$  satisfies the wave equation, then so does  $u(\mathbf{r},-t)$ . On the other hand, the diffusion equation involves a first time derivative and thus does not enjoy the same symmetry in time. In fact, if

$$\nabla^2 T(\mathbf{r},t) = \frac{1}{\alpha} \frac{\partial T(\mathbf{r},t)}{\partial t} \quad (7-853)$$

then

$$\nabla^2 T(\mathbf{r},-t) = -\frac{1}{\alpha} \frac{\partial T(\mathbf{r},-t)}{\partial t} \quad (7-854)$$

Because of the temporal asymmetry in the diffusion equation, the two Green's functions  $G_+(\mathbf{r},t;\mathbf{r}',t')$  and  $G_-(\mathbf{r},-t;\mathbf{r}',-t')$  must satisfy different equations:

$$\left( \nabla^2 - \frac{1}{\alpha} \frac{\partial}{\partial t} \right) G_+(\mathbf{r},t;\mathbf{r}',t') = -\delta(\mathbf{r} - \mathbf{r}') \delta(t - t') \quad (7-855)$$

$$\left( \nabla^2 + \frac{1}{\alpha} \frac{\partial}{\partial t} \right) G_-(\mathbf{r},-t;\mathbf{r}',-t') = -\delta(\mathbf{r} - \mathbf{r}') \delta(t - t') \quad (7-856)$$

The remainder of the proof of reciprocity is similar to Eqs. (7-765) to (7-768). An application of Green's theorem,

$$\int_V \{G_+ \nabla^2 G_- - G_- \nabla^2 G_+\} dV = \int_{S(V)} \left\{ G_+ \frac{\partial G_-}{\partial n} - G_- \frac{\partial G_+}{\partial n} \right\} dS \quad (7-857)$$

to Eqs. (7-855) and (7-856) yields

$$\begin{aligned} \int_V \{G_+ \nabla^2 G_- - G_- \nabla^2 G_+\} dV &= -G_+(r'', t; r', t') \delta(t - t') \\ &\quad + G_-(r', -t; r'', -t') \delta(t - t') \\ &\quad + \left( \frac{-1}{\alpha} \right) \int_V \left\{ G_+ \frac{\partial G_-}{\partial t} + G_- \frac{\partial G_+}{\partial t} \right\} dV = 0 \end{aligned} \quad (7-858)$$

The surface-integral contribution vanishes because  $G_+$  and  $G_-$  both satisfy the same homogeneous boundary conditions on  $S(V)$ . Both sides of Eq. (7-858) are now integrated with respect to time from  $t = -\infty$  to  $t = t$  where  $t > t'$  and  $t > t''$ . The result is

$$\begin{aligned} G_+(r'', t''; r', t') - G_-(r', -t'; r'', -t'') \\ = \frac{1}{\alpha} \int_{-\infty}^t dt \int_V \frac{\partial}{\partial t} (G_+ G_-) dV \\ = \frac{1}{\alpha} [G_+(r, t; r', t') G_-(r, -t; r'', -t'')] \Big|_{t=-\infty}^t = 0 \end{aligned} \quad (7-859)$$

since  $G_+$  vanishes at  $t = -\infty$  and  $G_-$  vanishes at  $t = t$  (because  $t > t''$  implies that  $-t < -t''$ ). Thus the reciprocity relation (7-852) is established.

### 7-30 GENERAL SOLUTION OF THE DIFFUSION EQUATION

The formal integral solution of the inhomogeneous diffusion equation is obtained, as usual, in terms of an appropriate Green's function. Thus consider

$$\left( \nabla'^2 - \frac{1}{\alpha} \frac{\partial}{\partial t'} \right) T(r', t') = -h(r', t') \quad (7-860)$$

together with the corresponding Green's-function equation,

$$\left( \nabla'^2 + \frac{1}{\alpha} \frac{\partial}{\partial t'} \right) G_-(r', -t'; r, -t) = -\delta(r - r') \delta(t - t') \quad (7-861)$$

where

$$\nabla'^2 = \frac{\partial^2}{\partial x'^2} + \frac{\partial^2}{\partial y'^2} + \frac{\partial^2}{\partial z'^2} \quad (7-862)$$

Green's theorem is now applied to Eqs. (7-860) and (7-861), and the result is integrated over  $dt'$  to give

$$\begin{aligned} \int_0^{t+\epsilon} dt' \int_{V'} \{G_- \nabla'^2 T - T \nabla'^2 G_-\} dV' \\ = \int_0^{t+\epsilon} dt' \int_{S(V')} \left\{ G_- \frac{\partial T}{\partial n'} - T \frac{\partial G_-}{\partial n'} \right\} dS' \end{aligned} \quad (7-863)$$

From this, we obtain

$$\begin{aligned} - \int_0^{t+\epsilon} dt' \int_{V'} G(r, t; r', t') h(r', t') dV' + T(r, t) \\ + \frac{1}{\alpha} \int_{V'} [G(r, t; r', t') T(r', t')] \Big|_{t'=0}^{t'=t+\epsilon} dV' \\ = \int_0^{t+\epsilon} dt' \int_{S(V')} \left\{ G \frac{\partial T}{\partial n'} - T \frac{\partial G}{\partial n'} \right\} dS' \end{aligned} \quad (7-864)$$

where  $\epsilon$  is an arbitrarily small positive number, and where the reciprocity relation,

$$G_-(r', -t'; r, -t) = G_-(r, t; r', t') = G(r, t; r', t') \quad (7-865)$$

has been employed.

By virtue of the causality requirement (7-850),

$$[G(r, t; r', t') T(r', t')]_{t'=t+\epsilon} = 0 \quad (7-866)$$

and this leaves us with

$$\begin{aligned} T(r, t) &= \int_0^{t+\epsilon} dt' \int_{V'} G(r, t; r', t') h(r', t') dV' \\ &\quad + \int_0^{t+\epsilon} dt' \int_{S(V')} \left\{ G \frac{\partial T}{\partial n'} - T \frac{\partial G}{\partial n'} \right\} dS' \\ &\quad + \frac{1}{\alpha} \int_{V'} G(r, t; r', 0) T(r', 0) dV' \end{aligned} \quad (7-867)$$

Equation (7-867) is the formal solution of the mixed initial- and boundary-value problem for the diffusion equation. Clearly,  $T(r, t)$  is determined by contributions from volume heat sources distributed throughout  $V'$  with a density  $h(r', t')$ , surface heat sources distributed over  $S(V')$ , and the initial temperature distribution  $T(r', 0)$ .

### 7-31 CONSTRUCTION OF THE INFINITE-MEDIUM GREEN'S FUNCTION FOR THE DIFFUSION EQUATION

Suppose we wish to solve

$$\left( \nabla^2 - \frac{1}{\alpha} \frac{\partial}{\partial t} \right) G(r, t; r', t') = -\delta(r - r') \delta(t - t') \quad (7-868)$$

in an unbounded medium of arbitrary dimension  $n < \infty$ . We can do so quite generally with the aid of the  $n$ -dimensional Fourier transform introduced in Sec. 5-18. Some simplification in the calculations will arise if we first observe that for a homogeneous isotropic medium of infinite extent, the Green's function should depend only on the distance  $R$  between the source and the observer and on the time  $\tau = t - t'$  which has elapsed since the source began to act. With these remarks,

$$G(r, t; r', t') = G(R, \tau) \quad (7-869)$$

where  $R = |r - r'|$  and  $\tau = t - t'$ , and the corresponding equation

for  $G(R, \tau)$  is

$$\left(\nabla^2 - \frac{1}{\alpha} \frac{\partial}{\partial \tau}\right) G(R, \tau) = -\delta(R) \delta(\tau) \quad (7-870)$$

Equation (7-870) is now multiplied on both sides by  $e^{-i\mathbf{K}\cdot\mathbf{R}} dV_{\mathbf{R}}$  and then integrated over all coordinate space, noting that

$$dV_{\mathbf{R}} = dx_1 dx_2 \cdots dx_n \quad (7-871)$$

$$\mathbf{K} \cdot \mathbf{R} = K_1(x_1 - x'_1) + K_2(x_2 - x'_2) + \cdots + K_n(x_n - x'_n) \quad (7-872)$$

$$R = [(x_1 - x'_1)^2 + \cdots + (x_n - x'_n)^2]^{1/2} \quad (7-873)$$

Since the  $x_i$ 's range over all possible values, the value  $R = 0$  is included in the range of integration, and consequently

$$\frac{\partial \tilde{G}}{\partial \tau} + \alpha K^2 \tilde{G}(\mathbf{K}, \tau) = \alpha \delta(\tau) \quad (7-874)$$

provided

$$\lim_{R \rightarrow \infty} [\nabla G(R, \tau) + i\mathbf{K}G(R, \tau)] = 0 \quad (7-875)$$

In Eq. (7-874)  $\tilde{G}(\mathbf{K}, \tau)$  is the  $n$ -dimensional Fourier transform of  $G(R, \tau)$  and is defined by

$$\tilde{G}(\mathbf{K}, \tau) = \int_{\text{All } \mathbf{R} \text{ space}} G(R, \tau) e^{-i\mathbf{K}\cdot\mathbf{R}} dV_{\mathbf{R}} \quad (7-876)$$

The corresponding inversion theorem for  $G(R, \tau)$  is

$$G(R, \tau) = \frac{1}{(2\pi)^n} \int_{\text{All } \mathbf{K} \text{ space}} \tilde{G}(\mathbf{K}, \tau) e^{i\mathbf{K}\cdot\mathbf{R}} dV_{\mathbf{K}} \quad (7-877)$$

Equation (7-874) is a simple ordinary differential equation which we can solve by reduction to quadrature as in Eq. (6-55). One finds

$$\frac{\partial}{\partial \tau} (e^{\alpha K^2 \tau} \tilde{G}) = e^{\alpha K^2 \tau} \left( \frac{\partial \tilde{G}}{\partial \tau} + \alpha K^2 \tilde{G} \right) = \alpha e^{\alpha K^2 \tau} \delta(\tau) \quad (7-878)$$

and therefore by integration,

$$e^{\alpha K^2 \tau} \tilde{G}(K, \tau) - \tilde{G}(K, 0) = \alpha H(\tau) \quad (7-879)$$

or

$$\tilde{G}(K, \tau) = \tilde{G}(K, 0) e^{-\alpha K^2 \tau} + \alpha H(\tau) e^{-\alpha K^2 \tau} \quad (7-880)$$

Causality requires that

$$G(R, \tau) = 0 \quad \tau \leq 0 \quad (7-881)$$

and by taking the Fourier transform of Eq. (7-881), we see that

$$\tilde{G}(K, \tau) = 0 \quad \tau \leq 0 \quad (7-882)$$

Hence

$$\tilde{G}(K, \tau) = \alpha H(\tau) e^{-\alpha K^2 \tau} \quad (7-883)$$

is the required solution of Eq. (7-874).

The Fourier inversion theorem (7-877) now yields

$$G(R, \tau) = \frac{\alpha H(\tau)}{(2\pi)^n} \int_{\text{All } \mathbf{K} \text{ space}} e^{i(\mathbf{K}\cdot\mathbf{R}) - \alpha K^2 \tau} dV_{\mathbf{K}} \quad (7-884)$$

where

$$dV_{\mathbf{K}} = dK_1 dK_2 \cdots dK_n \quad (7-885)$$

Our remaining efforts are directed toward evaluating Eq. (7-884) in closed form. To do this we notice that

$$G(R, \tau) = \frac{\alpha H(\tau)}{(2\pi)^n} \prod_{i=1}^n \int_{-\infty}^{+\infty} e^{i(K_i R_i) - \alpha K_i^2 \tau} dK_i \quad (7-886)$$

where

$$R_i = x_i - x'_i \quad (7-887)$$

and

$$K^2 = \sum_{i=1}^n K_i^2 \quad (7-888)$$

If we can evaluate

$$\int_{-\infty}^{+\infty} e^{i(K_i R_i) - \alpha K_i^2 \tau} dK_i = 2 \int_0^{\infty} e^{-\alpha K_i^2 \tau} \cos(K_i R_i) dK_i \quad (7-889)$$

then our problem is essentially solved. We notice that

$$\int_0^{\infty} e^{-\alpha K_i^2 \tau} \cos(K_i R_i) dK_i = \int_0^{\infty} dK_i \sum_{n=0}^{\infty} \frac{(-K_i^2 R_i^2)^n}{(2n)!} e^{-\alpha K_i^2 \tau} \quad (7-890)$$

and provided that Eq. (7-890) converges uniformly,

$$\int_0^{\infty} e^{-\alpha K_i^2 \tau} \cos(K_i R_i) dK_i = \sum_{n=0}^{\infty} \frac{(-R_i^2)^n}{(2n)!} \int_0^{\infty} e^{-\alpha K_i^2 \tau} K_i^{2n} dK_i \quad (7-891)$$

Therefore we must calculate

$$\int_0^{\infty} e^{-\alpha K_i^2 \tau} K_i^{2n} dK_i \quad (7-892)$$

For this purpose it is convenient to recall the result of Eqs. (6-398) to (6-402):

$$\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \quad (7-893)$$

Let  $x = \sqrt{\alpha \tau} K_i$  in Eq. (7-893) to obtain

$$I = \int_0^{\infty} e^{-\alpha K_i^2 \tau} dK_i = \frac{1}{2} \sqrt{\frac{\pi}{\alpha \tau}} \quad (7-894)$$

and then differentiate Eq. (7-894)  $n$  times with respect to  $\alpha\tau$ . The result is

$$\begin{aligned}\frac{d^n I}{d(\alpha\tau)^n} &= (-1)^n \int_0^\infty e^{-\alpha\tau K_i^2} K_i^{2n} dK_i \\ &= \frac{1}{2} \sqrt{\pi} \left[ \left(-\frac{1}{2}\right) \left(-\frac{3}{2}\right) \cdots \left(-\frac{2n-1}{2}\right) (\alpha\tau)^{-1/2-n} \right] \quad (7-895)\end{aligned}$$

Each time the right side of (Eq. 7-894) is differentiated with respect to  $\alpha\tau$ , a factor of  $-1$  is obtained; thus

$$\begin{aligned}\frac{d^n I}{d(\alpha\tau)^n} &= (-1)^n \int_0^\infty e^{-\alpha\tau K_i^2} K_i^{2n} dK_i \\ &= \frac{1}{2} \sqrt{\frac{\pi}{\alpha\tau}} (-1)^n \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^n (\alpha\tau)^n} \quad (7-896)\end{aligned}$$

It follows from Eq. (7-896) that

$$\int_0^\infty e^{-\alpha\tau K_i^2} K_i^{2n} dK_i = \frac{1}{2} \sqrt{\frac{\pi}{\alpha\tau}} \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^n (\alpha\tau)^n} \quad (7-897)$$

However,

$$1 \cdot 3 \cdot 5 \cdots (2n-1) = \frac{1 \cdot 2 \cdot 3 \cdots n \cdot (n+1) \cdots 2n}{2 \cdot 4 \cdot 6 \cdots 2n} \quad (7-898)$$

and

$$2 \cdot 4 \cdot 6 \cdots 2n = 2^n (1 \cdot 2 \cdot 3 \cdots n) = n! 2^n \quad (7-899)$$

Consequently

$$1 \cdot 3 \cdot 5 \cdots (2n-1) = \frac{(2n)!}{2^n n!} \quad (7-900)$$

and therefore

$$\int_0^\infty e^{-\alpha\tau K_i^2} K_i^{2n} dK_i = \frac{1}{2} \sqrt{\frac{\pi}{\alpha\tau}} \frac{(2n)!}{n! (4\alpha\tau)^n} \quad (7-901)$$

With this result, Eq. (7-891) gives us

$$\int_0^\infty e^{-\alpha\tau K_i^2} \cos(K_i R_i) dK_i = \sum_{n=0}^\infty \frac{(-R_i^2)^n}{(2n)!} \left[ \frac{1}{2} \sqrt{\frac{\pi}{\alpha\tau}} \frac{(2n)!}{n! (4\alpha\tau)^n} \right] \quad (7-902)$$

or

$$\begin{aligned}\int_0^\infty e^{-\alpha\tau K_i^2} \cos(K_i R_i) dK_i &= \frac{1}{2} \sqrt{\frac{\pi}{\alpha\tau}} \sum_{n=0}^\infty \frac{1}{n!} \left( \frac{-R_i^2}{4\alpha\tau} \right)^n \\ &= \frac{1}{2} \sqrt{\frac{\pi}{\alpha\tau}} e^{-R_i^2/4\alpha\tau} \quad (7-903)\end{aligned}$$

Thus Eq. (7-889) leads to

$$\int_{-\infty}^{+\infty} e^{i(K_i R_i) - \alpha\tau K_i^2} dK_i = \sqrt{\frac{\pi}{\alpha\tau}} e^{-R_i^2/4\alpha\tau} \quad (7-904)$$

and using this result, the  $n$ -dimensional infinite-medium Green's function (7-886) can be computed directly,

$$\begin{aligned}G(R, \tau) &= \frac{\alpha H(\tau)}{(2\pi)^n} \prod_{i=1}^n \sqrt{\frac{\pi}{\alpha\tau}} e^{-R_i^2/4\alpha\tau} \\ &= \frac{\alpha H(\tau)}{(2\pi)^n} \left( \frac{\pi}{\alpha\tau} \right)^{n/2} \exp \left( \frac{1}{4\alpha\tau} \sum_{i=1}^n R_i^2 \right) \\ &= \alpha H(\tau) \left( \frac{1}{4\pi\alpha\tau} \right)^{n/2} e^{-R^2/4\alpha\tau} \\ &= \alpha H(\tau) \left( \frac{1}{2\sqrt{\pi\alpha\tau}} \right)^n e^{-R^2/4\alpha\tau} \quad (7-905)\end{aligned}$$

where

$$R^2 = \sum_{i=1}^n R_i^2 = \sum_{i=1}^n (x_i - x'_i)^2 \quad (7-906)$$

To summarize, the  $n$ -dimensional infinite-medium Green's function for the diffusion equation is simply

$$G(R, \tau) = \alpha H(\tau) \left( \frac{1}{2\sqrt{\pi\alpha\tau}} \right)^n e^{-R^2/4\alpha\tau} \quad (7-907)$$

where  $R = |\mathbf{r} - \mathbf{r}'|$  is the distance between the source and the point of observation, and  $\tau = t - t'$  measures the time which has elapsed since the source was "turned on."

As an example of the application of Eq. (7-907), consider the problem of determining the temperature distribution in an infinitely long rod when the initial temperature distribution  $T(x', 0)$  is specified. Suppose that no heat sources are present and that  $G$  and its derivatives vanish at the infinitely remote end points of the rod. Then Eqs. (7-867) and (7-907) yield

$$T(x, t) = \frac{H(t)}{2\sqrt{\pi\alpha\tau}} \int_{-\infty}^{+\infty} e^{-(x-x')^2/4\alpha\tau} T(x', 0) dx' \quad (7-908)$$

for the solution of our initial-value problem.

## 7-32 PROBLEMS AND APPLICATIONS

1. Compute the solution of

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

subject to the boundary conditions

$$\begin{aligned}\phi(x, 0) &= \phi(x, b) = 0 & 0 \leq x \leq a \\ \phi(0, y) &= \phi_a(\text{const}) & 0 \leq y \leq b \\ \phi(a, y) &= 0 & 0 \leq y \leq b\end{aligned}$$

and find the limit of this solution as  $a \rightarrow \infty$ .

2. Find the solution of

$$\frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} = 0$$

that satisfies the following boundary conditions on the faces of a wedge:

$$\phi(r, \alpha) = f(r) \quad 0 \leq r < \infty$$

$$\phi(r, -\alpha) = g(r) \quad 0 \leq r < \infty$$

$$\lim_{r \rightarrow \infty} \phi(r, \theta) = 0 \quad -\alpha \leq \theta \leq +\alpha$$

Specialize this solution to the case in which the boundary conditions are the same on each wedge face: i.e.,

$$f(r) \equiv g(r) = \begin{cases} \phi_0 & 0 \leq r \leq a \\ 0 & r > a \end{cases}$$

and then let the wedge angle  $\alpha = \pi/2$ . Compare your result with Eq. (7-181).

3. Consider an infinitely long grounded rectangular cylinder whose generators lie along the  $z$  axis. Let its rectangular faces have coordinates  $x = 0$ ,  $x = a$ ,  $y = 0$ , and  $y = b$ . A point charge is placed at  $x = x'$ ,  $y = y'$ ,  $z = z'$  inside the cylinder. Compute the resulting potential inside the cylinder.
4. Repeat the calculation in Prob. 3 for the case of an infinitely long grounded *circular* cylinder.
5. A conducting sphere of radius  $r = a$  centered at  $r = 0$  is embedded in a homogeneous, isotropic dielectric of inductive capacity  $\epsilon_2$ . A point charge  $q$  is placed at  $z = z' > a$  on the  $z$  axis. Compute the electric potential everywhere outside the sphere, and calculate the surface charge density induced on the sphere.
6. Suppose we make the sphere in Prob. 5 a dielectric sphere with inductive capacity  $\epsilon_1$ . Compute the electric potential inside and outside this dielectric sphere.
7. Suppose the point source  $q$  in Prob. 6 is allowed to recede to infinity. As a result, the dielectric sphere of inductive capacity  $\epsilon_1$  finds itself embedded in a medium of inductive capacity  $\epsilon_2$  pervaded by a uniform constant electric field  $\mathbf{E}_0$  directed, say, along the  $z$  axis. Calculate the potentials and the electric fields inside and outside the sphere.
8. Compute the magnetic fields inside and outside an infinitely long straight wire of radius  $a$  and inductive capacity  $\mu_1$ , given that it carries a current  $I$ , and that it is embedded in an external medium of inductive capacity  $\mu_2$  containing a magnetic field  $\mathbf{B}_0$  directed transverse to the axis of the wire.
9. Calculate the time-dependent Green's function for the one-dimensional wave equation,

$$\left( \frac{\partial^2}{\partial x^2} - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) G(x, t; x', t') = -\delta(x - x') \delta(t - t')$$

assuming that  $G$  and  $\partial G / \partial t$  satisfy the causality condition,

$$G = 0 \quad t \leq t'$$

$$\frac{\partial G}{\partial t} = 0 \quad t \leq t'$$

10. Prove the reciprocity relation

$$G(x, t; x', t') = G(x', -t'; x, -t)$$

for the Green's function obtained in Prob. 9.

11. Obtain the formal integral solution of the mixed initial- and boundary-value problem for

$$\frac{\partial^2 u}{\partial x^2} - \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} = -\frac{1}{\rho c^2} f(x, t)$$

by using the Green's function found in Prob. 9 with an appropriate Green's theorem.

12. Solve the Cauchy initial-value problem for the wave equation,

$$\frac{\partial^2 u}{\partial x^2} - \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} = 0$$

$$u(x, 0) = F(x)$$

$$\frac{\partial u}{\partial t}(x, 0) = G(x)$$

in a one-dimensional medium of infinite extent. Use the results of Probs. 9 and 11 to obtain your answer, and then compare that answer with Eq. (7-460).

13. Consider the equation

$$(\nabla^2 + k^2)u(\mathbf{r}) = -F(\mathbf{r})u(\mathbf{r}) \quad (A)$$

in a volume  $V$ , and suppose that its solutions satisfy specified boundary conditions on the surface  $S(V)$  enclosing  $V$ . The values of  $k^2$  for which this equation can be solved, subject to the specified boundary conditions on  $u$ , are called the "eigenvalues" of the equation. These are denoted by  $k_n^2$ , and for each such  $k_n$  there is a corresponding eigenfunction  $u_n(\mathbf{r})$ . The spectrum of eigenvalues  $k_n$  may be a discrete set, or a continuum, or both, and the eigenfunctions must satisfy

$$(\nabla^2 + k_n^2)u_n(\mathbf{r}) = -F(\mathbf{r})u_n(\mathbf{r})$$

Assume that the eigenfunctions  $u_n(\mathbf{r})$  are normalized to unity so that

$$\int_V u_n^*(\mathbf{r})u_n(\mathbf{r}) dV = 1$$

where  $u_n^*(\mathbf{r})$  is the complex conjugate of  $u_n(\mathbf{r})$ .

Prove that

$$\int_V u_n^*(\mathbf{r})u_m(\mathbf{r}) dV = \delta_{nm}$$

when  $n$  and  $m$  are discrete parameters, and prove that

$$\int_V u_n^*(\mathbf{r}) u_m(\mathbf{r}) dV = \delta(n - m)$$

when  $n$  and  $m$  vary continuously.

14. Let us assume that the eigenfunctions  $u_n(\mathbf{r})$ , defined in Prob. 13, form a complete set. The Green's function for

$$(\nabla^2 + k^2)u(\mathbf{r}) = -F(\mathbf{r})u(\mathbf{r})$$

is defined as the solution of

$$(\nabla^2 + k^2)G(\mathbf{r}, \mathbf{r}') = -F(\mathbf{r})G(\mathbf{r}, \mathbf{r}') - \delta(\mathbf{r} - \mathbf{r}') \quad (B)$$

subject to the conditions that:

- $k^2 \neq k_n^2$  for any value of  $n$ ; i.e.,  $k$  must not be any one of the eigenvalues  $k_n$ .
- $G(\mathbf{r}, \mathbf{r}')$  must satisfy the same boundary conditions on  $S(V)$  as the eigenfunctions  $u_n(\mathbf{r})$ .

Show that the Green's function  $G(\mathbf{r}, \mathbf{r}')$  can be expanded in terms of the eigenfunctions  $u_n(\mathbf{r})$  as

$$G(\mathbf{r}, \mathbf{r}') = \sum_{n=1}^{\infty} \frac{u_n^*(\mathbf{r}') u_n(\mathbf{r})}{k_n^2 - k^2} \quad (C)$$

provided that the spectrum of eigenvalues  $\{k_n\}$  forms a countably infinite set. When the spectrum is continuous, show that

$$G(\mathbf{r}, \mathbf{r}') = \int_{V_K} \frac{u^*(\mathbf{r}', \mathbf{K}) u(\mathbf{r}, \mathbf{K})}{K^2 - k^2} dV_K \quad (D)$$

The volume of integrations  $V_K$  in Eq. (D) is the three-dimensional region composed of all the points  $\mathbf{K} = |\mathbf{K}|$  which satisfy

$$(\nabla^2 + K^2)u(\mathbf{r}, \mathbf{K}) = -F(\mathbf{r})u(\mathbf{r}, \mathbf{K}) \quad (E)$$

and which therefore must be eigenvalues of Eq. (A). The point  $k^2$  is specifically excluded from  $V_K$ .

15. Consider the three-dimensional Helmholtz equation in an unbounded domain, using the eigenvalue-eigenfunction point of view. For this purpose, let  $F(\mathbf{r}) \equiv 0$  in Eq. (A). Show that the normalized eigenfunctions are

$$u(\mathbf{r}, \mathbf{K}) = \frac{1}{(2\pi)^{3/2}} e^{i\mathbf{K} \cdot \mathbf{r}}$$

$$u^*(\mathbf{r}', \mathbf{K}) = \frac{1}{(2\pi)^{3/2}} e^{-i\mathbf{K} \cdot \mathbf{r}'}$$

and verify that the solution of Eq. (B) as given by Eq. (D) is the same as the result obtained in Eq. (7-630).

16. Use Eq. (C) to solve

$$\nabla^2 G(\mathbf{r}, \mathbf{r}') = -\delta(\mathbf{r} - \mathbf{r}')$$

subject to the boundary condition that  $G$  must vanish on the six faces of a rectangular box whose sides are at  $x = y = z = 0$ , and  $x = a, y = b, z = c$ .

17. Solve

$$(\nabla^2 - k^2)G(\mathbf{r}|\mathbf{r}') = -\delta(\mathbf{r} - \mathbf{r}')$$

in plane polar coordinates, and use your results to derive the addition theorem for hyperbolic Bessel functions,

$$K_0(kR) = I_0(kr_<)K_0(kr_>) + 2 \sum_{n=1}^{\infty} I_n(kr_<)K_n(kr_>) \cos n(\theta - \theta')$$

where

$$R^2 = r_>^2 + r_<^2 - 2r_>r_< \cos(\theta - \theta')$$

18. Prove that

$$K_0\left(\frac{sR}{c}\right) = \int_0^{\infty} \frac{H(t - R/c)}{\sqrt{t^2 - R^2/c^2}} e^{-st} dt$$

and use this result to solve

$$\left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) G(\mathbf{r}, t; \mathbf{r}', t') = -\delta(\mathbf{r} - \mathbf{r}') \delta(t - t')$$

in an unbounded plane region, assuming that

$$G = 0 \quad t \leq t'$$

$$\frac{\partial G}{\partial t} = 0 \quad t \leq t'$$

19. Show that the infinite-medium Green's function  $g$  which satisfies

$$\left(c_1^2 \frac{\partial^2}{\partial x^2} + c_2^2 \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial t^2}\right) g(x, y, t) = -\delta(t) \delta(x) \delta(y)$$

$$g(x, y, 0) = 0$$

$$\frac{\partial g}{\partial t}(x, y, 0) = 0$$

is given by

$$g(x, y, t) = \frac{1}{2\pi c_2^2} \frac{H[t - (R/c_1)(1 - \epsilon^2 \cos^2 \theta)]}{\sqrt{t^2 - (R^2/c_1^2)(1 - \epsilon^2 \cos^2 \theta)}}$$

where

$$R^2 = x^2 + y^2$$

$$x = y \tan \theta$$

and where  $c_1$  and  $c_2$  are constants such that

$$\epsilon = \left(1 - \frac{c_1^2}{c_2^2}\right)^{1/2}$$

Interpret this problem physically.

20. Solve the following boundary-value problem in the half space

$$-\infty < x < +\infty; y \geq 0:$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + k^2 u = 0$$

$$u(x, 0) = f(x) \quad -\infty < x < +\infty$$

21. Repeat the calculation in Prob. 20, this time using the boundary condition

$$\frac{\partial u(x, 0)}{\partial y} = g(x) \quad -\infty < x < +\infty$$

22. Consider a time-harmonic plane wave  $u_i = e^{-ikr \cos \theta}$  incident on a circular cylinder  $r = a$ , where  $r$  and  $\theta$  are two-dimensional polar coordinates such that  $0 \leq \theta < 2\pi$ . The scattered wave  $u_s$  must satisfy

$$(\nabla^2 + k^2)u_s = 0 \quad r \geq a$$

and the boundary condition

$$u_s(a, \theta) = -u_i(a, \theta) = -e^{-ika \cos \theta} \text{ on } r = a$$

Show that

$$u_s(r, \theta) = - \sum_{n=-\infty}^{+\infty} \frac{(-i)^n J_n(ka)}{H_n^{(2)}(ka)} H_n^{(2)}(kr) e^{in\theta}$$

23. Repeat the calculation in Prob. 22, assuming that the appropriate boundary condition on  $r = a$  is

$$\left(\frac{\partial u_s}{\partial r}\right)_{r=a} = - \left(\frac{\partial u_i}{\partial r}\right)_{r=a}$$

24. Consider a time-harmonic plane wave  $u_i = e^{ikr \cos \theta}$  incident on a sphere of radius  $r = a$ , where  $(r, \theta, \phi)$  are spherical polar coordinates. The scattered wave  $u_s$  must satisfy

$$(\nabla^2 + k^2)u_s = 0 \quad r \geq a$$

and the boundary condition

$$u_s(a, \theta, \phi) = -u_i(a, \theta, \phi) = e^{ika \cos \theta}$$

on the surface of the sphere  $r = a$ . Show that

$$u_s(r, \theta, \phi) = - \sum_{n=0}^{\infty} (2n+1) i^n P_n(\cos \theta) \frac{j_n(ka)}{h_n^{(1)}(ka)} h_n^{(1)}(kr)$$

25. Repeat the calculation in Prob. 24 for the boundary condition

$$\left(\frac{\partial u_s}{\partial r}\right)_{r=a} = - \left(\frac{\partial u_i}{\partial r}\right)_{r=a}$$

26. Consider the two-dimensional Helmholtz equation in plane polar coordinates  $(r, \theta)$ :

$$(\nabla^2 - k^2)G(r, \theta; r_1, \theta_1) = \frac{-\delta(r - r_1)}{r} \delta(\theta - \theta_1)$$

Show that the solution of this equation in a two-dimensional wedge-shaped region  $\{0 \leq r < \infty, 0 \leq \theta \leq \alpha\}$  is

$$G = \frac{1}{\alpha} \sum_{n=0}^{\infty} \frac{K_{n\pi/\alpha}(kr_>)}{\alpha} I_{n\pi/\alpha}(kr_<) \left\{ \cos \frac{n\pi}{\alpha} (\theta - \theta_1) - \cos \frac{n\pi}{\alpha} (\theta + \theta_1) \right\}$$

provided that

$$G(r, 0; r_1, \theta_1) = G(r, \alpha; r_1, \theta_1) = 0 \quad 0 \leq r < \infty$$

27. Repeat the calculations in Prob. 26, using the boundary conditions

$$\left(\frac{\partial G}{\partial \theta}\right)_{\theta=0} = \left(\frac{\partial G}{\partial \theta}\right)_{\theta=\alpha} = 0 \quad 0 \leq r < \infty$$

28. Consider the equation

$$(\nabla^2 + k^2)u(x, y, z) = 0$$

and assume that as  $k \rightarrow \infty$ ,

$$u \sim e^{ik\psi(x, y, z)} \sum_{n=0}^{\infty} \frac{V_n(x, y, z)}{(ik)^n}$$

where  $(\nabla\psi)^2 = 1$ . Let  $x = x(s)$ ,  $y = y(s)$ ,  $z = z(s)$  be the parametric equation of a spatial curve or ray which intersects  $\psi(x, y, z) = \text{constant}$  in the normal direction, i.e., along  $\nabla\psi$ . Show that in terms of the arc length  $s$  measured along a ray,

$$V_n(s) = V_n(s_0) e^{-\frac{1}{2} \int_{s_0}^s \nabla^2 \psi ds'} - \frac{1}{2} \int_{s_0}^s e^{-\frac{1}{2} \int_{s_0}^{\tau} \nabla^2 \psi ds'} \nabla^2 V_{n-1}(\tau) d\tau$$

29. Consider two fluid half spaces, 1 and 2, in contact with each other along the plane  $z = 0$ . The generation of an acoustic pulse by a point source at  $r = 0$ ,  $z = h$  in fluid 1 leads to the following boundary-value problem:

Solve

$$(\nabla^2 - k_1^2)u_1 = \frac{-\delta(r) \delta(z - h)}{2\pi r}$$

$$(\nabla^2 - k_2^2)u_2 = 0$$

subject to the boundary conditions

$$\rho_1 u_1 = \rho_2 u_2 \quad z = 0$$

$$\left(\frac{\partial u_1}{\partial z}\right)_{z=0} = \left(\frac{\partial u_2}{\partial z}\right)_{z=0}$$

$$\lim_{|z| \rightarrow \infty} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = 0 \quad 0 < r < \infty$$

Show that the solution to this problem is given by

$$u_1 = \frac{1}{2\pi} \int_0^\infty \left\{ \frac{\nu_1 \delta_1 \cosh \nu_1 z + \nu_2 \sinh \nu_1 z}{\nu_1(\nu_2 + \nu_1 \delta_1)} \right\} e^{-\nu_1 h} J_0(kr) k dk \quad 0 \leq z \leq h$$

$$u_1 = \frac{1}{2\pi} \int_0^\infty \left\{ \frac{\nu_1 \delta_1 \cosh \nu_1 h + \nu_2 \sinh \nu_1 h}{\nu_1(\nu_2 + \nu_1 \delta_1)} \right\} e^{-\nu_1 z} J_0(kr) k dk \quad z \geq h$$

$$u_2 = \frac{1}{2\pi} \int_0^\infty \frac{e^{\nu_2 z - \nu_1 h}}{\nu_1 \delta_1 + \nu_2} J_0(kr) k dk \quad z \leq 0$$

where

$$\delta_1 = \rho_2 / \rho_1$$

$$\nu_1 = \sqrt{k^2 - k_1^2}$$

$$\nu_2 = \sqrt{k^2 - k_2^2}$$

$$\operatorname{Re}(\nu_1) \geq 0$$

$$\operatorname{Re}(\nu_2) \geq 0$$

30. Consider the temperature  $T(x, t)$  in a semi-infinite bar  $0 < x < \infty$ , one end of which is subjected to a sinusoidal temperature variation  $T(0, t) = T_0 \cos \omega t$ . Show that the temperature distribution in this bar is given by

$$T(x, t) = T_0 e^{-x\sqrt{\omega/2\alpha}} \cos\left(\omega t - x\sqrt{\frac{\omega}{2\alpha}}\right)$$

The solution seems to be a damped temperature wave whose speed of propagation becomes infinite as  $\omega$  increases indefinitely. Can you explain this?

31. Solve

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

subject to the boundary conditions

$$T(0, t) = 0 \quad \text{for all } t$$

$$\left(\frac{\partial T}{\partial x}\right)_{x=a} = 0 \quad \text{for all } t$$

given the initial condition

$$T(x, 0) = T_0 \left(\frac{x}{a}\right) \quad 0 \leq x \leq a$$

32. Solve the diffusion equation,

$$\nabla^2 T = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

in plane polar coordinates  $(r, \theta)$  subject to the wedge boundary conditions

$$T(r, 0, t) = 0 \quad r > 0, t > 0$$

$$T(r, \alpha, t) = T_0 \quad r > 0, t > 0$$

and the initial condition

$$T(r, \theta, 0) = 0 \quad 0 < \theta < \alpha, r > 0$$

33. Solve

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

in the half space  $x > 0$ , given the boundary condition

$$T = T_0 \text{ at } x = 0 \quad t > 0$$

and the initial condition

$$T = 0 \text{ at } t = 0 \quad x > 0$$

34. Solve

$$(\nabla^2 - k)G(\mathbf{r}, \mathbf{r}') = -\delta(\mathbf{r} - \mathbf{r}')$$

in cylindrical coordinates  $(r, \theta, z)$  for an unbounded medium.

35. Solve

$$(\nabla^2 - k)G(\mathbf{r}, \mathbf{r}') = -\delta(\mathbf{r} - \mathbf{r}')$$

in spherical coordinates  $(r, \theta, \phi)$  for an unbounded medium.

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